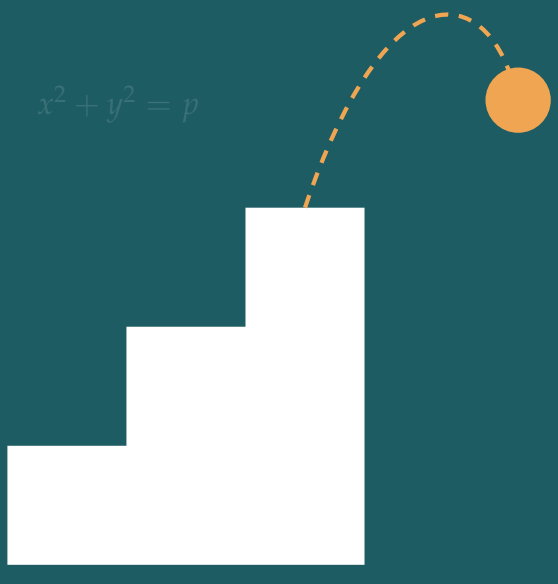
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2014 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2014 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2014 Worked Solutions

Worked solutions to all 39 questions of the 2014 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2014, Question 1

Pure • Differences of Squares

> **STEP I 2014, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/14-s1-q1

The key idea

Everything rests on one factorisation: $a^2 - b^2 = (a-b)(a+b)$. Writing n as a difference of two squares is the same as writing $n = de$ with $d = a - b$ and $e = a + b$, and since d and e differ by the even number $2b$, they must have the same parity. Once you see that, every part becomes a question about factor pairs of the same parity.

Solution

Throughout, a and b are non-negative integers. If $n = a^2 - b^2$ with $n \geq 1$ then $a > b$, so we may always list a way as (a, b) with $a > b \geq 0$.

Part (i)

$$3 = 2^2 - 1^2, \quad 5 = 3^2 - 2^2, \quad 8 = 3^2 - 1^2, \quad 12 = 4^2 - 2^2, \quad 16 = 5^2 - 3^2.$$

(For 16 the obvious $4^2 - 0^2$ is not allowed, as the squares must be non-zero.)

Part (ii)

Consecutive squares differ by an odd number: $(m+1)^2 - m^2 = 2m + 1$. So any odd number $2m + 1$, with $m \geq 0$, is

$$2m + 1 = (m+1)^2 - m^2.$$

Part (iii)

Squares two apart differ by a multiple of 4: $(k+1)^2 - (k-1)^2 = 4k$. For $k \geq 1$ both $k + 1$ and $k - 1$ are non-negative, so

$$4k = (k+1)^2 - (k-1)^2.$$

For $k = 0$ we simply have $0 = 0^2 - 0^2$ (or $1^2 - 1^2$).

Part (iv)

Every square leaves remainder 0 or 1 on division by 4: $(2m)^2 = 4m^2$ and $(2m+1)^2 = 4(m^2+m)+1$. So $a^2 - b^2$ leaves remainder

$$0-0=0, \quad 1-1=0, \quad 1-0=1 \quad \text{or} \quad 0-1 \equiv 3 \pmod 4,$$

and never 2. Hence no number of the form $4k+2$ is a difference of two squares.

(Equivalently: if $(a-b)(a+b) = 4k+2$, then $a-b$ and $a+b$ have the same parity. Both odd gives an odd product; both even gives a product divisible by 4. Neither is $4k+2$.)

Part (v)

Suppose $pq = a^2 - b^2 = (a-b)(a+b)$ with $a > b \geq 0$. Put $d = a-b$ and $e = a+b$. Then d and e are positive integers with $d \leq e$ and $de = pq$. Conversely, any factorisation $pq = de$ with $d \leq e$ and d, e of the same parity gives exactly one way, namely

$$a = \frac{e+d}{2}, \qquad b = \frac{e-d}{2},$$

which are non-negative integers. So the ways of writing pq correspond one-to-one with such factor pairs (d, e).

Ask yourself Why do d and e always have the same parity here, and why are there only two pairs?

Since p and q are odd primes, pq is odd, so every divisor of pq is odd and the parity condition holds automatically. Take $p \leq q$. By unique factorisation, the only factor pairs with $d \leq e$ are

$$(d, e) = (1, pq) \qquad \text{and} \qquad (d, e) = (p, q).$$

(This is still true when $p = q$: the divisors of p^2 are 1, p and p^2.) These give

$$pq = \left(\frac{pq+1}{2}\right)^2 - \left(\frac{pq-1}{2}\right)^2 \qquad \text{and} \qquad pq = \left(\frac{p+q}{2}\right)^2 - \left(\frac{q-p}{2}\right)^2.$$

The two ways are distinct, because their values of $d = a-b$ are 1 and p, and $p \geq 3$. So pq is a difference of two squares in exactly two ways. For example, $15 = 8^2 - 7^2 = 4^2 - 1^2$. When $p = q$ the second way is $p^2 - 0^2$, which is allowed here, as only part (i) demands non-zero squares.

The result does *not* hold when $q = 2$. Then $pq = 2p$ with p odd, say $p = 2m+1$, so $2p = 4m+2$. By part (iv), $2p$ cannot be written as a difference of two squares in any way at all. (The factor pairs $(1, 2p)$ and $(2, p)$ each have one odd and one even member.)

Part (vi)

As in part (v), the ways of writing 675 correspond to factor pairs $675 = de$ with $d \leq e$ and d, e of the same parity. Since 675 is odd, all its divisors are odd, so every factor pair counts.

Now $675 = 27 \times 25 = 3^3 \times 5^2$, which has $(3+1)(2+1) = 12$ divisors. They pair off as $d \leftrightarrow 675/d$, and no divisor is paired with itself because 675 is not a perfect square ($26^2 = 676$). So there are $12/2 = 6$ pairs:

(d,e)	$(1,675)$	$(3,225)$	$(5,135)$	$(9,75)$	$(15,45)$	$(25,27)$
(a,b)	$(338,337)$	$(114,111)$	$(70,65)$	$(42,33)$	$(30,15)$	$(26,1)$

So 675 can be written as a difference of two squares in exactly **6** distinct ways.

? **Ask yourself** Should (a,b) and $(-a,b)$, or a^2-b^2 and $-(b^2-a^2)$, count as different ways?

No. The question restricts everything to non-negative integers, and $a^2 - b^2 = 675 > 0$ forces $a > b$. So each way is a single pair (a,b) with $a > b \geq 0$, and these are what we have counted.

Checks

- Spot-check the table: $26^2 - 1^2 = 676 - 1 = 675$ and $30^2 - 15^2 = 900 - 225 = 675$.
- Parts (ii)–(iv) together say that $n \geq 1$ is a difference of two squares exactly when $n \not\equiv 2 \pmod 4$. A computer check of every n up to 2000 agrees, as does a check that pq has exactly two representations for all odd primes $p \leq q < 200$, and $2p$ has none.
- A search over all b finds exactly six representations of 675, the six in the table.

Common slips

- Forgetting the parity condition: counting a pair like $(2,p)$ for $2p$, which gives the non-integer $a = \frac{p+2}{2}$.
- Missing the case $p = q$ in part (v), or rejecting $p^2 - 0^2$ there. Zero is allowed outside part (i).
- Counting divisors (12) instead of factor pairs (6) in part (vi).

Beyond STEP

The pattern here, that the number of representations is controlled by the divisors of n, is typical of elementary number theory. A famous relative is Fermat's theorem on sums of two squares: an odd prime is $x^2 + y^2$ exactly when it is 1 mod 4, a statement proved neatly using the Gaussian integers $a + bi$. For more, read Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 2014, Question 2

Pure • Integration of Logarithms

STEP I 2014, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q2

The key idea

Everything rests on factorising $x^2-4=(x-2)(x+2)$, so that a logarithm of a quadratic splits into $\ln(2-x)+\ln(2+x)$, two pieces that part (i) (and its mirror image) integrates at once. The modulus signs then only ask us to keep track of where $\ln|x^2-4|$ is positive and where it is negative, and the sketch in (ii) shows exactly that.

Solution

Part (i)

For $x<2$ we have $2-x>0$, so everything is defined. We differentiate the right-hand side, using the product rule on $-(2-x)\ln(2-x)$. The derivative of $-(2-x)$ is 1, and by the chain rule the derivative of $\ln(2-x)$ is $-\frac{1}{2-x}$. So

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left[-(2-x)\ln(2-x)+(2-x)\right] &= 1\cdot\ln(2-x)-(2-x)\cdot\left(-\frac{1}{2-x}\right)-1\\ &= \ln(2-x)+1-1=\ln(2-x),\end{aligned}$$

and the stated antiderivative is correct.

(Alternatively, substitute $u=2-x$ and use $\int \ln u\,\mathrm{d}u = u\ln u-u$.)

In the same way, for $x>-2$,

$$\int \ln(2+x)\,\mathrm{d}x=(2+x)\ln(2+x)-(2+x)+c,$$

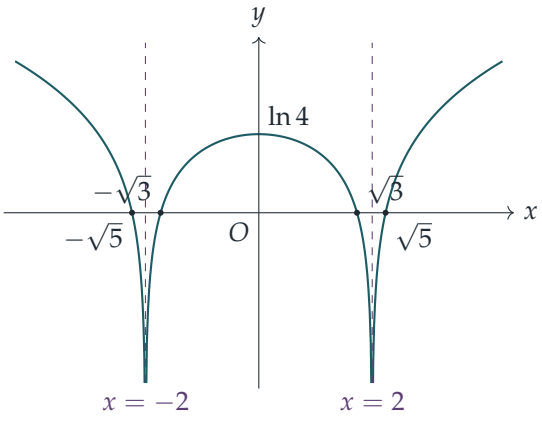
which we shall also need.

Part (ii)

Write $f(x)=\ln|x^2-4|$, defined for $x\neq\pm2$. Since $f(-x)=f(x)$, the curve is symmetric in the y-axis.

- For $|x| < 2$: $f(x) = \ln(4 - x^2)$, which is largest at $x = 0$, where $y = \ln 4$. It is zero when $4 - x^2 = 1$, i.e. $x = \pm\sqrt{3}$, and tends to $-\infty$ as $x \to \pm 2$.
- For $|x| > 2$: $f(x) = \ln(x^2 - 4)$, which rises from $-\infty$ near $x = \pm 2$, crosses the axis where $x^2 - 4 = 1$, i.e. $x = \pm\sqrt{5}$, and then grows slowly, like $2\ln|x|$.

The lines $x = \pm 2$ are vertical asymptotes.



Part (iii)

The finite region bounded by the positive x-axis, the y-axis and A is the arch between $x = 0$ and $x = \sqrt{3}$, where $y = \ln(4 - x^2) \geq 0$. Its area is

$$\int_0^{\sqrt{3}} \ln(4 - x^2)\,\mathrm{d}x = \int_0^{\sqrt{3}} \ln(2 - x)\,\mathrm{d}x + \int_0^{\sqrt{3}} \ln(2 + x)\,\mathrm{d}x,$$

using $\ln(4 - x^2) = \ln(2 - x) + \ln(2 + x)$, valid as both factors are positive here. By part (i),

$$\begin{aligned}\int_0^{\sqrt{3}} \ln(2 - x)\,\mathrm{d}x &= \Big[-(2 - x)\ln(2 - x) + (2 - x)\Big]_0^{\sqrt{3}} \\ &= -(2 - \sqrt{3})\ln(2 - \sqrt{3}) + (2 - \sqrt{3}) + 2\ln 2 - 2 \\ &= -(2 - \sqrt{3})\ln(2 - \sqrt{3}) - \sqrt{3} + 2\ln 2,\end{aligned}$$

and

$$\begin{aligned}\int_0^{\sqrt{3}} \ln(2 + x)\,\mathrm{d}x &= \Big[(2 + x)\ln(2 + x) - (2 + x)\Big]_0^{\sqrt{3}} \\ &= (2 + \sqrt{3})\ln(2 + \sqrt{3}) - (2 + \sqrt{3}) - 2\ln 2 + 2 \\ &= (2 + \sqrt{3})\ln(2 + \sqrt{3}) - \sqrt{3} - 2\ln 2.\end{aligned}$$

Ask yourself How do we combine $\ln(2 - \sqrt{3})$ with $\ln(2 + \sqrt{3})$?

Since $(2-\sqrt{3})(2+\sqrt{3}) = 4-3 = 1$, we have $\ln(2-\sqrt{3}) = -\ln(2+\sqrt{3})$. Adding the two integrals, the $\ln 2$ terms cancel and

$$\begin{aligned}\int_0^{\sqrt{3}} \ln(4-x^2)\,\mathrm{d}x &= (2-\sqrt{3})\ln(2+\sqrt{3}) + (2+\sqrt{3})\ln(2+\sqrt{3}) - 2\sqrt{3} \\ &= 4\ln(2+\sqrt{3}) - 2\sqrt{3},\end{aligned}$$

as required.

Part (iv)

For $|x| < 2$, B is the part of A between the asymptotes with the pieces below the axis reflected upwards. By the sketch, $\ln(4-x^2) \geq 0$ for $|x| \leq \sqrt{3}$ and $\ln(4-x^2) < 0$ for $\sqrt{3} < |x| < 2$. Using symmetry, the required area is

$$2\left[\int_0^{\sqrt{3}} \ln(4-x^2)\,\mathrm{d}x - \int_{\sqrt{3}}^{2} \ln(4-x^2)\,\mathrm{d}x\right].$$

The second integral is improper, because the integrand tends to $-\infty$ at $x = 2$. We integrate up to $x = 2-\varepsilon$ and let $\varepsilon \to 0^+$. In the antiderivative of $\ln(2-x)$ from (i), at $x = 2-\varepsilon$ we get

$$-\varepsilon\ln\varepsilon + \varepsilon \to 0,$$

by the given fact that $t\ln t \to 0$ as $t \to 0$. So the upper limit contributes 0 there, and

$$\begin{aligned}\int_{\sqrt{3}}^{2} \ln(2-x)\,\mathrm{d}x &= 0 - \left[-(2-\sqrt{3})\ln(2-\sqrt{3}) + (2-\sqrt{3})\right] \\ &= -(2-\sqrt{3})\ln(2+\sqrt{3}) - 2 + \sqrt{3},\end{aligned}$$

again using $\ln(2-\sqrt{3}) = -\ln(2+\sqrt{3})$. The other piece is an ordinary integral:

$$\begin{aligned}\int_{\sqrt{3}}^{2} \ln(2+x)\,\mathrm{d}x &= (4\ln 4 - 4) - \left[(2+\sqrt{3})\ln(2+\sqrt{3}) - (2+\sqrt{3})\right] \\ &= 8\ln 2 - 2 + \sqrt{3} - (2+\sqrt{3})\ln(2+\sqrt{3}).\end{aligned}$$

Adding,

$$\int_{\sqrt{3}}^{2} \ln(4-x^2)\,\mathrm{d}x = 8\ln 2 - 4 + 2\sqrt{3} - 4\ln(2+\sqrt{3}),$$

which is about -0.259: negative, as it should be. Therefore the area is

$$\begin{aligned}&2\Big[\big(4\ln(2+\sqrt{3}) - 2\sqrt{3}\big) - \big(8\ln 2 - 4 + 2\sqrt{3} - 4\ln(2+\sqrt{3})\big)\Big] \\ &\quad= 16\ln(2+\sqrt{3}) - 16\ln 2 + 8 - 8\sqrt{3} \\ &\quad= 16\ln\Big(\frac{2+\sqrt{3}}{2}\Big) + 8 - 8\sqrt{3} \approx 4.125.\end{aligned}$$

Checks

- Adding the two integrals over $[0, \sqrt{3}]$ and $[\sqrt{3}, 2]$ gives $\int_0^2 \ln(4 - x^2)\,\mathrm{d}x = 8\ln 2 - 4 \approx 1.545$. You can get this directly from (i) in one line, since at $x = 0$ the two antiderivatives give $-2\ln 2 + 2$ and $2\ln 2 - 2$, which cancel.
- The area in (iii) is $4\ln(2 + \sqrt{3}) - 2\sqrt{3} \approx 1.804$. That is less than the rectangle $\sqrt{3} \times \ln 4 \approx 2.40$ enclosing the arch, as it must be.
- Numerical integration of $\left|\ln|x^2 - 4|\right|$ over $(-2, 2)$ gives 4.1246, matching $16\ln\left(\frac{2+\sqrt{3}}{2}\right) + 8 - 8\sqrt{3} = 4.12457\ldots$.

Common slips

- Forgetting that the part of the curve between $x = \sqrt{3}$ and $x = 2$ lies below the axis, and so adding the signed integral instead of subtracting it.
- Leaving $\ln(2 - \sqrt{3})$ uncombined: the identity $(2 - \sqrt{3})(2 + \sqrt{3}) = 1$ is what makes the answers tidy.
- Simply substituting $x = 2$ into $(2 - x)\ln(2 - x)$ without comment: the expression is not defined there, and it is the limit $t \ln t \to 0$ that justifies the value 0.

Beyond STEP

The integral in (iv) is an *improper integral*: the integrand is unbounded, yet the area is finite because a logarithm blows up so slowly that $t \ln t \to 0$. At university you will make this precise, deciding exactly when $\int_0^1 g(t)\,\mathrm{d}t$ converges for unbounded g (for example, $\int_0^1 t^{-p}\,\mathrm{d}t$ converges exactly when $p < 1$). Abbott's *Understanding Analysis* treats limits like $t \ln t \to 0$ and the definition of the integral carefully.

WORKED SOLUTION

STEP I 2014, Question 3

Pure • Polynomial Equations and Inequalities

STEP I 2014, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q3

The key idea

Both integrals are elementary, so the condition is just a polynomial equation in a and b. The work is in spotting the common factor $b-a$ (which is non-zero, so we may cancel it) and, in part (iii), in noticing that the leftover expression is symmetric enough to be written neatly in terms of the sum $b+a$ and the difference $b-a$.

Solution

Evaluating the integrals, the condition is

$$\frac{b^3-a^3}{3} = \left(\frac{b^2-a^2}{2}\right)^2 = \frac{(b-a)^2(b+a)^2}{4}. \qquad (*)$$

Since $b^3-a^3 = (b-a)(b^2+ab+a^2)$ and $b-a>0$, we can divide $(*)$ by $b-a$ to get

$$4(b^2+ab+a^2) = 3(b-a)(b+a)^2. \qquad (**)$$

Part (i)

With $a=0$, $(**)$ becomes $4b^2 = 3b^3$. As $b>0$ we can divide by b^2, giving

$$b = \tfrac{4}{3}.$$

Part (ii)

With $a=1$, $(**)$ becomes

$$4(b^2+b+1) = 3(b-1)(b+1)^2 = 3(b^3+b^2-b-1),$$

that is, $3b^3+3b^2-3b-3-4b^2-4b-4=0$, or

$$3b^3-b^2-7b-7=0,$$

as required.

Now let $f(x) = 3x^3 - x^2 - 7x - 7$, so that

$$f'(x) = 9x^2 - 2x - 7 = (9x+7)(x-1).$$

The stationary points are at $x = -\frac{7}{9}$ (a local maximum, since the cubic has positive leading coefficient) and $x = 1$ (a local minimum), with

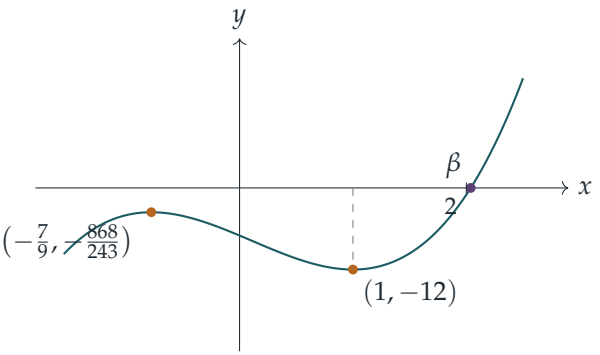
$$f(1) = 3 - 1 - 7 - 7 = -12, \qquad f\left(-\tfrac{7}{9}\right) = \frac{-343 - 147 + 1323 - 1701}{243} = -\frac{868}{243}.$$

Ask yourself How does knowing the value at the local maximum tell us how many roots there are?

The local maximum value is negative. For $x \leq 1$, f increases up to $x = -\frac{7}{9}$ and then decreases, so $f(x) \leq f(-\frac{7}{9}) < 0$ there: no roots with $x \leq 1$. For $x > 1$, f is strictly increasing (both factors of f' are positive) and $f(x) \to \infty$, so it crosses zero exactly once. Hence there is exactly one real root, and

$$f(2) = 24 - 4 - 14 - 7 = -1 < 0, \qquad f(3) = 81 - 9 - 21 - 7 = 44 > 0,$$

so by continuity it lies between 2 and 3.



The sketch shows the curve $y = 3x^3 - x^2 - 7x - 7$: both turning points lie below the x-axis, so it crosses only once, at $\beta \approx 2.04$.

Part (iii)

With $p = b + a$ and $q = b - a$ we have $p^2 = b^2 + 2ab + a^2$ and $q^2 = b^2 - 2ab + a^2$, so

$$3p^2 + q^2 = 4b^2 + 4ab + 4a^2 = 4(b^2 + ab + a^2).$$

The left-hand side of $(**)$ is therefore $3p^2 + q^2$, and the right-hand side is $3qp^2$. So

$$3p^2 + q^2 = 3p^2 q,$$

as required. Rearranging, $q^2 = 3p^2(q-1)$. Since $q > 0$, the left side is positive, so $q \neq 1$ and

$$p^2 = \frac{q^2}{3(q-1)}.$$

For the deduction:

- *Lower bound.* $q^2 > 0$ and $p^2 > 0$ (as $p = b + a > 0$), so $3(q-1) = q^2/p^2 > 0$, i.e. $q > 1$.
- *Upper bound.* Since $a \geq 0$, we have $p = q + 2a \geq q > 0$, so $p^2 \geq q^2$. Hence

  $$\frac{q^2}{3(q-1)} \geq q^2 \iff 1 \geq 3(q-1) \iff q \leq \tfrac{4}{3},$$

  where we divided by $q^2 > 0$ and multiplied by $3(q-1) > 0$, both of which preserve the inequality.

Therefore $1 < b - a \leq \frac{4}{3}$, with equality exactly when $a = 0$, which is part (i).

Checks

- Part (i) fits part (iii): $a = 0$ gives $q = b = \frac{4}{3}$, the top of the range.
- Part (ii) fits too: the root is $b \approx 2.0390$, so $q = b - 1 \approx 1.039$, inside $(1, \frac{4}{3}]$. Evaluating both integrals over $[1, 2.0390]$ gives equal values (about 2.4924) to many decimal places.
- Conversely, any q in $(1, \frac{4}{3}]$ gives $p = q/\sqrt{3(q-1)} \geq q$ and so a valid $a = \frac{1}{2}(p-q) \geq 0$; the bound is sharp.

Common slips

- Forgetting to cancel $b - a$ in part (ii), leaving a quartic with the spurious root $b = 1$.
- Showing only that $f(2) < 0 < f(3)$: this proves there is *a* root in $(2, 3)$, but not that it is the only real root. You need the turning points too.
- In part (iii), dividing an inequality by $q - 1$ before establishing that $q - 1 > 0$.

Beyond STEP

The Cauchy–Schwarz inequality for integrals gives $\left(\int_a^b x\,\mathrm{d}x\right)^2 \leq (b-a)\int_a^b x^2\,\mathrm{d}x$, with equality only when the integrand x is constant, which it is not. Combined with the given condition this says $\int_a^b x^2\,\mathrm{d}x < (b-a)\int_a^b x^2\,\mathrm{d}x$, so $b - a > 1$: the lower bound of part (iii) in one line. Cauchy–Schwarz, in its many forms, is one of the most used tools in first-year analysis; Bartle and Sherbert's *Introduction to Real Analysis* develops the Riemann integral and inequalities like this one carefully.

WORKED SOLUTION

STEP I 2014, Question 4

Pure • Related Rates of Change

STEP I 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q4

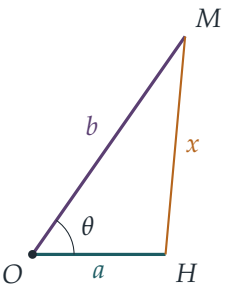
The key idea

The two hands and the line joining their tips form a triangle, so the cosine rule gives x in terms of θ. Because the clock is accurate, θ increases at a constant rate, so the rate of increase of x is that constant times $\mathrm{d}x/\mathrm{d}\theta$. Maximising this is then an ordinary one-variable calculus problem, and the stationary-point equation turns out to be a quadratic in $\cos\theta$ that factorises neatly.

Solution

The rate of increase of x

Let O be the pivot, H the tip of the hour hand and M the tip of the minute hand, so $OH = a$, $OM = b$, $\angle HOM = \theta$ and $HM = x$.



By the cosine rule,

$$x^2 = a^2 + b^2 - 2ab\cos\theta.$$

Note that $x \geq b - a > 0$, so we may divide by x freely. Differentiating with respect to time t,

$$2x\frac{\mathrm{d}x}{\mathrm{d}t} = 2ab\sin\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}t}.$$

The clock is accurate, so each hand turns at a constant angular speed, and the angle between them changes at a constant rate $\omega > 0$ (the difference of the two angular speeds; while $0 \leq \theta < \pi$ the minute hand is pulling ahead, so θ is increasing). Hence

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \omega ab \frac{\sin\theta}{x}.$$

So we must find where $f(\theta) = \dfrac{\sin\theta}{x}$ is greatest on $0 \leq \theta < \pi$, remembering that x depends on θ with $\dfrac{\mathrm{d}x}{\mathrm{d}\theta} = \dfrac{ab\sin\theta}{x}$.

Maximising

By the quotient rule,

$$f'(\theta) = \frac{x\cos\theta - \sin\theta \cdot \dfrac{ab\sin\theta}{x}}{x^2} = \frac{x^2\cos\theta - ab\sin^2\theta}{x^3}.$$

Write $c = \cos\theta$. Using $x^2 = a^2 + b^2 - 2abc$ and $\sin^2\theta = 1 - c^2$, the numerator is

$$\begin{aligned} x^2\cos\theta - ab\sin^2\theta &= (a^2+b^2)c - 2abc^2 - ab + abc^2 \\ &= -\big(abc^2 - (a^2+b^2)c + ab\big) \\ &= -(ac-b)(bc-a) \\ &= (b-ac)(bc-a). \end{aligned}$$

Ask yourself How do we know this stationary point is a maximum, and the only one?

Since $c \leq 1 < b/a$, we have $ac < b$, so the factor $b - ac$ is always positive. Also $x^3 > 0$. So $f'(\theta)$ has the same sign as $bc - a$, that is, of $\cos\theta - \dfrac{a}{b}$:

- when $\cos\theta > a/b$ (smaller θ), $f'(\theta) > 0$ and f is increasing;
- when $\cos\theta < a/b$ (larger θ), $f'(\theta) < 0$ and f is decreasing.

As $\cos\theta$ is decreasing on $[0, \pi)$ and $0 < a/b < 1$, there is exactly one θ in the range with $\cos\theta = a/b$, and f rises up to it and falls after it. So this gives the global maximum of the rate of increase of x. There,

$$x^2 = a^2 + b^2 - 2ab \cdot \frac{a}{b} = b^2 - a^2,$$

so the rate of increase of x is greatest when $x = (b^2 - a^2)^{1/2}$, as required.

Geometrically, $x^2 + a^2 = b^2$ says the triangle has a right angle at H: the rate is greatest when the line joining the tips is perpendicular to the hour hand.

The case $b = 2a$

Now $\cos\theta = a/b = \frac{1}{2}$, so the maximum rate occurs when $\theta = \pi/3$.

Measure t in minutes after mid-day. The minute hand turns through 2π in 60 minutes and the hour hand through 2π in 720 minutes, both clockwise from the vertical. So the angle between them is

$$\theta = 2\pi\left(\frac{1}{60} - \frac{1}{720}\right)t = 2\pi \cdot \frac{11}{720}\,t = \frac{11\pi}{360}\,t,$$

which is valid while this is less than π. Starting from $\theta = 0$ at mid-day, θ first reaches $\pi/3$ when

$$\frac{11\pi}{360}\,t = \frac{\pi}{3} \quad\Longrightarrow\quad t = \frac{120}{11} = 10\tfrac{10}{11} \text{ minutes.}$$

This is about 10.91 minutes (10 minutes and roughly 55 seconds), a little less than 11 minutes, as required.

Checks

- At $\theta = 0$ the rate is 0 (the hands are together and $x = b - a$ is at its minimum), and as $\theta \to \pi$ it tends to 0 again (where $x \to a + b$ is at its maximum). So a maximum strictly in between is exactly what we should expect.
- With $a = 1$, $b = 2$, tracking the positions of the hand tips minute by minute and estimating the rate of change of their distance, the greatest rate appears at $t \approx 10.909$ minutes, where $x \approx 1.732 = \sqrt{3} = \sqrt{b^2 - a^2}$.
- The answer is just under 11 because if the hour hand stood still the minute hand would need exactly 10 minutes to reach 60°; the hour hand creeps forward, so a little extra time is needed, and $120/11$ is the same kind of number as the familiar "the hands next coincide after $720/11$ minutes".

Common slips

- Maximising x itself, or $\mathrm{d}x/\mathrm{d}\theta$ without noticing that x also depends on θ in the quotient-rule step.
- Using the minute hand's speed alone ($2\pi/60$ per minute) for $\mathrm{d}\theta/\mathrm{d}t$, which gives exactly 10 minutes; the angle between the hands grows at the *difference* of the two speeds.
- Forgetting to discard the root $\cos\theta = b/a$, which is impossible because $b > a$.

Beyond STEP

This is a related-rates problem: one quantity changes at a known rate and we track how a geometrically linked quantity responds. The same idea, describing positions

by angles that vary uniformly in time, is the starting point for planetary motion: the distance between two planets on circular orbits behaves exactly like x here, and the moments of fastest separation are found the same way. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this kind of kinematics with vectors.

WORKED SOLUTION

STEP I 2014, Question 5

Pure • Cubic Inequalities

STEP I 2014, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q5

The key idea

Part (i) is a single-variable cubic whose minimum on $x \geq 0$ turns out to be exactly zero, touching the axis at $x = a$. That one inequality, $(x+2a)^3 \geq 27a^2x$, is an AM–GM inequality in disguise: the left side involves the sum $x + a + a$ and the right side the product $x \cdot a \cdot a$. Parts (ii) and (iii) are about choosing what to call x and a, and tracking exactly when equality holds.

Solution

Part (i)

Differentiate:

$$f'(x) = 3(x+2a)^2 - 27a^2 = 3\big((x+2a)^2 - 9a^2\big) = 3(x-a)(x+5a),$$

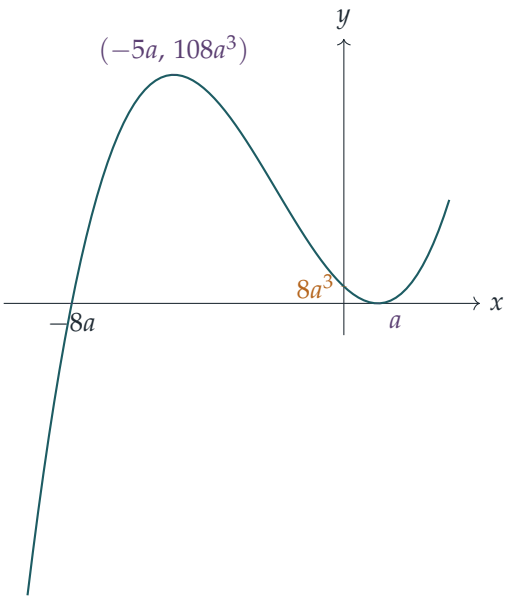
so the stationary points are at $x = a$ and $x = -5a$. There

$$f(a) = (3a)^3 - 27a^3 = 0, \qquad f(-5a) = (-3a)^3 + 135a^3 = 108a^3,$$

and $f(0) = 8a^3$. Since f is a cubic with positive leading coefficient, $x = -5a$ is a local maximum and $x = a$ a local minimum. Because the minimum value is 0, the graph touches the x-axis at $x = a$, so $(x-a)^2$ is a factor; comparing constant terms then gives

$$f(x) = (x-a)^2(x+8a),$$

which you can confirm by expanding both sides to $x^3 + 6ax^2 - 15a^2x + 8a^3$.



For $x \geq 0$ the graph starts at height $8a^3 \geq 0$, decreases to 0 at $x = a$, and then increases for ever. So the least value of f on $x \geq 0$ is 0, and

$$f(x) \geq 0 \quad \text{for } x \geq 0,$$

with equality only at $x = a$. (The factorised form says the same thing directly: for $x \geq 0$, $(x-a)^2 \geq 0$ and $x + 8a > 0$ unless $x = a = 0$.) If $a = 0$ the picture degenerates to $f(x) = x^3$, which is still non-negative for $x \geq 0$, with equality only at $x = 0 = a$. In every case

$$(x+2a)^3 \geq 27a^2x \quad (x, a \geq 0), \qquad \text{with equality if and only if } x = a.$$

Part (ii)

We want xy^2, so take $a = y \geq 0$ in part (i). For $x \geq 0$, $y \geq 0$,

$$27xy^2 \leq (x+2y)^3 \leq 3^3 = 27,$$

the second step because $0 \leq x + 2y \leq 3$ and cubing is increasing. Hence $xy^2 \leq 1$ throughout the region.

The bound is attained: equality needs $x = y$ (from part (i)) and $x + 2y = 3$, which give $x = y = 1$, and indeed $1 \cdot 1^2 = 1$. So the greatest value of xy^2 is 1, achieved only at $x = 1$, $y = 1$.

Part (iii)

Ask yourself Part (i) involves only two numbers, x and a. How can it say anything about three?

Group q and r together: put $x = p$ and $2a = q + r$, so $a = \frac{1}{2}(q + r) \geq 0$. Part (i) gives

$$(p + q + r)^3 \geq 27p\left(\frac{q + r}{2}\right)^2.$$

Now $\left(\frac{q+r}{2}\right)^2 - qr = \frac{(q-r)^2}{4} \geq 0$, and $p \geq 0$, so

$$(p + q + r)^3 \geq 27p\left(\frac{q + r}{2}\right)^2 \geq 27pqr,$$

which is the required inequality.

Equality. If $(p + q + r)^3 = 27pqr$ then both inequalities in the chain are equalities.

- The first is equality exactly when $x = a$, that is $p = \frac{1}{2}(q + r)$.
- The second is equality exactly when $\frac{27}{4}p(q - r)^2 = 0$, that is $p = 0$ or $q = r$.

If $p = 0$, the first condition gives $q + r = 0$, so $q = r = 0$ as well. If $q = r$, the first condition gives $p = q$. Either way $p = q = r$. Conversely, if $p = q = r$ then both sides equal $27p^3$. So equality holds if and only if

$$p = q = r.$$

Checks

- Expanding $(x - a)^2(x + 8a)$ recovers $(x + 2a)^3 - 27a^2x$; with $a = 1$, $f(0) = 8$, $f(1) = 0$, $f(3) = 4 \cdot 11 = 44$, all non-negative.
- In part (ii), try other points on the line $x + 2y = 3$: $(x, y) = (2, \frac{1}{2})$ gives $\frac{1}{2}$, and $(0.5, 1.25)$ gives about 0.78, both below 1. A computer search over a fine grid of the triangle also gives a maximum of 1 at $(1, 1)$.
- In part (iii), $p = 1, q = 2, r = 3$ gives $216 \geq 162$; a computer search over small whole numbers found equality only when $p = q = r$.

Common slips

- Showing $f(a) = 0$ but not explaining why this is the *least* value on $x \geq 0$: you need the shape of the graph (or the factorisation), including the endpoint value $f(0) = 8a^3 \geq 0$.
- In part (ii), forgetting to check that the bound 1 is actually reached; an upper bound alone is not a greatest value.

- In the equality case of part (iii), concluding only $q = r$ and missing the alternative $p = 0$, which has to be ruled out (or shown to force $p = q = r = 0$).

Beyond STEP

Part (iii) is the three-term AM–GM inequality, $\frac{1}{3}(p+q+r) \geq \sqrt[3]{pqr}$, and the method here, merging two variables into their average and using a two-variable inequality, is the germ of the classical proofs of AM–GM for n terms. Part (ii) is a small constrained optimisation problem, the sort that university courses attack with Lagrange multipliers, though inequalities like AM–GM often give the answer far more cleanly. For much more in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2014, Question 6

Pure • Logistic Map Recurrences

STEP I 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 8.7, *Chaos You Can Solve*: https://beyondstep.keithmcnulty.org/ch08/solvable-chaos.html

The key idea

The expression $4x(1-x)$ looks awkward until you put $x = \sin^2\theta$: then $1 - x = \cos^2\theta$ and $4\sin^2\theta\cos^2\theta = (2\sin\theta\cos\theta)^2$ is a double-angle formula in disguise. Each step of the recurrence simply doubles the angle. Part (ii) is then about recognising that *any* quadratic recurrence can be turned into this one by a linear change of variable, as long as one condition on the coefficients holds.

Solution

Part (i)

With $u_0 = \sin^2\theta$ we have $1 - u_0 = \cos^2\theta$, so

$$u_1 = 4\sin^2\theta\cos^2\theta = (2\sin\theta\cos\theta)^2 = \sin^2 2\theta.$$

This has exactly the same form as u_0, with θ replaced by 2θ, so the same calculation gives

$$u_2 = 4\sin^2 2\theta\ \cos^2 2\theta = \sin^2 4\theta.$$

The angle doubles at each step, so we conjecture

$$u_n = \sin^2(2^n\theta) \qquad (n \geq 0).$$

Proof by induction. For $n = 0$ this reads $u_0 = \sin^2\theta$, which is true. Suppose $u_k = \sin^2(2^k\theta)$

for some $k \geq 0$. Then $1 - u_k = \cos^2(2^k\theta)$, and

$$\begin{aligned} u_{k+1} &= 4\sin^2(2^k\theta)\cos^2(2^k\theta) \\ &= \big(2\sin(2^k\theta)\cos(2^k\theta)\big)^2 \\ &= \sin^2(2^{k+1}\theta), \end{aligned}$$

using $\sin 2A = 2\sin A\cos A$ with $A = 2^k\theta$. So the result for k implies the result for $k+1$, and by induction $u_n = \sin^2(2^n\theta)$ for all $n \geq 0$.

Part (ii)

Put $v_n = \alpha u_n + \beta$ with $\alpha \neq 0$ (we need $\alpha \neq 0$ so that we can recover u_n from v_n). Substituting into the recurrence for v_n:

$$\alpha u_{n+1} + \beta = -p(\alpha u_n + \beta)^2 + q(\alpha u_n + \beta) + r.$$

Expanding and dividing by α,

$$u_{n+1} = -p\alpha\, u_n^2 + (q - 2p\beta)\, u_n + \frac{-p\beta^2 + (q-1)\beta + r}{\alpha}.$$

We want this to be $u_{n+1} = 4u_n - 4u_n^2$, so we compare coefficients.

- u_n^2: $p\alpha = 4$, so $\alpha = \dfrac{4}{p}$ (fine, since $p \neq 0$, and then $\alpha \neq 0$ too).
- u_n: $q - 2p\beta = 4$, so $\beta = \dfrac{q-4}{2p}$.
- constant: $-p\beta^2 + (q-1)\beta + r = 0$.

The first two fix α and β; the third is then a condition on p, q, r. Substituting $\beta = (q-4)/(2p)$ and multiplying through by $4p$:

$$\begin{aligned} & 0 = -p \cdot \frac{(q-4)^2}{4p^2} + (q-1)\frac{q-4}{2p} + r \\ \iff \quad & 0 = -(q-4)^2 + 2(q-1)(q-4) + 4pr \\ & \quad = (q-4)\big(-(q-4) + 2(q-1)\big) + 4pr \\ & \quad = (q-4)(q+2) + 4pr \\ & \quad = q^2 - 2q - 8 + 4pr. \end{aligned}$$

So if $4pr = 8 + 2q - q^2$, then with $\alpha = 4/p$ and $\beta = (q-4)/(2p)$ all three coefficients match, and the sequence $u_n = (v_n - \beta)/\alpha$ satisfies $(*)$, as required.

The particular sequence. Here $p = 1$, $q = 2$, $r = 2$. The condition holds: $4pr = 8$ and $8 + 2q - q^2 = 8 + 4 - 4 = 8$. We get

$$\alpha = 4, \qquad \beta = \frac{2-4}{2} = -1, \qquad \text{so} \qquad v_n = 4u_n - 1.$$

Ask yourself Part (i) needs $u_0 = \sin^2\theta$. Can we find a suitable θ here?

Since $v_0 = 1$, we need $u_0 = (1+1)/4 = \frac{1}{2}$, and $\frac{1}{2} = \sin^2\frac{\pi}{4}$. So $\theta = \pi/4$ works, and part (i) gives $u_n = \sin^2(2^n\pi/4) = \sin^2(2^{n-2}\pi)$. Therefore

$$v_n = 4\sin^2\left(2^{n-2}\pi\right) - 1 = 1 - 2\cos\left(2^{n-1}\pi\right),$$

using $2\sin^2 A = 1 - \cos 2A$. Evaluating:

- $n = 0$: $v_0 = 4\sin^2\frac{\pi}{4} - 1 = 1$;
- $n = 1$: $v_1 = 4\sin^2\frac{\pi}{2} - 1 = 3$;
- $n \geq 2$: 2^{n-2} is an integer, so $\sin(2^{n-2}\pi) = 0$ and $v_n = -1$.

So the sequence is

$$1,\ 3,\ -1,\ -1,\ -1,\ \ldots$$

that is, $v_0 = 1$, $v_1 = 3$ and $v_n = -1$ for all $n \geq 2$.

Checks

Iterate the recurrence directly: $v_1 = -1 + 2 + 2 = 3$, $v_2 = -9 + 6 + 2 = -1$, and $v_3 = -1 - 2 + 2 = -1$; since -1 maps to itself, the sequence stays at -1, as the formula says. For part (i), try $\theta = \pi/6$: $u_0 = \frac{1}{4}$, $u_1 = 4 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{4} = \sin^2\frac{\pi}{3}$, and $u_2 = 4 \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{4} = \sin^2\frac{2\pi}{3}$, as predicted. The formula $u_n = \sin^2(2^n\theta)$ was also confirmed by computer for many random angles and the first several terms.

Common slips

- Writing $u_n = \sin^2(2n\theta)$: the angle doubles each time, so it grows like 2^n, not $2n$.
- Forgetting to check that the starting value $u_0 = (v_0 - \beta)/\alpha$ lies in $[0, 1]$, so that it really can be written as $\sin^2\theta$.
- Treating the constant-term equation as a third unknown to solve for, rather than as the condition the question asks you to reach.

Beyond STEP

The recurrence $u_{n+1} = 4u_n(1 - u_n)$ is the logistic map at its most extreme parameter, a standard example of chaos: since $u_n = \sin^2(2^n\theta)$, writing θ/π in binary shows that each step just shifts the binary digits one place, so two nearby starting values separate exponentially fast. The substitution in part (ii) is a *conjugacy*, the idea that two dynamical systems are really the same after a change of coordinates. Strogatz's *Nonlinear Dynamics and Chaos* is a very readable introduction to these ideas.

WORKED SOLUTION

STEP I 2014, Question 7

Pure • Vectors in a Triangle

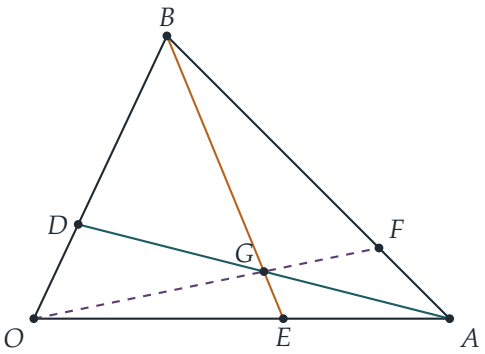
STEP I 2014, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q7

The key idea

Write every point as a combination of **a** and **b**. A point on a line through two known points has a one-parameter vector equation, so G can be written in two ways, one from each line. Because **a** and **b** are not parallel (the triangle is non-degenerate), we may compare coefficients, which gives two simple equations for the two parameters.



Solution

First we locate D and E. Since $BD = r\,DO$, the point D is the fraction $\frac{1}{1+r}$ of the way from O to B; since $OE = s\,EA$, the point E is the fraction $\frac{s}{1+s}$ of the way from O to A. So

$$\mathbf{d} = \frac{1}{1+r}\,\mathbf{b}, \qquad \mathbf{e} = \frac{s}{1+s}\,\mathbf{a}.$$

Part (i)

A general point of the line AD is

$$\mathbf{a} + \lambda(\mathbf{d} - \mathbf{a}) = (1-\lambda)\,\mathbf{a} + \frac{\lambda}{1+r}\,\mathbf{b},$$

and a general point of the line BE is

$$\mathbf{b} + \mu(\mathbf{e} - \mathbf{b}) = \frac{\mu s}{1+s}\,\mathbf{a} + (1-\mu)\,\mathbf{b}.$$

At G these are equal.

Ask yourself Why are we allowed to equate the coefficients of **a** and of **b** separately?

Because O, A, B form a genuine triangle, **a** and **b** are non-zero and not parallel. If $\alpha\mathbf{a} + \beta\mathbf{b} = \alpha'\mathbf{a} + \beta'\mathbf{b}$ with $\alpha \neq \alpha'$, then **a** would be a multiple of **b**, which is impossible. So

$$1 - \lambda = \frac{\mu s}{1+s}, \qquad \frac{\lambda}{1+r} = 1 - \mu.$$

The second equation gives $\lambda = (1+r)(1-\mu)$. Substituting into the first,

$$\begin{aligned} 1 - (1+r)(1-\mu) &= \frac{\mu s}{1+s} \\ \mu\left[(1+r) - \frac{s}{1+s}\right] &= r \\ \mu\,\frac{(1+r)(1+s) - s}{1+s} &= r \\ \mu\,\frac{1+r+rs}{1+s} &= r, \end{aligned}$$

so

$$\mu = \frac{r(1+s)}{1+r+rs}.$$

This is allowed because $1 + r + rs > 0$, as $r, s > 0$. Putting this value of μ into the equation of BE, the coefficient of **a** is

$$\frac{\mu s}{1+s} = \frac{rs}{1+r+rs},$$

and the coefficient of **b** is

$$1 - \mu = \frac{1+r+rs-r-rs}{1+r+rs} = \frac{1}{1+r+rs}.$$

Hence

$$\mathbf{g} = \frac{rs}{1+r+rs}\,\mathbf{a} + \frac{1}{1+r+rs}\,\mathbf{b},$$

as required.

Part (ii)

Every point of the line OG is a scalar multiple $k\mathbf{g}$ of **g**, since the line passes through the origin:

$$k\mathbf{g} = \frac{k}{1+r+rs}\,(rs\,\mathbf{a} + \mathbf{b}).$$

A point $\alpha\mathbf{a}+\beta\mathbf{b}$ lies on the line AB exactly when $\alpha+\beta=1$, because AB consists of the points $(1-\nu)\mathbf{a}+\nu\mathbf{b}$. So at F we need $k(rs+1)=1+r+rs$, and

$$\mathbf{f}=\frac{rs}{1+rs}\,\mathbf{a}+\frac{1}{1+rs}\,\mathbf{b}.$$

On the other hand, $AF=t\,FB$ means that F is the fraction $\frac{t}{1+t}$ of the way from A to B:

$$\mathbf{f}=\frac{1}{1+t}\,\mathbf{a}+\frac{t}{1+t}\,\mathbf{b}.$$

Comparing the coefficients of $\mathbf{a}$ (allowed for the same reason as before) gives $1+t=\dfrac{1+rs}{rs}$, so

$$t=\frac{1}{rs}.$$

The coefficients of $\mathbf{b}$ agree too: $\dfrac{t}{1+t}=\dfrac{1/(rs)}{(1+rs)/(rs)}=\dfrac{1}{1+rs}$.

Checks

- **Medians.** With $r=s=1$, D and E are midpoints, so G should be the centroid. The formula gives $\mathbf{g}=\frac{1}{3}\mathbf{a}+\frac{1}{3}\mathbf{b}$, which is right, and $t=1$, so F is the midpoint of AB, as it must be, since the three medians are concurrent.
- **Ceva's theorem.** The answer says $r\cdot s\cdot t=1$, that is, $\dfrac{BD}{DO}\cdot\dfrac{OE}{EA}\cdot\dfrac{AF}{FB}=1$, which is exactly Ceva's theorem for the concurrent lines AD, BE, OF.
- **A numerical case.** Take $\mathbf{a}=(5,0)$, $\mathbf{b}=(0,5)$, $r=2$, $s=3$. Then $D=(0,\frac{5}{3})$, $E=(\frac{15}{4},0)$ and the formula gives $G=(\frac{30}{9},\frac{5}{9})=(\frac{10}{3},\frac{5}{9})$. This lies on AD, whose equation is $x+3y=5$, and on BE, whose equation is $4x+3y=15$. The line OG meets $x+y=5$ at $F=(\frac{30}{7},\frac{5}{7})$, and $\overrightarrow{AF}=(-\frac{5}{7},\frac{5}{7})$, $\overrightarrow{FB}=(-\frac{30}{7},\frac{30}{7})$, so $AF:FB=1:6$, giving $t=\frac{1}{6}=\frac{1}{rs}$. The formulas were also confirmed by computer for two thousand random triangles and ratios, using exact arithmetic.

Common slips

- Getting the ratios the wrong way round: $BD=r\,DO$ puts D at $\frac{1}{1+r}$ of the way from O to B, not $\frac{r}{1+r}$.
- Comparing coefficients without saying why it is valid: it needs $\mathbf{a}$ and $\mathbf{b}$ to be non-parallel.
- In part (ii), forgetting to rescale $\mathbf{g}$: G itself is not on AB, since its coefficients add up to less than 1.

Beyond STEP

Coefficients that add up to 1, like those of **f**, are *barycentric coordinates*. They describe points by weights at the vertices, and they make results such as Ceva's and Menelaus' theorems almost automatic. In the language of linear algebra, comparing coefficients works because **a** and **b** are linearly independent and so form a basis of the plane. Strang's *Introduction to Linear Algebra* develops bases and coordinates carefully.

WORKED SOLUTION

STEP I 2014, Question 8

Pure • Envelope of Lines

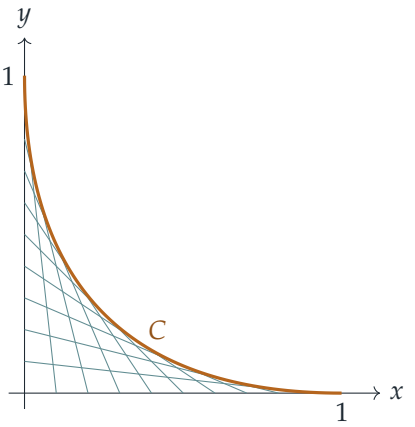
STEP I 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q8

The key idea

Write each line in intercept form, $\frac{x}{a} + \frac{y}{1-a} = 1$; then the intersection of two of them falls out of a pair of linear equations, and comes out symmetric in a and b. Letting $b \to a$ gives the point where a line meets its "neighbour", and these points trace out a curve which every line L_a touches. The curve C is the *envelope* of the family, and part (iii) confirms this from the other direction.



Solution

The line through $(a, 0)$ and $(0, 1-a)$ has intercepts a and $1-a$, so

$$L_a : \quad \frac{x}{a} + \frac{y}{1-a} = 1, \qquad \text{that is} \qquad (1-a)x + ay = a(1-a).$$

Part (i)

We solve

$$(1-a)x + ay = a(1-a),$$
$$(1-b)x + by = b(1-b).$$

To eliminate y, multiply the first equation by b and the second by a, and subtract:

$$\big[b(1-a) - a(1-b)\big]x = ab(1-a) - ab(1-b), \qquad \text{i.e.} \qquad (b-a)x = ab(b-a).$$

Since $a \neq b$ we may divide by $b-a$, giving $x = ab$. Substituting into the first equation,

$$ay = a(1-a) - (1-a)ab = a(1-a)(1-b),$$

and since $a \neq 0$, $y = (1-a)(1-b)$. So L_a and L_b meet at

$$\big(ab,\ (1-a)(1-b)\big).$$

(The answer is symmetric in a and b, as it must be. You can check it lies on L_b too: $(1-b)ab + b(1-a)(1-b) = b(1-b)\big(a + 1 - a\big) = b(1-b)$.)

Part (ii)

Both coordinates are polynomials in b, so they are continuous and the limit as $b \to a$ is just the value at $b = a$:

$$(x, y) = \big(a^2,\ (1-a)^2\big).$$

Since $0 < a < 1$, we have $0 < x = a^2 < 1$, and because $a > 0$ the positive square root gives $\sqrt{x} = a$. Hence

$$y = (1-a)^2 = (1-\sqrt{x})^2,$$

so the limiting point lies on C, as required.

Part (iii)

Take any point of C. Its x-coordinate lies in $(0,1)$, so we can write it as t^2 with $t = \sqrt{x} \in (0,1)$, and the point is $\big(t^2, (1-t)^2\big)$.

Differentiating $y = (1-\sqrt{x})^2$ by the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2(1-\sqrt{x}) \cdot \left(-\frac{1}{2\sqrt{x}}\right) = -\frac{1-\sqrt{x}}{\sqrt{x}},$$

which at $x = t^2$ is $-\frac{1-t}{t}$. The tangent there is

$$y - (1-t)^2 = -\frac{1-t}{t}\,(x - t^2).$$

Ask yourself How do we recognise this line as one of the L_a?

A line L_a is pinned down by its two intercepts, which must be a and $1-a$. So find the intercepts of the tangent.

- Putting $y = 0$: $-(1-t)^2 = -\frac{1-t}{t}(x - t^2)$. Since $1 - t \neq 0$, we can cancel to get $x - t^2 = t(1-t)$, so $x = t$.
- Putting $x = 0$: $y = (1-t)^2 + \frac{1-t}{t}\,t^2 = (1-t)^2 + t(1-t) = 1 - t$.

So the tangent passes through $(t, 0)$ and $(0, 1-t)$ with $0 < t < 1$: it is exactly L_t. Since every point of C arises from some $t \in (0,1)$, every tangent to C is of the form L_a (with $a = \sqrt{x}$ at the point of contact). This completes the proof.

Notice that the tangent at $\left(a^2, (1-a)^2\right)$ is L_a, and this is precisely the point that part (ii) produced from L_a: each line touches C at its own limiting intersection point.

Checks

- With $a = \frac{1}{2}$, $b = \frac{1}{3}$, part (i) gives $\left(\frac{1}{6}, \frac{1}{3}\right)$. On $L_{1/2}$: $\frac{1}{2}\cdot\frac{1}{6} + \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{4} = \frac{1}{2}\cdot\frac{1}{2}$; on $L_{1/3}$: $\frac{2}{3}\cdot\frac{1}{6} + \frac{1}{3}\cdot\frac{1}{3} = \frac{2}{9} = \frac{1}{3}\cdot\frac{2}{3}$.
- At $x = \frac{1}{4}$ the curve has $y = \frac{1}{4}$ and gradient -1, so the tangent is $x + y = \frac{1}{2}$, which is $L_{1/2}$, through $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$.
- The curve itself is part of the parabola $\sqrt{x} + \sqrt{y} = 1$, symmetric in x and y, matching the symmetry $a \leftrightarrow 1 - a$ of the family of lines.

Common slips

- Forgetting to justify $\sqrt{x} = a$ in part (ii): it needs $a > 0$, which is given.
- In part (iii), showing only that L_a touches C. The question asks the converse: start from an arbitrary tangent of C and show it is some L_a.

Beyond STEP

The curve C is the *envelope* of the family of lines L_a: the curve that touches every member. In general, the envelope of a family $F(x, y, a) = 0$ is found by eliminating a between $F = 0$ and $\partial F/\partial a = 0$, which is the calculus version of the "let $b \to a$" limit in part (ii). Envelopes appear in optics as caustics, such as the bright curve of light in a coffee cup. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP I 2014, Question 9

Mechanics • Projectile with Horizontal Force

STEP I 2014, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

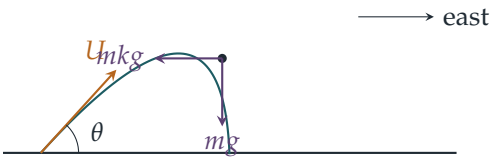
step.maths.org/questions/14-s1-q9

The key idea

Both forces are constant, so the acceleration is constant in each direction and we can treat the horizontal and vertical motions separately, exactly as for an ordinary projectile. The vertical motion doesn't feel the westward force at all. The new feature is that the horizontal velocity is steadily reduced, so the particle stops moving east at some moment and then drifts back west. The whole shape of the path depends on *when* that moment comes compared with the top of the flight and the landing.

Solution

Model the particle as a point mass moving in a vertical plane, with no forces other than its weight and the given horizontal force. Take x east and y vertically upwards, with the origin at the point of projection.



The times T_H and T_L

Newton's second law in each direction, after dividing by m, gives

$$\ddot{x} = -kg, \qquad \ddot{y} = -g.$$

The initial velocity is $(U\cos\theta,\ U\sin\theta)$, so integrating twice,

$$\dot{x} = U\cos\theta - kgt, \qquad x = Ut\cos\theta - \tfrac{1}{2}kgt^2,$$
$$\dot{y} = U\sin\theta - gt, \qquad y = Ut\sin\theta - \tfrac{1}{2}gt^2.$$

The greatest height is reached when $\dot{y} = 0$, and the particle lands when $y = 0$ with $t > 0$:

$$\boxed{T_H = \frac{U\sin\theta}{g}, \qquad T_L = \frac{2U\sin\theta}{g}.}$$

These are the same as with no westward force, because that force has no vertical component.

The meaning of T

The given time $T = \dfrac{U\cos\theta}{kg}$ is exactly when $\dot{x} = 0$. Before T the particle is moving east; after T it moves west. So, if the particle is still in the air at time T, that is the moment when it is *furthest east*. Comparing the three times,

$$\frac{T_H}{T} = \frac{U\sin\theta}{g}\cdot\frac{kg}{U\cos\theta} = k\tan\theta, \qquad \frac{T_L}{T} = 2k\tan\theta.$$

It will also help to know where the particle lands:

$$x(T_L) = T_L\left(U\cos\theta - \tfrac{1}{2}kgT_L\right) = \frac{2U^2\sin\theta}{g}(\cos\theta - k\sin\theta) = \frac{2U^2\sin\theta\cos\theta}{g}(1 - k\tan\theta).$$

This is positive (east of the start) if $k\tan\theta < 1$ and negative (west) if $k\tan\theta > 1$.

> **Ask yourself** The path is no longer symmetric, so what does it actually look like?

The acceleration $(-kg, -g)$ is constant, so the path is still a parabola, but with its axis tilted: it is the parabola we'd get under a "gravity" of $g\sqrt{1+k^2}$ pointing down and to the west. The top of the flight is no longer the turning point of the parabola, and the three cases below are about where the furthest-east point (time T) falls.

The three cases

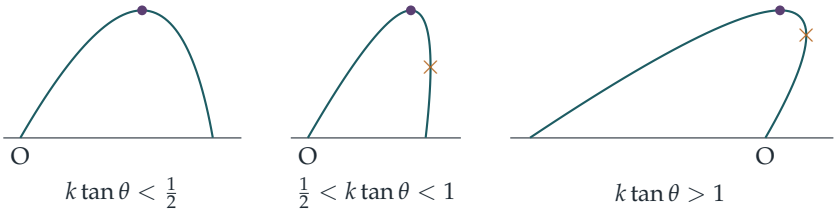
Case $k\tan\theta < \frac{1}{2}$. Then $T_H < T_L < T$. The particle lands before its eastward velocity has run out, so it moves east throughout. The path looks like an ordinary trajectory, rising to its highest point and then coming down more steeply than it went up, since the horizontal speed keeps falling. It lands east of the start.

Case $\frac{1}{2} < k\tan\theta < 1$. Now $T_H < T < T_L$. The particle passes its highest point while still moving east, carries on east while descending until time T, where its velocity is vertically downwards, and then curls back westwards before landing. Since $k\tan\theta < 1$ it still lands east of the start, but west of its furthest-east point.

Case $k\tan\theta > 1$. Now $T < T_H$. The eastward motion stops while the particle is still rising, at which point it is moving vertically upwards. It reaches its greatest height while already moving west and comes down to land *west* of the point of projection.

The sketches show the three cases with $\theta = 60^\circ$ and the same U and g, with $k\tan\theta \approx$

0.35, 0.6 and 1.8 respectively. The dot marks the highest point, at T_H, the cross marks the furthest-east point, at T, and O is the point of projection.



$k\tan\theta < \frac{1}{2}$ $\qquad$ $\frac{1}{2} < k\tan\theta < 1$ $\qquad$ $k\tan\theta > 1$

The case $k\tan\theta = 1$

Now $T = T_H$, so at the top of the flight *both* velocity components vanish together and the particle is momentarily at rest. Also $x(T_L) = 0$, so it lands back at the point of projection.

In fact the motion is along a straight line. When $\tan\theta = 1/k$, the initial velocity $U(\cos\theta, \sin\theta)$ is a positive multiple of $(k, 1)$, while the acceleration $(-kg, -g) = -g(k, 1)$ points exactly the opposite way. A particle whose acceleration is constant and directly opposed to its velocity moves in a straight line: it travels up the line of projection, decelerating at $g\sqrt{1+k^2}$, stops at time T_H, and slides straight back down the same line to land where it started at time $T_L = 2T_H$. You can confirm this from the equations: with $k = \cot\theta$,

$$x\tan\theta = Ut\sin\theta - \tfrac{1}{2}gt^2 = y \quad \text{for all } t,$$

so the particle never leaves the line $y = x\tan\theta$.

Checks

- Setting $k \to 0$ makes $T \to \infty$, so we are always in the first case, and $x(T_L)$ becomes the familiar range $U^2\sin 2\theta/g$.
- The landing position $\frac{2U^2\sin\theta\cos\theta}{g}(1 - k\tan\theta)$ changes sign exactly at $k\tan\theta = 1$, matching the boundary between the second and third cases.
- With $U = g = 1$, $\theta = 60°$ and $k = 0.45$, the particle lands at $x \approx 0.191$ but reaches $x = \frac{\cos^2\theta}{2k} \approx 0.278$ at time $T \approx 1.11$, between $T_H \approx 0.87$ and $T_L \approx 1.73$, so it does turn back before landing. A step-by-step numerical integration of the motion agrees with all three times and the landing point.

Common slips

- Letting the westward force change the vertical motion: T_H and T_L don't involve k at all.
- Assuming the highest point is the "middle" of the path. With the horizontal force

the path is a tilted parabola, so the highest point and the furthest-east point are different.

- Drawing the second case as if the particle lands west of the start: it turns back, but not far enough, since $k\tan\theta < 1$.

Beyond STEP

Constant acceleration in any direction always gives a parabola. Here the effective gravity is simply tilted, and rotating the axes to line up with it turns the problem back into an ordinary projectile. That idea, choosing coordinates adapted to the forces, is central to first-year mechanics, as are charged particles in uniform electric fields, which behave in just this way. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

WORKED SOLUTION

STEP I 2014, Question 10

Mechanics • Collisions and Restitution

STEP I 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q10

The key idea

Both parts are energy for the free flight and Newton's experimental law for each impact. In part (ii) the trick is to notice that at the moment of the ball-on-ball collision the big ball has just reversed direction, while the small ball is still coming down at full speed: the two meet with a large closing speed, and a heavy lower ball can hurl the light one far higher than it started. After that it is a two-equation collision problem, and the given numbers make the ratio of speeds come out cleanly.

Solution

Model both balls as uniform rigid spheres moving vertically, with no air resistance, and take g as the acceleration due to gravity. Each impact is instantaneous, so gravity does not change any momentum during it.

Part (i)

The ball touches the ground when its centre is at height R, so it has fallen a distance H. By conservation of energy its speed just before impact is

$$u = \sqrt{2gH}.$$

The ground is fixed, so Newton's experimental law gives a rebound speed of eu upwards. The ball then rises until its speed is zero; by energy again it rises a further

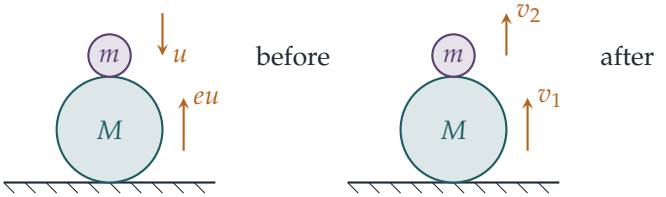
$$\frac{(eu)^2}{2g} = \frac{e^2 \cdot 2gH}{2g} = e^2 H.$$

Its centre starts this rise at height R, so it reaches a height $R + He^2$ above the ground, as required.

Part (ii)

Both balls fall freely from rest together, so at every instant they have the same speed. The lower ball reaches the ground after its centre has fallen H, so at that moment both balls are moving downwards at $u = \sqrt{2gH}$. The lower ball bounces and, as in part (i), leaves the ground at speed eu upwards; immediately afterwards it meets the upper ball, which is still moving down at u.

Take upwards as positive, and let the velocities just after the collision between the balls be v_1 (ball of mass M) and v_2 (ball of mass m).



Conservation of momentum gives

$$Meu - mu = Mv_1 + mv_2,$$

and Newton's experimental law (speed of separation is e times speed of approach, where the approach speed is $eu + u$) gives

$$v_2 - v_1 = e(1+e)u.$$

Substituting $v_1 = v_2 - e(1+e)u$ into the momentum equation,

$$\begin{aligned}(M+m)v_2 &= Meu - mu + Me(1+e)u \\ &= \big(Me(2+e) - m\big)u,\end{aligned}$$

so

$$v_2 = \frac{Me(2+e) - m}{M+m}\,u.$$

? **Ask yourself** Where is the centre of the small ball when the collision happens?

The collision is immediately after the bounce, so the big ball is still touching the ground: its centre is at height R, its top at $2R$, and the small ball's centre at $2R + r$. From there the small ball rises a further $v_2^2/(2g)$, so

$$h = 2R + r + \frac{v_2^2}{2g}.$$

With the given values, $2R + r = 0.45$, so $v_2^2/(2g) = 4.05$; and $u^2/(2g) = H = 1.8$. Dividing,

$$\frac{v_2^2}{u^2} = \frac{4.05}{1.8} = \frac{9}{4}, \qquad \text{so} \qquad v_2 = \tfrac{3}{2}u,$$

taking the positive root because the small ball moves upwards after the collision.

With $e = \frac{2}{3}$ we have $e(2+e) = \frac{2}{3} \cdot \frac{8}{3} = \frac{16}{9}$. Writing $k = M/m$ and dividing top and bottom by m,

$$\frac{\frac{16}{9}k - 1}{k+1} = \frac{3}{2} \implies \tfrac{32}{9}k - 2 = 3k + 3 \implies \tfrac{5}{9}k = 5,$$

so

$$\frac{M}{m} = 9.$$

Checks

- With $M/m = 9$ we get $v_2 = \frac{3}{2}u$ and $v_1 = v_2 - \frac{10}{9}u = \frac{7}{18}u$. Momentum before is $9m \cdot \frac{2}{3}u - mu = 5mu$, and after is $9m \cdot \frac{7}{18}u + m \cdot \frac{3}{2}u = 5mu$. Both agree.
- Since $v_1 < v_2$ the balls separate after the impact, as they must.
- Numerically, $u = \sqrt{2 \times 9.8 \times 1.8} \approx 5.94\,\mathrm{m\,s^{-1}}$ and $v_2 \approx 8.91\,\mathrm{m\,s^{-1}}$, which rises 4.05 m, giving $h = 4.5$.
- If $M \gg m$, $v_2 \to e(2+e)u$; with $e = 1$ this is $3u$, the well-known factor of nine in height for a perfectly elastic "stacked ball" drop.

Common slips

- Measuring the final height from the ground as $v_2^2/(2g)$ alone, forgetting that the collision happens with the small ball's centre at $2R + r$.
- Using $e \cdot 2u$, or eu, as the separation speed: the approach speed is $eu + u$, because the balls are moving towards each other.
- Sign errors: fix upwards as positive and give the upper ball velocity $-u$ before the collision.

Beyond STEP

Stacked-ball drops are a lovely demonstration of how momentum can be funnelled into a light object: with a tower of several balls, each lighter than the one beneath, the top ball can be launched to many times its release height. The cleanest way to analyse such collisions is in the zero-momentum (centre-of-mass) frame, where each ball's velocity is simply reversed and scaled by e. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops collisions and the centre-of-mass frame carefully.

WORKED SOLUTION

STEP I 2014, Question 11

Mechanics • Connected Pulley Systems

STEP I 2014, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q11

The key idea

Write Newton's second law for each particle, using one tension per string, and get the accelerations of all the particles from the string constraints. In System II the important fact is that the pulley P_1 is *light*: the net force on it must be zero (otherwise it would have infinite acceleration), so the upper string's tension is exactly twice the lower one's. Once that is in place, P_1 and its two particles act on M just as a single mass 4μ would.

Solution

Throughout, the particles are modelled as point masses, the strings as light and inextensible, and the pulleys as smooth and light, so the tension is the same all along each string.

Part (i)

Let the tension in the string be T. Since the string is inextensible, if M accelerates downwards at a_1 then m accelerates upwards at a_1. Newton's second law for each particle, taking the direction of motion as positive, gives

$$\begin{aligned} M: &\quad Mg - T = Ma_1, \\ m: &\quad T - mg = ma_1. \end{aligned}$$

Adding, $(M-m)g = (M+m)a_1$, so

$$a_1 = \frac{M-m}{M+m}\,g,$$

as required. Substituting back into the equation for m,

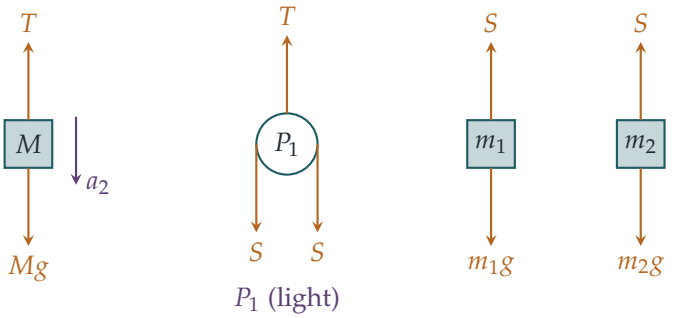
$$T = m(g+a_1) = m \cdot \frac{2M}{M+m}\,g = \frac{2Mmg}{M+m}.$$

Both straight parts of the string leave the pulley vertically downwards and each pulls with tension T, so the force on the pulley due to the tension is

$$2T = \frac{4Mmg}{M+m} \quad \text{vertically downwards.}$$

Part (ii)

Let T be the tension in the string over the fixed pulley P and S the tension in the string over P_1.



The light pulley. The forces on P_1 are T upwards and $2S$ downwards (its own weight is zero). Since its mass is zero, $T - 2S = 0 \cdot (\text{acceleration}) = 0$, so

$$T = 2S.$$

Ask yourself How do the accelerations of m_1 and m_2 relate to that of M?

The upper string is inextensible, so if M accelerates downwards at a_2 then P_1 accelerates *upwards* at a_2. The lower string only fixes the motion of m_1 and m_2 *relative to* P_1: if m_1 accelerates downwards at b relative to P_1, then m_2 accelerates upwards at b relative to P_1. Adding on the motion of P_1, the actual accelerations are

$$m_1 : \ b - a_2 \text{ downwards}, \qquad m_2 : \ b + a_2 \text{ upwards}.$$

Here b is a second unknown, which we will eliminate.

Equations of motion.

$$\begin{aligned} M : &\quad Mg - 2S = Ma_2, \\ m_1 : &\quad m_1 g - S = m_1(b - a_2), \\ m_2 : &\quad S - m_2 g = m_2(b + a_2). \end{aligned}$$

Divide the m_1 equation by m_1 and the m_2 equation by m_2:

$$g - \frac{S}{m_1} + a_2 = b = \frac{S}{m_2} - g - a_2.$$

Hence
$$2(g+a_2) = S\left(\frac{1}{m_1}+\frac{1}{m_2}\right) = S\,\frac{m_1+m_2}{m_1 m_2} = \frac{S}{\mu},$$
so $S = 2\mu(g+a_2)$ and $T = 2S = 4\mu(g+a_2)$. Substituting into the equation for M,
$$Mg - 4\mu g - 4\mu a_2 = Ma_2 \quad\Longrightarrow\quad (M+4\mu)a_2 = (M-4\mu)g,$$
and therefore
$$a_2 = \frac{M-4\mu}{M+4\mu}\,g,$$
as required. Compare this with part (i): M behaves exactly as if it were hanging against a single particle of mass 4μ.

When are a_1 and a_2 equal? Put $f(x) = \dfrac{M-x}{M+x}$ for $x > 0$, so that $a_1 = f(m)\,g$ and $a_2 = f(4\mu)\,g$. Since
$$f(x) = \frac{2M}{M+x} - 1$$
and $M > 0$, f is strictly decreasing, hence one-to-one, on $x > 0$. So $a_1 = a_2$ if and only if $m = 4\mu$. With $m = m_1 + m_2$ this becomes
$$m_1+m_2 = \frac{4m_1m_2}{m_1+m_2} \iff (m_1+m_2)^2 = 4m_1m_2 \iff (m_1-m_2)^2 = 0 \iff m_1 = m_2,$$
where the first step multiplies by $m_1+m_2 > 0$, which is reversible. This completes the proof in both directions.

Checks

- If $m_1 = m_2$, the lower part of System II balances, so $b = 0$: P_1 and its particles move as one body of mass m_1+m_2, and System II reduces to System I. That matches $4\mu = m_1+m_2$ in this case.
- If $m_2 \to 0$ then $\mu \to 0$ and $a_2 \to g$: nothing heavy restrains M, since m_1 simply falls and takes no tension. Sensible.
- The tension $T = 4\mu(g+a_2)$ in part (ii) can also be found from part (i): if P_1 were fixed, the force on it would be $4m_1m_2g/(m_1+m_2) = 4\mu g$. Accelerating upwards at a_2 just replaces g by $g+a_2$.
- Numerically, with $M = 3$, $m_1 = 1$, $m_2 = 2$ we get $\mu = \frac{2}{3}$ and $a_2 = \frac{1}{17}g \approx 0.576\ \mathrm{m\,s^{-2}}$ (for $g = 9.8\ \mathrm{m\,s^{-2}}$). Solving the four equations directly gives the same value.

Common slips

- Forgetting that P_1 moves, and writing the accelerations of m_1 and m_2 as $\pm b$ instead of $b - a_2$ and $b + a_2$.

- Using $T = S$ across the light pulley. Each side of P_1 pulls down with S, so the string above it must carry $2S$.
- Concluding $a_1 = a_2 \Rightarrow m = 4\mu$ without saying why: it needs f to be one-to-one.

Beyond STEP

This set-up is an Atwood machine within an Atwood machine, and $\mu = m_1 m_2/(m_1 + m_2)$ is the *reduced mass*, the same quantity that appears when a two-body problem such as a planet and its star is reduced to a single body. Systems with constraints like these are handled most cleanly by Lagrangian mechanics, which works directly with the independent coordinates (here the height of P_1 and the position of m_1 relative to it) and never needs the tensions. Kleppner and Kolenkow, *An Introduction to Mechanics*, works through pulley systems carefully and goes on to reduced mass in the two-body problem.

WORKED SOLUTION

STEP I 2014, Question 12

Probability and Statistics • Discrete Random Variables

STEP I 2014, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q12

The key idea

Squaring removes the modulus signs, so $\mathrm{E}(X^2)$ needs only $\mathrm{E}(S)$ and $\mathrm{E}(S^2)$ for the die score S, averaged over the two coin outcomes. The integer condition then turns into a divisibility question about $(k-1)^2$, which leaves very few single-digit candidates. Once k is known, X takes only five values, and the two-round game is a short enumeration of pairs.

Solution

Finding $\mathrm{E}(X^2)$

Let S be the number shown on the die, so

$$\mathrm{P}(S=1)=\tfrac{1}{2}, \qquad \mathrm{P}(S=2)=\tfrac{1}{3}, \qquad \mathrm{P}(S=3)=\tfrac{1}{6}.$$

Then

$$\mathrm{E}(S)=\tfrac{1}{2}+\tfrac{2}{3}+\tfrac{3}{6}=\tfrac{5}{3}, \qquad \mathrm{E}(S^2)=\tfrac{1}{2}+\tfrac{4}{3}+\tfrac{9}{6}=\tfrac{10}{3}.$$

The coin is fair and independent of the die. Since $|a|^2=a^2$, given heads $X^2=(kS-1)^2$, and given tails $X^2=(k-S)^2$. So

$$\begin{aligned}\mathrm{E}(X^2) &= \tfrac{1}{2}\,\mathrm{E}\left[(kS-1)^2\right]+\tfrac{1}{2}\,\mathrm{E}\left[(k-S)^2\right]\\ &= \tfrac{1}{2}\left[k^2\mathrm{E}(S^2)-2k\mathrm{E}(S)+1\right]+\tfrac{1}{2}\left[k^2-2k\mathrm{E}(S)+\mathrm{E}(S^2)\right]\\ &= \tfrac{1}{2}\left[\tfrac{10}{3}k^2-\tfrac{10}{3}k+1+k^2-\tfrac{10}{3}k+\tfrac{10}{3}\right]\\ &= \frac{13k^2-20k+13}{6}.\end{aligned}$$

On the other hand,

$$k+\frac{13(k-1)^2}{6}=\frac{6k+13k^2-26k+13}{6}=\frac{13k^2-20k+13}{6},$$

so $\mathrm{E}(X^2)=k+\frac{13}{6}(k-1)^2$, as required.

Finding k

Since k is an integer, $\mathrm{E}(X^2)$ is an integer exactly when 6 divides $13(k-1)^2$. As 13 is coprime to 6, this means $6 \mid (k-1)^2$, that is, both 2 and 3 divide $(k-1)^2$. Because 2 and 3 are prime, each must then divide $k-1$ itself, so $6 \mid k-1$. With $1 \le k \le 9$ this leaves $k = 1$ or $k = 7$.

Ask yourself Both candidates make $\mathrm{E}(X^2)$ an integer. Which condition rules one out?

The condition on $\mathrm{E}(X)$ does. If $k = 1$, both rules give $X = |S-1| = S-1$, so $\mathrm{E}(X) = \frac{5}{3} - 1 = \frac{2}{3}$, which is not an integer.

If $k = 7$, then $7S - 1 > 0$ and $7 - S > 0$ for every face, so the modulus signs can be dropped:

$$\mathrm{E}(X) = \tfrac{1}{2}\left(7 \cdot \tfrac{5}{3} - 1\right) + \tfrac{1}{2}\left(7 - \tfrac{5}{3}\right) = \tfrac{16}{3} + \tfrac{8}{3} = 8,$$

and $\mathrm{E}(X^2) = 7 + \frac{13}{6} \cdot 36 = 85$. Both are positive integers, so $k = 7$.

The distribution of X

With $k = 7$, heads gives $X = 7S - 1 \in \{6, 13, 20\}$ and tails gives $X = 7 - S \in \{6, 5, 4\}$. Each (coin, die) combination has probability $\frac{1}{2}\mathrm{P}(S = s)$, and the value 6 arises both from heads with $S = 1$ and from tails with $S = 1$:

x	4	5	6	13	20
$\mathrm{P}(X = x)$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{12}$

The probabilities add to 1.

The two-round game

Let X_1, X_2 be the outcomes of the two rounds. They are independent, since each round uses a fresh roll and toss, and each has the distribution above. We need the pairs with sum above 25 and with sum exactly 25.

Ask yourself How can we list the pairs with total over 25 without missing any?

Sort them by the larger value. If neither outcome is 20, the largest possible total is $13 + 13 = 26$, and $13 + 6 = 19$ is already too small, so $(13, 13)$ is the only such pair. If one outcome is 20, the other must exceed 5, so it is 6, 13 or 20; it equals 5 for a total of exactly 25. Hence

$$\begin{aligned}\mathrm{P}(X_1 + X_2 > 25) &= \mathrm{P}(13,13) + 2\,\mathrm{P}(20,6) + 2\,\mathrm{P}(20,13) + \mathrm{P}(20,20) \\ &= \tfrac{1}{36} + 2 \cdot \tfrac{1}{12} \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{12} \cdot \tfrac{1}{6} + \tfrac{1}{144} \\ &= \frac{4 + 12 + 4 + 1}{144} = \frac{7}{48},\end{aligned}$$

where the factor 2 counts the two orders. Similarly

$$\mathrm{P}(X_1 + X_2 = 25) = 2\,\mathrm{P}(20,5) = 2 \cdot \tfrac{1}{12} \cdot \tfrac{1}{6} = \tfrac{1}{36}.$$

So the amount the gambler expects to be paid, in pounds, is

$$\frac{7w}{48} + \frac{1}{36}.$$

The game favours the casino when this is less than the £1 stake:

$$\frac{7w}{48} + \frac{1}{36} < 1 \iff \frac{7w}{48} < \frac{35}{36} \iff w < \frac{35 \cdot 48}{36 \cdot 7} = \frac{20}{3}.$$

As w is an integer, the largest possible value is $w = 6$. Since $\frac{20}{3}$ is not an integer, the answer is the same whether or not a fair game (expected payment exactly £1) is allowed.

Checks

- From the table, $\mathrm{E}(X) = \frac{4+10+36+26+20}{12} = 8$ and $\mathrm{E}(X^2) = \frac{16+50+216+338+400}{12} = 85$, matching the formulas.
- With $w = 6$ the expected payment is $\frac{7}{8} + \frac{1}{36} = \frac{65}{72} \approx$ £0.90, below the stake; with $w = 7$ it is $\frac{151}{144} \approx$ £1.05, above it.
- A computer simulation of 2×10^5 two-round games gave a total over 25 in about 14.5% of games ($\frac{7}{48} \approx 14.6\%$) and exactly 25 in about 2.8% ($\frac{1}{36} \approx 2.8\%$).

Common slips

- Listing $X = 6$ twice in the distribution, once for heads and once for tails, instead of combining them into $\mathrm{P}(X = 6) = \frac{1}{2}$.
- Forgetting the factor 2 for the two orders of an unequal pair such as $(20, 6)$.
- Treating "sum equals 25" as a win of £w or as a loss: the gambler just gets the stake back.

Beyond STEP

The calculation of $\mathrm{E}(X^2)$ is an instance of the law of total expectation, $\mathrm{E}(Y) = \sum_i \mathrm{P}(A_i)\,\mathrm{E}(Y \mid A_i)$, which first-year probability courses use constantly. The two-round game is about the distribution of a sum of independent random variables, a convolution, which generating functions handle neatly. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

WORKED SOLUTION

STEP I 2014, Question 13

Probability and Statistics • Triangular Distribution

STEP I 2014, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

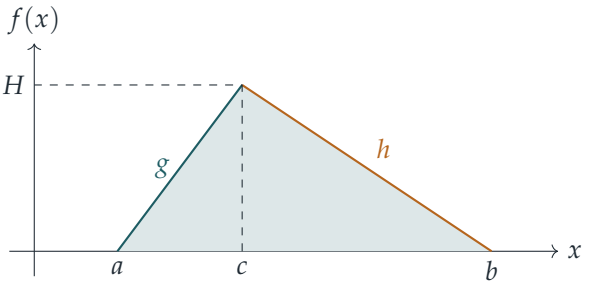
step.maths.org/questions/14-s1-q13

The key idea

The graph of the density is a triangle with base from a to b and apex above c, and its area must be 1. That fixes the height of the apex, and then each linear piece is pinned down by two points. For the mean and median, shifting the variable so that each linear piece starts at zero keeps the integrals tidy, and for the median the first question is simply which side of c it lies on.

Solution

Throughout we take $a < c < b$, so that both pieces are genuine (the question needs $c - a$ and $b - c$ to be non-zero for its formulas to make sense).



The two linear pieces

Let $H = g(c) = h(c)$ be the height of the apex. The region under the graph of f is a triangle with base $b - a$ and height H, and the total probability is 1, so

$$\tfrac{1}{2}(b-a)H = 1, \qquad H = \frac{2}{b-a}.$$

Since g is linear with $g(a) = 0$, we have $g(x) = k(x-a)$ for some constant k, and $g(c) = H$ gives $k(c-a) = \dfrac{2}{b-a}$. Hence

$$g(x) = \frac{2(x-a)}{(b-a)(c-a)},$$

as required. In the same way $h(x) = m(b-x)$ with $h(c) = H$, so

$$h(x) = \frac{2(b-x)}{(b-a)(b-c)}.$$

As required, g is increasing and h is decreasing, because $b-a$, $c-a$ and $b-c$ are all positive.

Part (i)

Write $\mathrm{E}(X) = I_1 + I_2$, where $I_1 = \int_a^c x\,g(x)\,\mathrm{d}x$ and $I_2 = \int_c^b x\,h(x)\,\mathrm{d}x$.

In I_1 substitute $x = a + t$, so t runs from 0 to $c-a$:

$$\begin{aligned} I_1 &= \frac{2}{(b-a)(c-a)} \int_0^{c-a} (a+t)\,t\,\mathrm{d}t = \frac{2}{(b-a)(c-a)} \left[\frac{a(c-a)^2}{2} + \frac{(c-a)^3}{3}\right] \\ &= \frac{(c-a)\big(3a + 2(c-a)\big)}{3(b-a)} = \frac{(c-a)(a+2c)}{3(b-a)}. \end{aligned}$$

In I_2 substitute $x = b - s$, so s runs from 0 to $b-c$:

$$\begin{aligned} I_2 &= \frac{2}{(b-a)(b-c)} \int_0^{b-c} (b-s)\,s\,\mathrm{d}s = \frac{2}{(b-a)(b-c)} \left[\frac{b(b-c)^2}{2} - \frac{(b-c)^3}{3}\right] \\ &= \frac{(b-c)\big(3b - 2(b-c)\big)}{3(b-a)} = \frac{(b-c)(b+2c)}{3(b-a)}. \end{aligned}$$

Adding, the numerator is

$$\begin{aligned} (c-a)(a+2c) + (b-c)(b+2c) &= \big(ac + 2c^2 - a^2 - 2ac\big) + \big(b^2 + 2bc - bc - 2c^2\big) \\ &= b^2 - a^2 + bc - ac \\ &= (b-a)(b+a+c). \end{aligned}$$

Dividing by $3(b-a)$,

$$\mathrm{E}(X) = \tfrac{1}{3}(a+b+c),$$

as required. (This is the x-coordinate of the centroid of the triangle, which is no coincidence: the mean is the balance point of the region under the density.)

Part (ii)

First find how much probability lies to the left of the apex. This is the area of the left-hand triangle:

$$\mathrm{P}(X \le c) = \tfrac{1}{2}(c-a)H = \frac{c-a}{b-a}.$$

Ask yourself How do we know whether the median is to the left or the right of c?

The median M satisfies $\mathrm{P}(X \le M) = \frac{1}{2}$. If $\mathrm{P}(X \le c) \ge \frac{1}{2}$ the median is at or to the left of c; otherwise it is to the right. Now $\frac{c-a}{b-a} \ge \frac{1}{2}$ exactly when $c \ge \frac{1}{2}(a+b)$, that is, when the apex is at or to the right of the midpoint of $[a, b]$. So the cases are decided by comparing c with $\frac{1}{2}(a+b)$.

Case $c \ge \frac{1}{2}(a+b)$. Then $a < M \le c$, and for $a \le m \le c$ the area of the small triangle from a to m is

$$\mathrm{P}(X \le m) = \int_a^m g(x)\,\mathrm{d}x = \frac{(m-a)^2}{(b-a)(c-a)}.$$

Setting this equal to $\frac{1}{2}$ and taking the positive root (since $M > a$),

$$M = a + \sqrt{\tfrac{1}{2}(b-a)(c-a)}.$$

Case $c < \frac{1}{2}(a+b)$. Now the median is to the right of c, and it is easier to work from the right-hand end. For $c \le m \le b$,

$$\mathrm{P}(X \ge m) = \int_m^b h(x)\,\mathrm{d}x = \frac{(b-m)^2}{(b-a)(b-c)}.$$

Setting this equal to $\frac{1}{2}$ and using $M < b$,

$$M = b - \sqrt{\tfrac{1}{2}(b-a)(b-c)}.$$

When $c = \frac{1}{2}(a+b)$ the distribution is symmetric, and both formulas give $M = \frac{1}{2}(a+b)$ (for instance the first gives $a + \sqrt{\frac{1}{4}(b-a)^2} = a + \frac{1}{2}(b-a)$), so the two cases join up.

Checks

- With $a = 0$, $b = 4$, $c = 1$: $\mathrm{P}(X \le c) = \frac{1}{4} < \frac{1}{2}$, so we are in the second case and $M = 4 - \sqrt{6} \approx 1.5505$. The mean is $\frac{5}{3}$. A simulation of 200 000 values by computer gave sample median 1.5505 and sample mean 1.668.
- With $a = 0$, $b = 4$, $c = 3$: the first case applies and $M = \sqrt{6} \approx 2.449$, the mirror image of the previous example, as it should be.
- The median always lies between the mode c and the midpoint $\frac{1}{2}(a+b)$. For instance in the first case $M \le c$, and $M \ge \frac{1}{2}(a+b)$ because $(b-a)(c-a) \ge \frac{1}{2}(b-a)^2$ when $c \ge \frac{1}{2}(a+b)$.

Common slips

- Using the wrong piece of the density for the median: check first which side of c the median lies on.

- Taking the negative square root, which puts M outside $[a, b]$.
- Expanding the mean integrals directly in x without shifting: it works, but the algebra is much heavier and errors creep in.

Beyond STEP

The triangular distribution is the distribution of the sum of two independent uniform random variables when c is the midpoint, and it is widely used as a rough model when only a minimum, maximum and most likely value are known (as in project planning). Computing the distribution of sums of independent variables leads to convolution, a central tool in first-year probability. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read about it.

WORKED SOLUTION

STEP II 2014, Question 1

Pure • Triangle Geometry and Quadratics

STEP II 2014, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q1

The key idea

Walk from A to B along the zig-zag $A \to P \to Q \to B$. The three legs all have length x, and together they must cover a horizontal distance of exactly 1 and climb a net height of 0. Resolving horizontally and vertically gives $x\cos\theta$ and $x\sin\theta$, and $\cos^2\theta + \sin^2\theta = 1$ removes the unknown angle. Everything after that is a quadratic in x and its discriminant.

Solution

Put A at the origin and B at $(1, 0)$, with C above AB. Since P is at distance x from A along the ray from A through C, and Q is at distance x from B along the ray from B through C,

$$P = (x\cos\alpha,\ x\sin\alpha), \qquad Q = (1 - x\cos\beta,\ x\sin\beta).$$

Part (i)

Horizontally. The horizontal components of $\overrightarrow{AP}$, $\overrightarrow{PQ}$ and $\overrightarrow{QB}$ are $x\cos\alpha$, $x\cos\theta$ and $x\cos\beta$, and they add up to $AB = 1$. So

$$x\cos\theta = 1 - x\cos\alpha - x\cos\beta,$$

which is the first result. (In coordinates: it is the difference of the x-coordinates of Q and P.)

Vertically. P is at height $x\sin\alpha$ and Q at height $x\sin\beta$, so

$$x\sin\theta = x\sin\beta - x\sin\alpha.$$

(This is never negative: $\alpha \le \beta < 180^\circ - \alpha$ because the angles of the triangle add up to less than 180°, and on that interval $\sin\beta \ge \sin\alpha$. So Q is never lower than P.)

Eliminating θ. Squaring and adding, and using $\cos^2\theta + \sin^2\theta = 1$,

$$x^2 = \big(1 - x(\cos\alpha + \cos\beta)\big)^2 + x^2(\sin\beta - \sin\alpha)^2.$$

Expanding the right-hand side, the coefficient of x^2 is

$$\begin{aligned}(\cos\alpha+\cos\beta)^2+(\sin\beta-\sin\alpha)^2 &= (\cos^2\alpha+\sin^2\alpha)+(\cos^2\beta+\sin^2\beta)\\ &\quad+2(\cos\alpha\cos\beta-\sin\alpha\sin\beta)\\ &= 2+2\cos(\alpha+\beta).\end{aligned}$$

So $x^2 = 1-2(\cos\alpha+\cos\beta)x+\big(2+2\cos(\alpha+\beta)\big)x^2$. Taking x^2 to the right-hand side gives

$$\big(1+2\cos(\alpha+\beta)\big)x^2-2(\cos\alpha+\cos\beta)x+1=0, \qquad (*)$$

as required.

P and Q on the sides produced. If P lies on AC produced, it is still on the ray from A through C, at distance x from A, so its coordinates are still $(x\cos\alpha, x\sin\alpha)$. Likewise Q is still $(1-x\cos\beta, x\sin\beta)$. The derivation above used nothing else: the displacement from P to Q is

$$\big(1-x\cos\alpha-x\cos\beta,\ x\sin\beta-x\sin\alpha\big),$$

and $PQ^2 = x^2$ gives exactly the same equation $(*)$. (Now x may exceed AC and BC, and $1-x\cos\alpha-x\cos\beta$ may be negative, so θ may be obtuse. This doesn't matter, because the signs disappear on squaring.)

Part (ii)

The equation $(*)$ is linear exactly when $1+2\cos(\alpha+\beta)=0$, that is, $\cos(\alpha+\beta)=-\frac{1}{2}$. Since $0<\alpha+\beta<180^\circ$, the only possibility is

$$\alpha+\beta = 120^\circ.$$

Otherwise $(*)$ is a genuine quadratic, and a quarter of its discriminant is

$$\begin{aligned}\tfrac{1}{4}\Delta &= (\cos\alpha+\cos\beta)^2-\big(1+2\cos\alpha\cos\beta-2\sin\alpha\sin\beta\big)\\ &= \cos^2\alpha+\cos^2\beta-1+2\sin\alpha\sin\beta\\ &= 1-\sin^2\alpha-\sin^2\beta+2\sin\alpha\sin\beta\\ &= 1-(\sin\beta-\sin\alpha)^2,\end{aligned}$$

using $\cos^2\alpha = 1-\sin^2\alpha$ and $\cos^2\beta-1=-\sin^2\beta$.

> **Ask yourself** Why must $(\sin\beta-\sin\alpha)^2$ be less than 1?

Both angles lie strictly between 0° and 180°, so $0<\sin\alpha\le 1$ and $0<\sin\beta\le 1$. The difference of two numbers in $(0,1]$ lies strictly between -1 and 1. Hence $(\sin\beta-\sin\alpha)^2<1$, so $\Delta>0$ and $(*)$ has two distinct real roots.

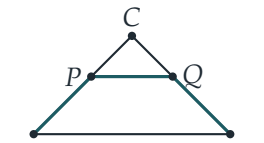
Part (iii)

(a) $\alpha=\beta=45^\circ$**.** Here $\cos(\alpha+\beta)=\cos 90^\circ=0$ and $\cos\alpha+\cos\beta=\sqrt{2}$, so $(*)$ is

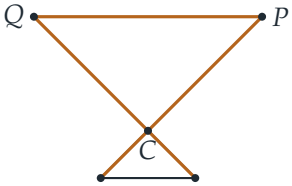
$$x^2-2\sqrt{2}\,x+1=0, \qquad x=\sqrt{2}\pm 1.$$

The triangle is right-angled and isosceles, with $C = (\frac{1}{2}, \frac{1}{2})$ and $AC = BC = \frac{1}{\sqrt{2}} \approx 0.707$.

- $x = \sqrt{2} - 1 \approx 0.414 < AC$: P and Q lie on the sides, at $P = (1 - \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}})$ and $Q = (\frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}})$. By symmetry PQ is parallel to AB, so $\theta = 0$. Check: $PQ = \sqrt{2} - 1 = x$.
- $x = \sqrt{2} + 1 \approx 2.414 > AC$: P and Q lie on AC produced and BC produced, at $P = (1 + \frac{1}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}})$ and $Q = (-\frac{1}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}})$. The two produced lines have crossed at C, so P is now to the *right* of Q. PQ is again horizontal, with length $1 + \sqrt{2} = x$.



$x = \sqrt{2} - 1$

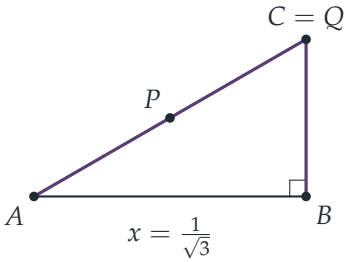


$x = \sqrt{2} + 1$

(b) $\alpha = 30°$**,** $\beta = 90°$**.** Now $\alpha + \beta = 120°$, so this is the linear case of part (ii). With $\cos\alpha + \cos\beta = \frac{\sqrt{3}}{2}$, $(*)$ becomes

$$-\sqrt{3}\,x + 1 = 0, \qquad x = \frac{1}{\sqrt{3}}.$$

The triangle has its right angle at B, with $BC = \tan 30° = \frac{1}{\sqrt{3}}$ and $AC = \frac{2}{\sqrt{3}}$. So $QB = BC$ means Q *is* C, and $AP = \frac{1}{\sqrt{3}} = \frac{1}{2}AC$ puts P at the midpoint of AC. Then $PQ = PC = \frac{1}{\sqrt{3}} = x$, as it should be. This is the only possibility.



$x = \frac{1}{\sqrt{3}}$

Where has the second root gone? If $\alpha + \beta = 120°$, the lines AC and BC meet at 60°. For very large x, the points P and Q far out along them are then almost exactly at distance x apart, forming a nearly equilateral triangle with C. As $\alpha + \beta \to 120°$, one root of the quadratic runs off to infinity.

Checks

- In (a), $(\sqrt{2}-1)(\sqrt{2}+1) = 1$, matching the product of the roots, $1/(1+2\cos 90^\circ) = 1$.
- For a random triangle, say $\alpha = 40^\circ$, $\beta = 70^\circ$, solve $(*)$, place P and Q and measure PQ: it equals x for both roots. We confirmed this by computer for many random triangles, to about 10^{-10}.
- In (b), $x = 1/\sqrt{3} \approx 0.577$ is exactly BC, and the midpoint of AC is $(\frac{1}{2}, \frac{1}{2\sqrt{3}})$, at distance $\frac{1}{\sqrt{3}}$ from $C = (1, \frac{1}{\sqrt{3}})$.

Common slips

- Getting the sign of the $\cos\beta$ term wrong: $\overrightarrow{QB}$ runs *towards* B, so its horizontal component is $+x\cos\beta$, even though Q is to the left of B.
- In part (ii), allowing $\alpha + \beta = 240^\circ$. The angles of a triangle add up to less than 180°.
- In (b), trying to use the quadratic formula when the leading coefficient is zero, or concluding that there is no solution.

Beyond STEP

Resolving a closed polygon of vectors into components, then eliminating an angle with $\cos^2 + \sin^2 = 1$, is a standard technique in kinematics and linkage problems: our zig-zag is a simple three-bar linkage. The root that "escapes to infinity" when the quadratic becomes linear is a first glimpse of projective geometry. There, parallel lines meet at a point at infinity, and a degree-two equation always has two roots once these are counted. Coxeter and Greitzer, *Geometry Revisited*, is a lovely way into classical geometry at this level.

WORKED SOLUTION

STEP II 2014, Question 2

Pure • Integral Inequalities

STEP II 2014, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q2

This question has a full guided treatment in *Beyond STEP*, Section 3.7, $\pi^2 \leq 10$: https://beyondstep.keithmcnulty.org/ch03/wirtinger.html

The key idea

Both sides of $(*)$ are integrals we can work out exactly for any explicit f, so every function that vanishes at both ends turns $(*)$ into an inequality between numbers involving π. The only real choice is to pick the free constants so that $f(0) = f(\pi) = 0$. A function that does *not* vanish at an end can be large while changing slowly, which is why the end conditions matter.

Solution

Part (i)

With $f(x) = \sin nx$ we have $f'(x) = n\cos nx$. Using $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$ and $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$,

$$\int_0^\pi \sin^2 nx\,\mathrm{d}x = \tfrac{1}{2}\Big[x - \tfrac{1}{2n}\sin 2nx\Big]_0^\pi = \frac{\pi}{2},$$
$$\int_0^\pi n^2\cos^2 nx\,\mathrm{d}x = \tfrac{n^2}{2}\Big[x + \tfrac{1}{2n}\sin 2nx\Big]_0^\pi = \frac{n^2\pi}{2},$$

since $\sin 2n\pi = 0$ for an integer n. As $n \geq 1$, $n^2 \geq 1$, so $\frac{\pi}{2} \leq \frac{n^2\pi}{2}$ and $(*)$ holds (with equality exactly when $n = 1$).

Counterexamples. Take $f(x) = \pi - x$, which has $f(\pi) = 0$ but $f(0) = \pi \neq 0$. Then $f'(x) = -1$ and

$$\int_0^\pi (\pi-x)^2\,\mathrm{d}x = \Big[-\tfrac{1}{3}(\pi-x)^3\Big]_0^\pi = \frac{\pi^3}{3}, \qquad \int_0^\pi 1\,\mathrm{d}x = \pi.$$

Since $\pi^2 > 3$, we have $\frac{\pi^3}{3} > \pi$, so $(*)$ fails. Similarly $f(x) = x$ has $f(0) = 0$ but $f(\pi) = \pi \neq 0$, and gives the same two integrals $\frac{\pi^3}{3} > \pi$, so $(*)$ fails again. So neither end condition can be dropped, even when the other one holds.

Part (ii)

The quadratic. We need $f(0) = c = 0$ and $f(\pi) = a\pi^2 + b\pi = 0$, so $b = -a\pi$ and $f(x) = ax(x - \pi)$. Taking $a = 0$ would give $f \equiv 0$ and the useless inequality $0 \leq 0$, so take $a \neq 0$. Both sides of $(*)$ are multiplied by a^2 when f is multiplied by a, so we may as well take $a = -1$:

$$f(x) = x(\pi - x), \qquad f'(x) = \pi - 2x.$$

Substituting $x = \frac{\pi}{2} + u$ makes both integrals symmetric: $f = \frac{\pi^2}{4} - u^2$ and $f' = -2u$, with u running over $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Writing $h = \frac{\pi}{2}$,

$$\int_0^\pi f^2 \,\mathrm{d}x = \int_{-h}^{h} \left(h^2 - u^2\right)^2 \mathrm{d}u = 2\left(h^5 - \tfrac{2}{3}h^5 + \tfrac{1}{5}h^5\right) = \frac{16h^5}{15} = \frac{\pi^5}{30},$$
$$\int_0^\pi f'^2 \,\mathrm{d}x = \int_{-h}^{h} 4u^2 \,\mathrm{d}u = \frac{8h^3}{3} = \frac{\pi^3}{3}.$$

So $(*)$ gives $\dfrac{\pi^5}{30} \leq \dfrac{\pi^3}{3}$, and dividing by the positive number $\frac{\pi^3}{30}$,

$$\pi^2 \leq 10,$$

as required.

The trigonometric function. Now $f(0) = q + r$ and $f(\pi) = p \sin\frac{\pi}{2} + q\cos\frac{\pi}{2} + r = p + r$. Both vanish exactly when $p = q = -r$. Again $r = 0$ gives $f \equiv 0$, and scaling doesn't matter, so take $r = -1$:

$$f(x) = \sin\tfrac{1}{2}x + \cos\tfrac{1}{2}x - 1, \qquad f'(x) = \tfrac{1}{2}\left(\cos\tfrac{1}{2}x - \sin\tfrac{1}{2}x\right).$$

? **Ask yourself** How do we square f without a mess of cross terms?

Group it as $(s + c) - 1$ with $s = \sin\frac{1}{2}x$, $c = \cos\frac{1}{2}x$. Then $(s + c)^2 = 1 + 2sc = 1 + \sin x$, so

$$f^2 = (s + c)^2 - 2(s + c) + 1 = 2 + \sin x - 2\sin\tfrac{1}{2}x - 2\cos\tfrac{1}{2}x.$$

Using $\int_0^\pi \sin x \,\mathrm{d}x = 2$, $\int_0^\pi \sin\frac{1}{2}x \,\mathrm{d}x = \left[-2\cos\frac{1}{2}x\right]_0^\pi = 2$ and $\int_0^\pi \cos\frac{1}{2}x \,\mathrm{d}x = \left[2\sin\frac{1}{2}x\right]_0^\pi = 2$,

$$\int_0^\pi f^2 \,\mathrm{d}x = 2\pi + 2 - 4 - 4 = 2\pi - 6.$$

In the same way $(c - s)^2 = 1 - \sin x$, so

$$\int_0^\pi f'^2 \,\mathrm{d}x = \frac{1}{4}\int_0^\pi (1 - \sin x) \,\mathrm{d}x = \frac{\pi - 2}{4}.$$

Now $(*)$ says $2\pi - 6 \le \frac{1}{4}(\pi - 2)$, that is $8\pi - 24 \le \pi - 2$, so

$$7\pi \le 22, \qquad \text{i.e.} \qquad \pi \le \frac{22}{7}.$$

Comparing. Squaring (both sides are positive), the second inequality gives

$$\pi^2 \le \frac{484}{49} < \frac{490}{49} = 10.$$

So the trigonometric choice gives the smaller upper bound, $\pi^2 \le \frac{484}{49} \approx 9.878$, and hence the better estimate. Both are upper bounds, so a smaller one is closer to the truth.

Checks

- The true value is $\pi^2 \approx 9.8696$, below both 10 and $\frac{484}{49} \approx 9.8776$, as it must be.
- Numerically, for the trigonometric f the two sides of $(*)$ are $2\pi - 6 \approx 0.2832$ and $\frac{1}{4}(\pi - 2) \approx 0.2854$: very close, which is why the bound $\pi \le \frac{22}{7}$ is so sharp. For the quadratic they are $\frac{\pi^5}{30} \approx 10.20$ and $\frac{\pi^3}{3} \approx 10.34$.
- In part (i), $n = 1$ gives equality, so $\sin x$ is the extreme case of $(*)$. The trigonometric f is close in shape to $\sin x$, while $x(\pi - x)$ is a little less so, which explains which bound is better.

Common slips

- Choosing a counterexample such as $f = 1$ that breaks *both* end conditions. It is valid, but $\pi - x$ and x show more: each condition is needed on its own.
- Forgetting $\int_0^\pi \sin x \, dx = 2$ (not 0) when integrating f^2 in the trigonometric case.
- Saying 10 is the better estimate because it is "rounder". Both are upper bounds, so the smaller one wins.

Beyond STEP

Inequality $(*)$ is Wirtinger's inequality, and the best constant 1 comes from $\sin x$: it is the smallest eigenvalue of $-f'' = \lambda f$ with $f(0) = f(\pi) = 0$. Testing $(*)$ on chosen functions is the Rayleigh–Ritz method, which always gives upper bounds for that eigenvalue and underlies how engineers estimate vibration frequencies. Fourier series give a slick proof of $(*)$ itself. For the analysis behind this, see Abbott, *Understanding Analysis*; for the eigenvalue side, Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 2014, Question 3

Pure • Tangents and Envelopes

STEP II 2014, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q3

The key idea

The tangent at P is just a straight line, with gradient y' and y-intercept $y - xy'$, so part (i) turns the constant-distance property into a differential equation. Differentiating that equation makes a factor y'' appear, and the two ways of making the product zero give the two families of curves: straight lines ($y'' = 0$) and a circle. Part (iii) then asks us to glue pieces of these together smoothly.

Solution

Part (i)

If $m = 0$ the line is $y = c$ and the distance is clearly c, so suppose $m \neq 0$. The line meets the y-axis at $A = (0, c)$, and the line makes an acute angle θ with the x-axis, where $\tan\theta = |m|$. Let N be the foot of the perpendicular from O to the line. In the right-angled triangle ONA, the angle at A between the line and the y-axis is $\frac{\pi}{2} - \theta$, so the angle AON is θ, and

$$ON = OA\cos\theta = \frac{c}{\sqrt{1+\tan^2\theta}} = \frac{c}{\sqrt{1+m^2}} = c(m^2+1)^{-\frac{1}{2}}.$$

(The same value comes from minimising $x^2 + (mx+c)^2$, a quadratic in x with minimum value $\frac{c^2}{1+m^2}$ at $x = -\frac{mc}{1+m^2}$.)

If $c < 0$, reflecting in the origin turns the line into $y = mx + |c|$ without changing distances from O, so in general the distance is $|c|(1+m^2)^{-\frac{1}{2}}$.

Part (ii)

At $P = (x, y)$ the tangent has gradient y' and passes through P, so its equation in coordinates (X, Y) is

$$Y = y'X + (y - xy').$$

This is a line of the form in part (i) with $m = y'$ and $c = y - xy'$ (it is non-vertical by assumption). Its distance from O is therefore

$$a = \frac{|y - xy'|}{\sqrt{1 + (y')^2}},$$

and squaring gives

$$(y - xy')^2 = a^2\big(1 + (y')^2\big). \qquad (*)$$

Differentiating. Here a is a constant, and $\frac{\mathrm{d}}{\mathrm{d}x}(y - xy') = y' - y' - xy'' = -xy''$. So differentiating $(*)$ gives

$$2(y - xy')(-xy'') = 2a^2y'y'', \qquad \text{that is,} \qquad y''\left[\, x(y - xy') + a^2y' \,\right] = 0.$$

Ask yourself How does a product being zero turn into two separate cases for the curve?

At each point, either $y'' = 0$ or the bracket is zero. On a stretch of the curve where $y'' = 0$ the curve is straight; on a stretch where $y'' \neq 0$ the bracket must vanish. We take the two cases in turn. (We assume $a > 0$; if $a = 0$ every tangent passes through O and C is simply a line through the origin.)

Case 1: $y'' = 0$. Then $y' = m$ is constant and $y = mx + k$ for constants m, k. Substituting into $(*)$, where $y - xy' = k$, gives $k^2 = a^2(1 + m^2)$, so

$$y = mx \pm a(1 + m^2)^{\frac{1}{2}}.$$

Case 2: the bracket is zero. Then $xy - x^2y' + a^2y' = 0$, so

$$(x^2 - a^2)\, y' = xy.$$

First, $x^2 \neq a^2$: if $x = \pm a$ then $xy = 0$ forces $y = 0$, and $(*)$ becomes $a^2(y')^2 = a^2\big(1 + (y')^2\big)$, which is impossible. So we may divide:

$$y' = \frac{xy}{x^2 - a^2}, \qquad y - xy' = \frac{y(x^2 - a^2) - x^2y}{x^2 - a^2} = \frac{-a^2y}{x^2 - a^2}.$$

Substituting both into $(*)$ and multiplying through by $(x^2 - a^2)^2/a^2$:

$$\begin{aligned} & a^2y^2 = (x^2 - a^2)^2 + x^2y^2 \\ \iff \quad & y^2(a^2 - x^2) = (a^2 - x^2)^2 \\ \iff \quad & (a^2 - x^2)\,(x^2 + y^2 - a^2) = 0. \end{aligned}$$

Since $a^2 - x^2 \neq 0$, we conclude $x^2 + y^2 = a^2$.

So either $y = mx \pm a(1 + m^2)^{\frac{1}{2}}$ for some m, or $x^2 + y^2 = a^2$, as required. Both do have the property: the lines are at distance a from O by part (i), and every tangent to a circle centred at O with radius a is at distance a from O.

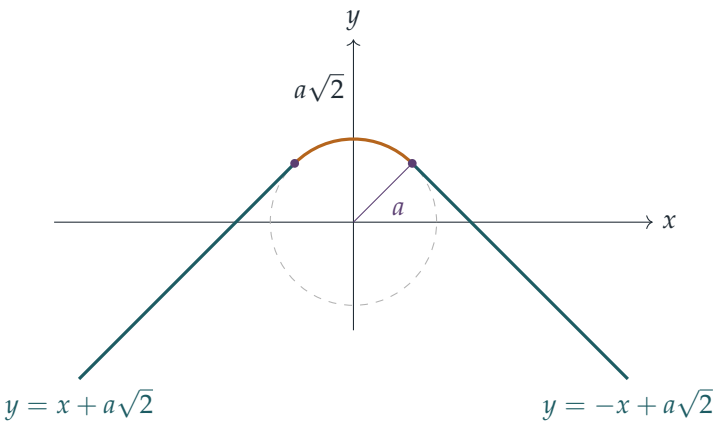
Part (iii)

At a point where a straight piece meets the arc, C must have a tangent, so the line must have the same gradient as the arc there: each line is the tangent to the circle at the end of the arc. Also, every tangent must be non-vertical, so the arc must not reach the points $(\pm a, 0)$, and since C is the graph of a function of x it uses only the upper or only the lower half of the circle.

An example: take the arc of $x^2 + y^2 = a^2$ from $\left(-\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ to $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ over the top, continued by the tangents at its ends:

$$y = \begin{cases} x + a\sqrt{2}, & x \le -\frac{a}{\sqrt{2}}, \\ \sqrt{a^2 - x^2}, & |x| \le \frac{a}{\sqrt{2}}, \\ -x + a\sqrt{2}, & x \ge \frac{a}{\sqrt{2}}. \end{cases}$$

The lines have $m = \pm 1$ and intercept $a\sqrt{2} = a(1 + m^2)^{\frac{1}{2}}$, as Case 1 requires. At $x = \pm\frac{a}{\sqrt{2}}$ the arc has gradient $-\frac{x}{y} = \mp 1$, matching the lines, so the gradient is continuous and finite everywhere.



Any arc of the upper (or lower) semicircle not containing $(\pm a, 0)$, continued by its end tangents, works in the same way.

Checks

- In part (i), $m = 0$ gives distance c, and as $|m| \to \infty$ the line becomes steep and the distance tends to 0, as it should for a line through $(0, c)$.
- On the circle, $y' = -x/y$ gives $y - xy' = a^2/y$, and both sides of $(*)$ equal a^4/y^2.
- For the example with $a = 1$, the distance from O to the tangent was checked by computer at 1001 points between $x = -5$ and $x = 5$ and was 1 every time; at $x = 2$, for instance, the tangent is $y = -x + \sqrt{2}$, at distance $\sqrt{2}/\sqrt{2} = 1$.

Common slips

- Dividing the differentiated equation by y'' and losing the straight-line solutions.
- Forgetting that a is constant, or differentiating xy' as y' instead of $y' + xy''$.
- In part (iii), joining the lines to the arc at a corner, or using a whole semicircle, whose ends have vertical tangents.

Beyond STEP

The circle here is the *envelope* of the family of lines $y = mx \pm a\sqrt{1+m^2}$, and $(*)$ is a Clairaut equation, $y = xy' + f(y')$: its general solution is a family of straight lines and its *singular solution* is their envelope. As a quadratic in y', $(*)$ has discriminant $4a^2(x^2 + y^2 - a^2)$, so the two gradient choices merge exactly on the circle; that is why solutions can be patched together there, as in part (iii), and why uniqueness of solutions fails. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness for first-order equations.

WORKED SOLUTION

STEP II 2014, Question 4

Pure • Integration by Substitution

STEP II 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q4

The key idea

The substitution $u = 1/x$ swaps the two ends of the interval $[1/b, b]$, so it turns the integral I back into an integral over the same range. If the new integrand is the old one with a sign change, we get $I = -I$. If it's something closely related, adding the two forms of I can give an integrand that is much easier to integrate. In the final part the same trick, with $u = a^2/x$ so that $[0, \infty)$ maps to itself, gives an equation that we can solve for the integral we want.

Solution

Part (i)

Call the integral I. With $x = 1/u$ we have $\mathrm{d}x = -\frac{1}{u^2}\,\mathrm{d}u$ and $\ln x = -\ln u$. When $x = 1/b$, $u = b$, and when $x = b$, $u = 1/b$. Also

$$(a^2 + x^2)(a^2x^2 + 1) = \left(a^2 + \frac{1}{u^2}\right)\left(\frac{a^2}{u^2} + 1\right) = \frac{(a^2u^2 + 1)(a^2 + u^2)}{u^4}.$$

So

$$\begin{aligned} I &= \int_b^{1/b} \frac{\frac{1}{u}(-\ln u)}{(a^2u^2 + 1)(a^2 + u^2)/u^4}\left(-\frac{1}{u^2}\right)\mathrm{d}u \\ &= \int_b^{1/b} \frac{u \ln u}{(a^2 + u^2)(a^2u^2 + 1)}\,\mathrm{d}u = -\int_{1/b}^{b} \frac{u \ln u}{(a^2 + u^2)(a^2u^2 + 1)}\,\mathrm{d}u. \end{aligned}$$

The last integral is just I again with the dummy variable renamed. So $I = -I$, which gives $I = 0$, as required. (The integrand is continuous on $[1/b, b]$ because $b > 0$, so I is a finite number and this step is valid.)

Part (ii)

Call this integral I. The same substitution gives $\frac{\mathrm{d}x}{x} = u \cdot \left(-\frac{1}{u^2}\right)\mathrm{d}u = -\frac{\mathrm{d}u}{u}$, and the limits swap as before:

$$I = \int_b^{1/b} \arctan\left(\frac{1}{u}\right)\left(-\frac{1}{u}\right)\mathrm{d}u = \int_{1/b}^{b} \frac{\arctan(1/u)}{u}\,\mathrm{d}u.$$

This time the sign doesn't simply flip, so we add this form to the original one:

$$2I = \int_{1/b}^{b} \frac{\arctan u + \arctan(1/u)}{u}\,\mathrm{d}u.$$

? **Ask yourself** What is $\arctan u + \arctan(1/u)$ when $u > 0$?

If $\theta = \arctan u$ with $u > 0$, then $0 < \theta < \frac{\pi}{2}$ and $\tan(\frac{\pi}{2} - \theta) = \cot\theta = 1/u$. Since $\frac{\pi}{2} - \theta$ also lies in $(0, \frac{\pi}{2})$, it is $\arctan(1/u)$. So $\arctan u + \arctan(1/u) = \frac{\pi}{2}$ for all $u > 0$. (For $u < 0$ the sum is $-\frac{\pi}{2}$, but every u in $[1/b, b]$ is positive.) Hence

$$2I = \frac{\pi}{2}\int_{1/b}^{b} \frac{\mathrm{d}u}{u} = \frac{\pi}{2}\left(\ln b - \ln(1/b)\right) = \pi \ln b,$$

so $I = \dfrac{\pi \ln b}{2}$, as required. This holds whether $b > 1$ or $0 < b < 1$.

Part (iii)

Let $J = \displaystyle\int_0^{\infty} \frac{\mathrm{d}x}{(a^2 + x^2)^2}$. This converges, since the integrand is continuous and at most $1/x^4$ for large x. We want a substitution that maps $(0, \infty)$ to itself and turns $a^2 + x^2$ into a multiple of $a^2 + u^2$, so we take $u = a^2/x$, i.e. $k = a^2$. Then $x = a^2/u$, $\mathrm{d}x = -\frac{a^2}{u^2}\,\mathrm{d}u$, $x \to 0^+$ corresponds to $u \to \infty$, and

$$a^2 + x^2 = a^2 + \frac{a^4}{u^2} = \frac{a^2(u^2 + a^2)}{u^2}.$$

Therefore

$$J = \int_{\infty}^{0} \frac{u^4}{a^4(a^2 + u^2)^2}\left(-\frac{a^2}{u^2}\right)\mathrm{d}u = \frac{1}{a^2}\int_0^{\infty} \frac{u^2}{(a^2 + u^2)^2}\,\mathrm{d}u.$$

Now write $u^2 = (a^2 + u^2) - a^2$ in the numerator:

$$\frac{u^2}{(a^2 + u^2)^2} = \frac{1}{a^2 + u^2} - \frac{a^2}{(a^2 + u^2)^2}.$$

Both pieces have convergent integrals over $(0, \infty)$, so we may split the integral. Using the given result,

$$J = \frac{1}{a^2}\left(\frac{\pi}{2a} - a^2 J\right) = \frac{\pi}{2a^3} - J.$$

Hence $2J = \dfrac{\pi}{2a^3}$, so

$$\int_0^\infty \frac{\mathrm{d}x}{(a^2+x^2)^2} = \frac{\pi}{4a^3},$$

as required.

Checks

- In (ii), $b = 1$ gives an interval of zero length, and indeed $\frac{\pi}{2}\ln 1 = 0$. Replacing b by $1/b$ swaps the limits, so the integral should change sign, and $\ln(1/b) = -\ln b$ agrees.
- In (iii), the answer should scale correctly: putting $x = at$ gives $J = a^{-3}\int_0^\infty (1+t^2)^{-2}\,\mathrm{d}t$, so J must be a constant times a^{-3}. With $a = 1$ the substitution $x = \tan\theta$ gives $\int_0^{\pi/2}\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{4}$, which matches.
- Numerically, the integral in (ii) with $b = 2$ is $1.08879\ldots = \frac{\pi}{2}\ln 2$. The integral in (i) evaluates to 0 (to many decimal places) for several random choices of a and b.

Common slips

- Forgetting that $u = 1/x$ also reverses the limits. The minus sign from $\mathrm{d}x = -\mathrm{d}u/u^2$ and the swap of limits must both be tracked.
- Using $\arctan u + \arctan(1/u) = \frac{\pi}{2}$ without saying that it needs $u > 0$.
- In (iii), getting $\frac{1}{a^2}\int_0^\infty \frac{u^2}{(a^2+u^2)^2}\,\mathrm{d}u$ and then stopping, without spotting that it can be rewritten in terms of J and the given integral.

Beyond STEP

Using a symmetry of the domain to relate an integral to itself is a common trick. It comes back in contour integration and in integrals such as $\int_0^{\pi/2}\ln\sin x\,\mathrm{d}x$. The integrals in (iii) also come from a general family: differentiating $\int_0^\infty(\lambda+x^2)^{-1}\mathrm{d}x = \frac{\pi}{2}\lambda^{-1/2}$ with respect to λ gives the same answer at once ("differentiation under the integral sign"). Real analysis courses explain when steps like this, and the handling of improper integrals, are allowed. See Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP II 2014, Question 5

Pure • Homogeneous Differential Equations

STEP II 2014, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q5

The key idea

Both right-hand sides are ratios of linear expressions. In (i) every term has degree one, so the gradient depends only on y/x, and putting $y = xu$ turns the equation into a separable one. In (ii) the constant terms spoil this, but they can be removed by moving the origin to the point where the two lines in the numerator and denominator meet; after that it is (i) all over again.

Solution

If $y = xu$, the product rule gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = u + x\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Part (i)

Dividing the top and bottom of the right-hand side by x, the equation becomes

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2u+1}{u-2}, \qquad \text{so} \qquad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2u+1-u(u-2)}{u-2} = \frac{1+4u-u^2}{u-2}.$$

Separating the variables,

$$\int \frac{u-2}{1+4u-u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x.$$

? **Ask yourself** How do we integrate the left-hand side without partial fractions?

Differentiate the denominator: $\frac{\mathrm{d}}{\mathrm{d}u}(1+4u-u^2) = 4-2u = -2(u-2)$. So the numerator is $-\frac{1}{2}$ times the derivative of the denominator, and

$$-\tfrac{1}{2}\ln\left|1+4u-u^2\right| = \ln|x| + C.$$

Multiplying by -2 and exponentiating, $x^2\,|1+4u-u^2| = A$ for a positive constant A, that is, $x^2(1+4u-u^2) = K$ for some constant K (the sign is fixed along the curve). Now substitute $u = y/x$, so that $x^2(1+4y/x-y^2/x^2) = x^2+4xy-y^2$:

$$x^2+4xy-y^2 = K.$$

At $(1,1)$ we get $K = 1+4-1 = 4$, so the solution curve is

$$\boxed{x^2+4xy-y^2 = 4}$$

(At $(1,1)$ we have $u = 1$, where $1+4u-u^2 = 4 \neq 0$ and $u \neq 2$, so nothing we divided by vanishes at the starting point.)

Part (ii)

The lines $x-2y-4 = 0$ and $2x+y-3 = 0$ meet where $y = 3-2x$ and $x-2(3-2x)-4 = 0$, i.e. at $(2,-1)$. So take $a = 2$, $b = -1$: $x = X+2$, $y = Y-1$. Then $\mathrm{d}y/\mathrm{d}x = \mathrm{d}Y/\mathrm{d}X$, and

$$\begin{aligned} x-2y-4 &= (X+2)-2(Y-1)-4 = X-2Y, \\ 2x+y-3 &= 2(X+2)+(Y-1)-3 = 2X+Y, \end{aligned}$$

so the constants disappear:

$$\frac{\mathrm{d}Y}{\mathrm{d}X} = \frac{X-2Y}{2X+Y}.$$

This has the same shape as (i), so put $Y = XV$:

$$V+X\frac{\mathrm{d}V}{\mathrm{d}X} = \frac{1-2V}{2+V} \quad\Longrightarrow\quad X\frac{\mathrm{d}V}{\mathrm{d}X} = \frac{1-2V-V(2+V)}{2+V} = \frac{1-4V-V^2}{2+V}.$$

Again the numerator is a multiple of the derivative of the denominator, since $\frac{\mathrm{d}}{\mathrm{d}V}(1-4V-V^2) = -2(2+V)$. Hence

$$\int \frac{2+V}{1-4V-V^2}\,\mathrm{d}V = \int \frac{1}{X}\,\mathrm{d}X \quad\Longrightarrow\quad -\tfrac{1}{2}\ln\left|1-4V-V^2\right| = \ln|X|+C,$$

and as before $X^2(1-4V-V^2) = K$, i.e.

$$X^2-4XY-Y^2 = K.$$

The point $(1,1)$ corresponds to $X = -1$, $Y = 2$, giving $K = 1-4(-1)(2)-4 = 5$. Returning to x and y,

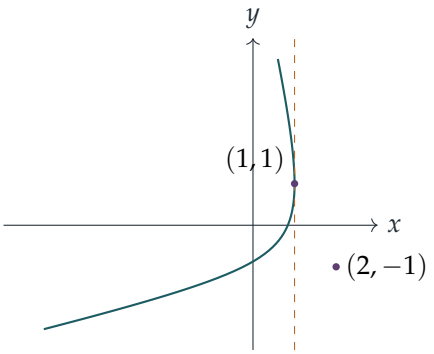
$$(x-2)^2-4(x-2)(y+1)-(y+1)^2 = 5,$$

and expanding,

$$\begin{aligned} &(x^2-4x+4)-4(xy+x-2y-2)-(y^2+2y+1) = 5 \\ \Longleftrightarrow\quad &\boxed{x^2-4xy-y^2-8x+6y+6 = 0} \end{aligned}$$

Ask yourself At $(1,1)$ the denominator $2x+y-3$ is zero. Does the solution still make sense?

Yes: the gradient is infinite there, so the curve has a vertical tangent at $(1,1)$. The implicit equation describes the curve perfectly well; it just isn't the graph of a function $y(x)$ through that point. Equivalently, x is a smooth function of y. Solving the quadratic for x gives $x = 2y+4-\sqrt{5y^2+10y+10}$ on the branch through $(1,1)$, which is defined for every y since $5y^2+10y+10 = 5(y+1)^2+5 > 0$, and which has its largest value $x=1$ at $y=1$.



The figure shows the branch through $(1,1)$ (teal), touching the vertical line $x=1$ (dashed) there; $(2,-1)$ is the centre of the hyperbola.

Checks

- Differentiate the answers implicitly. In (i), $2x+4y+(4x-2y)y' = 0$ gives $y' = \dfrac{x+2y}{y-2x}$, as required. In (ii), $2x-4y-8+(6-4x-2y)y' = 0$ gives $y' = \dfrac{x-2y-4}{2x+y-3}$.
- Both curves pass through $(1,1)$: $1+4-1=4$ and $1-4-1-8+6+6=0$.
- Stepping each differential equation numerically away from $(1,1)$ by computer, the left-hand sides of the boxed equations stayed equal to 4 and 0 to about 10^{-13}. In (i) the gradient at $(1,1)$ is -3.

Common slips

- Forgetting the u in $\mathrm{d}y/\mathrm{d}x = u + x\,\mathrm{d}u/\mathrm{d}x$, which leaves a wrong separable equation.
- Applying the condition in the wrong coordinates in (ii): the point $(1,1)$ becomes $(X,Y) = (-1,2)$, not $(1,1)$.
- Reaching for partial fractions in u: the roots of $1+4u-u^2$ are $2\pm\sqrt{5}$, which is messy, whereas the integrand is already of the form f'/f.

Beyond STEP

Equations whose right-hand side depends only on y/x are called homogeneous, and the shift of origin in (ii) is the standard trick for ratios of linear functions. Both equations here are also exact: writing (i) as $(2y + x)\,\mathrm{d}x - (y - 2x)\,\mathrm{d}y = 0$, the two coefficients satisfy the cross-derivative test, and the conserved quantity $x^2 + 4xy - y^2$ can be read off directly. Exact equations, integrating factors and phase portraits of linear systems (which is what the hyperbolae really are) are covered in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 2014, Question 6

Pure • Trigonometric Series

STEP II 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q6

The key idea

Multiplying each $\cos rx$ by $2\sin\frac{1}{2}x$ turns it into a difference of two sines, so the sum telescopes. That sum is exactly $S_n'(x)$, which is why the identity is there: it tells us where S_n is stationary. At such a point the new term $\frac{1}{n}\sin nx$ turns out to be non-negative, so adding it cannot create a negative minimum, and an induction starting from $S_1 = \sin x$ finishes the job.

Solution

The cosine sum

By the factor formula $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$ with $A = (r+\frac{1}{2})x$, $B = (r-\frac{1}{2})x$,

$$\sin(r+\tfrac{1}{2})x - \sin(r-\tfrac{1}{2})x = 2\cos rx\,\sin\tfrac{1}{2}x.$$

Summing from $r = 1$ to n, the right-hand sides add to $2\sin\frac{1}{2}x\,(\cos x + \cdots + \cos nx)$, while on the left everything cancels except the first and last terms:

$$2\sin\tfrac{1}{2}x\sum_{r=1}^{n}\cos rx = \sin(n+\tfrac{1}{2})x - \sin\tfrac{1}{2}x.$$

Dividing by $2\sin\frac{1}{2}x \neq 0$ gives the required result. Note that differentiating term by term,

$$S_n'(x) = \sum_{r=1}^{n}\cos rx,$$

so the identity is a closed form for S_n'.

Part (i)

Here $S_2(x) = \sin x + \frac{1}{2}\sin 2x$, so

$$S_2'(x) = \cos x + \cos 2x = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1).$$

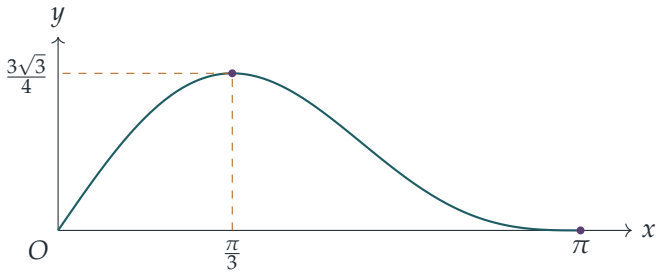
On $[0, \pi]$ this vanishes when $\cos x = \frac{1}{2}$, i.e. $x = \frac{\pi}{3}$, and when $\cos x = -1$, i.e. $x = \pi$. The values are

$$S_2(\tfrac{\pi}{3}) = \tfrac{\sqrt{3}}{2} + \tfrac{1}{2} \cdot \tfrac{\sqrt{3}}{2} = \tfrac{3\sqrt{3}}{4}, \qquad S_2(\pi) = 0.$$

Since $\cos x + 1 \geq 0$, the sign of S_2' is that of $2\cos x - 1$: positive on $[0, \frac{\pi}{3})$ and negative on $(\frac{\pi}{3}, \pi)$. So $x = \frac{\pi}{3}$ is a maximum, and S_2 decreases from there to 0 at $x = \pi$, arriving with zero gradient. Near π, putting $x = \pi - t$,

$$S_2 = \sin t - \tfrac{1}{2}\sin 2t \approx \left(t - \tfrac{t^3}{6}\right) - \left(t - \tfrac{4t^3}{6}\right) = \tfrac{1}{2}t^3,$$

so the curve flattens out like a cubic as it reaches the axis. Also $S_2(0) = 0$ with $S_2'(0) = 2$.



Part (ii)

Suppose $S_n'(x_0) = 0$ with $0 < x_0 < \pi$. Then $\sin \frac{1}{2}x_0 \neq 0$, so the identity applies and gives $\sin(n + \frac{1}{2})x_0 = \sin \frac{1}{2}x_0$. Expanding the left-hand side,

$$\sin nx_0 \cos \tfrac{1}{2}x_0 + \cos nx_0 \sin \tfrac{1}{2}x_0 = \sin \tfrac{1}{2}x_0 \iff \sin nx_0 \cos \tfrac{1}{2}x_0 = (1 - \cos nx_0) \sin \tfrac{1}{2}x_0.$$

Since $0 < \frac{1}{2}x_0 < \frac{\pi}{2}$, $\cos \frac{1}{2}x_0 > 0$, and dividing by it gives

$$\sin nx_0 = (1 - \cos nx_0) \tan \tfrac{1}{2}x_0,$$

as required. Hence, for $n \geq 2$,

$$S_n(x_0) - S_{n-1}(x_0) = \frac{1}{n}\sin nx_0 = \frac{1}{n}(1 - \cos nx_0) \tan \tfrac{1}{2}x_0 \geq 0,$$

because $1 - \cos nx_0 \geq 0$ and $\tan \frac{1}{2}x_0 > 0$. So $S_n(x_0) \geq S_{n-1}(x_0)$.

Now suppose $S_{n-1}(x) > 0$ for all $x \in (0, \pi)$, and, for a contradiction, that $S_n(c) \leq 0$ for some $c \in (0, \pi)$.

Ask yourself How do we know there is a stationary point to use?

S_n is continuous on the closed interval $[0, \pi]$, so it attains a minimum value m there, and $m \leq S_n(c) \leq 0$. If $m < 0$, the minimum is not at an end, since $S_n(0) = S_n(\pi) = 0$; if $m = 0$, then c itself is an interior point where it is attained. Either way there is

$x_0 \in (0, \pi)$ with $S_n(x_0) = m \leq 0$, and since S_n is differentiable, an interior minimum is a stationary point: $S_n'(x_0) = 0$. The previous result then gives

$$0 \geq S_n(x_0) \geq S_{n-1}(x_0) > 0,$$

a contradiction. So $S_n(x) > 0$ for all $x \in (0, \pi)$.

Part (iii)

We prove by induction that $S_n(x) > 0$ for $0 < x < \pi$. For $n = 1$, $S_1(x) = \sin x > 0$ on $(0, \pi)$. If it holds for S_{n-1}, then part (ii) shows it holds for S_n. So it holds for every $n \geq 1$. At the end points every term $\frac{1}{r}\sin rx$ is zero, so $S_n(0) = S_n(\pi) = 0$. Therefore $S_n(x) \geq 0$ for all $n \geq 1$ and $0 \leq x \leq \pi$.

Checks

- Put $n = 1$ in the cosine sum: $\dfrac{\sin \frac{3}{2}x - \sin \frac{1}{2}x}{2 \sin \frac{1}{2}x}$ should be $\cos x$, and it is, by the factor formula.
- Part (i) agrees with the identity: $S_2' = 0$ means $\sin \frac{5}{2}x = \sin \frac{1}{2}x$, whose solutions in $(0, \pi]$ are $x = \frac{\pi}{3}$ and $x = \pi$.
- The maximum $\frac{3\sqrt{3}}{4} \approx 1.299$ matches a direct evaluation of S_2 on a fine grid. A computer check found every stationary point of S_n in $(0, \pi)$ for $2 \leq n \leq 40$ and confirmed the identity of part (ii) and $S_n(x_0) \geq S_{n-1}(x_0)$ at each, and that $S_n > 0$ inside the interval for $n \leq 60$.

Common slips

- Arguing "$S_n \geq S_{n-1}$ everywhere". That is false (for example $\sin 2x < 0$ on $(\frac{\pi}{2}, \pi)$); the inequality holds only at stationary points of S_n, which is why the minimum argument is needed.
- Forgetting the end points when locating the minimum: you must use $S_n(0) = S_n(\pi) = 0$ to place a non-positive minimum inside the interval.
- Dividing by $\cos \frac{1}{2}x_0$ without saying why it is positive.

Beyond STEP

The result of part (iii) is the Fejér–Jackson–Gronwall inequality. As $n \to \infty$, $S_n(x)$ converges to $\frac{1}{2}(\pi - x)$ on $(0, \pi]$: $\sum \frac{1}{r}\sin rx$ is the Fourier series of a sawtooth wave, and near $x = 0$ the partial sums overshoot (the Gibbs phenomenon). Fourier series, and the delicate question of when they converge, are a central topic in a first analysis or methods course. T. W. Körner's *Fourier Analysis* is a wonderful, wide-ranging account.

WORKED SOLUTION

STEP II 2014, Question 7

Pure • Modulus Function Graphs

> **STEP II 2014, Question 7**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/14-s2-q7

The key idea

A sum of modulus functions is piecewise linear: it changes gradient only at the points where one of the moduli is zero. So the graph of $|x-a|+|x-b|$ is a "bath" with a flat floor between a and b and walls of gradient ± 2. Counting solutions of the later equations then reduces to comparing gradients: a graph of gradient 2 always overtakes one of gradient 1, and two graphs with the same gradient either coincide or never meet.

Solution

Throughout, write $f(x) = |x-a|+|x-b|$ with $a < b$.

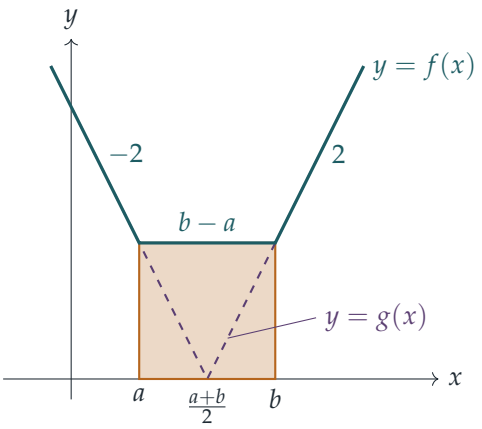
Part (i)

Removing the moduli region by region:

$$f(x) = \begin{cases} a+b-2x, & x \le a, \\ b-a, & a \le x \le b, \\ 2x-a-b, & x \ge b. \end{cases}$$

So the gradient is -2 for $x < a$, 0 for $a < x < b$ and 2 for $x > b$. The graph is continuous, with a flat floor at height $b-a$ above $[a,b]$.

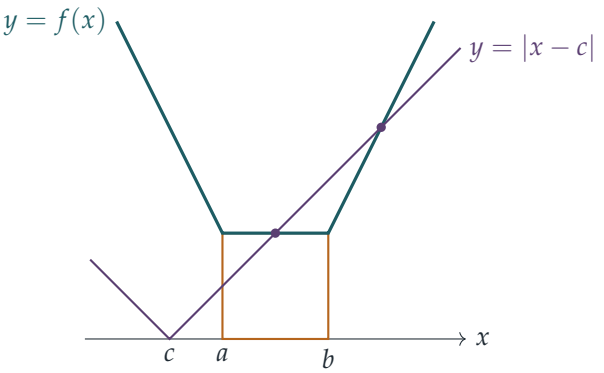
The graph of $g(x) = |2x-a-b| = 2|x-\frac{a+b}{2}|$ is a V with vertex at $\left(\frac{a+b}{2}, 0\right)$ and gradients ± 2. For $x \ge b$ we have $2x-a-b \ge b-a > 0$, so $g(x) = 2x-a-b = f(x)$, and similarly $g(x) = a+b-2x = f(x)$ for $x \le a$. Hence the two graphs coincide outside (a,b). Inside, g dips down to 0 while f stays at $b-a$. (In fact $f(x) = \max\{b-a,\, g(x)\}$.)



The quadrilateral has vertices $(a,0)$, $(b,0)$, $(b,b-a)$ and $(a,b-a)$, using $f(a)=f(b)=b-a$. Its base has length $b-a$, its vertical sides have length $b-a$, and all its angles are right angles. So it is a **square** of side $b-a$. Notice too that the V of g runs from the midpoint of the square's base up through its two top corners, since $g(a)=g(b)=b-a$.

Part (ii)

Now compare $y=f(x)$ with $y=|x-c|$, a V with vertex $(c,0)$ on the axis and arms of gradient ±1. The key fact is that f is always at least $b-a>0$, and its walls (gradient ±2) are steeper than the arms of the V. The sketch shows the case $c<a$, where the V cuts the bath twice.



Case $a<c<b$: no solutions. On $[a,b]$, the distance $|x-c|$ is at most $\max\{c-a,\,b-c\}<b-a=f(x)$, so the V stays strictly below the floor of the bath. For $x\geq b$,

$$f(x)-|x-c|=(2x-a-b)-(x-c)=x-b+(c-a)\geq c-a>0,$$

and the same argument (or the symmetry below) deals with $x\leq a$. So the V lies entirely inside the bath, and there are no solutions.

Case $c=a$ or $c=b$: one solution. Take $c=a$. On $[a,b]$ we have $|x-a|=x-a\leq b-a$, with equality only at $x=b$. For $x>b$, $f(x)-(x-a)=x-b>0$; for $x<a$,

$f(x) - (a - x) = b - x > 0$. So the only solution is $x = b$: the right arm of the V passes exactly through the top corner $(b, b - a)$ of the square. Similarly, when $c = b$ the only solution is $x = a$.

Case $c < a$: two solutions. The left arm $y = c - x$ lives on $x \le c < a$, where f has gradient -2. At $x = c$ we have $f(c) > 0 = c - c$, and as we move left f rises faster than the arm, so $f(x) > c - x$ for all $x \le c$. The left arm gives no solutions.

For the right arm $y = x - c$ ($x \ge c$), let $k(x) = f(x) - (x - c)$. Subtracting 1 from the gradients of f, k has gradient -3 on (c, a), -1 on (a, b) and $+1$ for $x > b$. So k is strictly decreasing on $[c, b]$ and strictly increasing on $[b, \infty)$. Also

$$k(c) = f(c) > 0, \qquad k(b) = (b - a) - (b - c) = c - a < 0, \qquad k(x) \to \infty \text{ as } x \to \infty.$$

Hence k has exactly one zero in (c, b) and exactly one in (b, ∞). Explicitly, these are

$$x = a + b - c \quad \text{and} \quad x = \begin{cases} b - a + c, & 2a - b \le c < a, \\ \frac{1}{3}(a + b + c), & c < 2a - b, \end{cases}$$

the second one meeting the floor of the bath or its left wall respectively.

Case $c > b$: two solutions. The reflection $x \mapsto a + b - x$ in the line $x = \frac{a+b}{2}$ leaves f unchanged (it swaps $|x - a|$ and $|x - b|$) and sends $|x - c|$ to $|x - c'|$ with $c' = a + b - c < a$. So this case is the mirror image of the previous one.

In summary, there are 0 solutions if $a < c < b$, 1 solution if $c = a$ or $c = b$, and 2 solutions if $c < a$ or $c > b$.

Part (iii)

Let $F(x) = |x - c| + |x - d|$. By part (i), F is a second bath, with floor at height $d - c$ over $[c, d]$ and walls of gradient ± 2. Since $d - c < b - a$, its floor is lower and narrower than that of f. Write

$$m = \tfrac{1}{2}(a + b), \qquad n = \tfrac{1}{2}(c + d)$$

for the midpoints of the two floors, and let $h(x) = f(x) - F(x)$.

Far to the left (for $x \le \min\{a, c\}$) both graphs are on their left walls, so $h(x) = (a + b - 2x) - (c + d - 2x) = 2(m - n)$. Far to the right (for $x \ge \max\{b, d\}$), $h(x) = 2(n - m)$. Between these, h is continuous and piecewise linear.

> **Ask yourself** Can the two graphs share a whole segment, giving infinitely many solutions?

Only where they have the same gradient. Both floors are flat, but they lie at different heights $b - a \ne d - c$, so they can't overlap. Otherwise the gradients agree only when both graphs are on their left walls ($x < \min\{a, c\}$) or both on their right walls ($x > \max\{b, d\}$), and there $h = \pm 2(m - n)$. So the graphs share a segment exactly when $m = n$.

Case $a + b = c + d$: infinitely many solutions. Now $h = 0$ for $x \le \min\{a, c\}$ and $x \ge \max\{b, d\}$. Since $d - c < b - a$ and the midpoints agree, $a < c < d < b$: the small

bath sits centrally inside the big one. For $c \le x \le b$ the graph of F is on its floor or its right wall, so $F(x) \le 2b - c - d = b - a = f(x)$, with equality only at $x = b$. The same holds on $[a, d]$ by symmetry. So the solution set is exactly $x \le a$ together with $x \ge b$.

Case $a + b < c + d$: exactly one solution. Here $h = 2(m - n) < 0$ far to the left and $h = 2(n - m) > 0$ far to the right. Where can h decrease? We need the gradient of f to be less than that of F. There are two ways.

- f on its left wall while F is not, which needs $c < x < a$. Then $c < a$, and since $d - c < b - a$ also $d < b$, giving $c + d < a + b$, which is false here.
- f on its floor while F is on its right wall, which needs $d < x < b$.

So h is non-decreasing on $(-\infty, d]$. On $[d, \infty)$, h is non-increasing on $[d, b]$ (if $d < b$) and then constant at $2(n - m)$ for $x \ge \max\{b, d\}$. So $h(x) \ge 2(n - m) > 0$ for all $x \ge d$. On $(-\infty, d]$, h climbs from the value $2(m - n) < 0$ up to $h(d) > 0$ without decreasing, and it is not constant on any interval where it is zero (by the box above). So h has exactly one zero there. That gives exactly one solution in total.

Case $a + b > c + d$: exactly one solution, by the reflection $x \mapsto -x$, which replaces a, b, c, d by $-b, -a, -d, -c$ and reverses the inequality between the sums.

In summary, with $a < b$, $c < d$ and $d - c < b - a$: if $a + b = c + d$ (the two intervals share a midpoint) there are infinitely many solutions, namely every $x \le a$ and every $x \ge b$. Otherwise there is exactly one solution.

Checks

- Part (ii) with $a = 1$, $b = 3$, $c = 0$: the solutions $x = a + b - c = 4$ and $x = b - a + c = 2$ give $|3| + |1| = 4$ and $1 + 1 = 2$, as required. With $c = -5$ the second solution is $\frac{1}{3}(1 + 3 - 5) = -\frac{1}{3}$, and indeed $\frac{4}{3} + \frac{10}{3} = \frac{14}{3} = |-\frac{1}{3} + 5|$.
- Part (iii) with $a = 0$, $b = 10$, $c = 6$, $d = 7$: h is -3 on the far left, rises to 9 on $[6, 7]$, then falls back to 3 and stays there. Its only zero is $x = \frac{3}{2}$, where both sides equal 10.
- A computer check over tens of thousands of random integer choices of a, b, c, d found exactly the solution counts above in every case.

Common slips

- In (ii), forgetting the boundary cases $c = a$ and $c = b$, where the V just touches a corner of the square.
- In (iii), assuming h is monotonic. It isn't: when the small bath sits off-centre inside the big one, h decreases on (d, b). What saves the count is that h is already positive there.
- In (iii), missing the infinite case when the midpoints coincide.

Beyond STEP

The function $|x-a|+|x-b|$ is the total distance from x to two points. Minimising sums of distances like this is the idea behind the median in statistics and the ℓ^1 ("taxicab") norm. Functions like f are convex, and the arguments here about gradients increasing from piece to piece are the start of convex analysis, which underlies linear programming and optimisation. Bartle and Sherbert, *Introduction to Real Analysis*, makes the continuity and monotonicity arguments used here rigorous.

WORKED SOLUTION

STEP II 2014, Question 8

Pure • Greatest Binomial Coefficient

STEP II 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q8

The key idea

To find where a sequence of positive terms peaks, compare each term with the next: look at the ratio c_{r+1}/c_r. For a binomial expansion this ratio is a simple fraction in r that decreases steadily, so the coefficients rise while it exceeds 1 and fall once it drops below 1. The largest coefficient sits where the ratio crosses 1, and every later part is a matter of reading off the integer part of one expression.

Solution

The coefficients and their ratio

By the binomial theorem,

$$c_r = \binom{n}{r} a^{n-r} b^r, \qquad 0 \le r \le n.$$

All of these are positive, since $a, b > 0$. For $0 \le r \le n-1$,

$$\frac{c_{r+1}}{c_r} = \frac{\binom{n}{r+1}}{\binom{n}{r}} \cdot \frac{b}{a} = \frac{(n-r)\,b}{(r+1)\,a},$$

using $\binom{n}{r+1} = \binom{n}{r}\frac{n-r}{r+1}$. Write

$$L = \frac{b(n+1)}{a+b}.$$

Because $(r+1)a > 0$,

$$\begin{aligned} c_{r+1} \ge c_r &\iff (n-r)b \ge (r+1)a \iff nb - a \ge r(a+b) \\ &\iff r \le \frac{nb-a}{a+b} = \frac{b(n+1)}{a+b} - 1 = L - 1, \end{aligned}$$

and the same chain holds with $=$ throughout. So

$$c_{r+1} > c_r \text{ if } r < L-1, \qquad c_{r+1} = c_r \text{ if } r = L-1, \qquad c_{r+1} < c_r \text{ if } r > L-1. \qquad (*)$$

The bounds on m

Upper bound. If $m = 0$ then $m \le L$ because $L > 0$. If $m \ge 1$, then $c_m \ge c_{m-1}$ because c_m is greatest, and by $(*)$ with $r = m-1$ this gives $m - 1 \le L - 1$, that is, $m \le L$.

Lower bound. If $m = n$ then $m \ge L - 1$ because $L < n + 1$ (as $\frac{b}{a+b} < 1$). If $m \le n - 1$, then $c_m \ge c_{m+1}$, and by $(*)$ with $r = m$ this forces $m \ge L - 1$.

So in every case

$$\frac{b(n+1)}{a+b} - 1 \le m \le \frac{b(n+1)}{a+b},$$

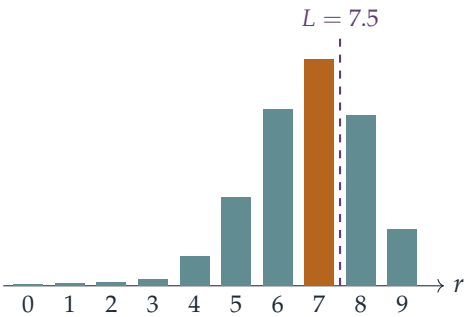
as required.

Ask yourself Why can't there be three values of m, or two that are not next to each other?

The interval $[L-1, L]$ has length 1. If L is not an integer, the interval contains exactly one integer, $\lfloor L \rfloor$, so m is unique. If L is an integer, it contains exactly the two consecutive integers $L - 1$ and L, so m is one of these. (Indeed $(*)$ with $r = L - 1$ shows $c_{L-1} = c_L$, so both really are greatest.) Either way, the unique value or the larger of the two is

$$G(n,a,b) = \left\lfloor \frac{b(n+1)}{a+b} \right\rfloor.$$

This formula is all we need from now on.



The coefficients of $(1+3x)^9$, to scale: they rise while $r < L - 1 = 6.5$ and then fall, peaking at $r = 7$.

Part (i)

For $(n,a,b) = (9,1,3)$, $L = \frac{3\times 10}{4} = 7.5$, so $G(9,1,3) = 7$.

For $(n,a,b) = (9,2,3)$, $L = \frac{3\times 10}{5} = 6$, an integer, so the greatest coefficient occurs at both $r = 5$ and $r = 6$, and $G(9,2,3) = 6$. (Check: $c_5 = 126 \cdot 2^4 \cdot 3^5 = 489\,888 = 84 \cdot 2^3 \cdot 3^6 = c_6$.)

Part (ii)

With $a = b$, $L = \frac{n+1}{2}$.

- $n = 2k$: $L = k + \frac{1}{2}$, so $G(2k, a, a) = k$ (the middle coefficient, unique).
- $n = 2k - 1$: $L = k$, an integer, so the two middle coefficients $r = k - 1$ and $r = k$ tie, and $G(2k - 1, a, a) = k$.

This matches the symmetry of the rows of Pascal's triangle.

Part (iii)

For fixed n and b, $L = \frac{b(n+1)}{a+b}$ is a decreasing function of a, so $\lfloor L \rfloor$ is a non-increasing function of a. Over the positive integers it is therefore greatest at $a = 1$, where $G(n, 1, b) = \left\lfloor \frac{b(n+1)}{b+1} \right\rfloor$. (Other values of a may tie with it, but none does better.)

Part (iv)

Now

$$\frac{b(n+1)}{b+1} = (n+1) - \frac{n+1}{b+1}.$$

Since $\frac{n+1}{b+1} > 0$, this is less than $n + 1$, so $G(n, 1, b) \leq n$. (This also follows from $m \leq n$.) Moreover,

$$G(n, 1, b) = n \iff (n+1) - \frac{n+1}{b+1} \geq n \iff \frac{n+1}{b+1} \leq 1 \iff b \geq n.$$

So the greatest possible value of $G(n, 1, b)$ is n, and it is achieved exactly when $b \geq n$. When $b = n$ the last two coefficients tie, since $c_{n-1} = n \cdot b^{n-1} = b^n = c_n$; for $b > n$ the last coefficient b^n is the unique largest.

Checks

- The coefficients of $(1 + 3x)^9$ are $1, 27, 324, 2268, 10206, 30618, 61236, 78732, 59049, 19683$: the largest is at $r = 7$, as found in (i).
- For $(1 + x)^4$ ($n = 2k$, $k = 2$) the coefficients $1, 4, 6, 4, 1$ peak at $r = 2 = k$; for $(1 + x)^3$ ($k = 2$) they are $1, 3, 3, 1$, with the tie at $r = 1, 2$ and larger value $2 = k$.
- In (iv) with $n = 3$: $(1 + 2x)^3$ has coefficients $1, 6, 12, 8$, peak at $r = 2$; $(1 + 3x)^3$ has $1, 9, 27, 27$, so $G = 3$ is reached once $b = 3$.
- A computer check of $G(n, a, b) = \lfloor b(n+1)/(a+b) \rfloor$ over all $n < 30$ and $a, b < 15$ agreed in every case.

Common slips

- Forgetting the end cases $m = 0$ and $m = n$ when proving the bounds: there is no c_{-1} or c_{n+1} to compare with.
- Taking the smaller of the two tied values when L is an integer: G is defined as the larger, so $G(9, 2, 3) = 6$, not 5.
- In (iv), answering only $b = n$: every $b \geq n$ gives $G = n$.

Beyond STEP

The ratio test used here, comparing consecutive terms, is the same idea as the ratio test for convergence of series, and it locates the mode of the binomial distribution $\mathrm{B}(n, p)$ with $p = \frac{b}{a+b}$: the mode is $\lfloor (n+1)p \rfloor$. Sequences like the binomial coefficients, whose terms rise and then fall, are called unimodal, and log-concavity ($c_r^2 \geq c_{r-1}c_{r+1}$) is the usual tool for proving this. For the probability side, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP II 2014, Question 9

Mechanics • Rigid Body Equilibrium

STEP II 2014, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q9

The key idea

There are three unknown forces (the tension, the normal reaction and the friction at A) and three equations: resolve horizontally, resolve vertically, and take moments. Taking moments about the corner A is the clever choice, because both contact forces at the wall act there and drop out, leaving a single equation linking the tension to the weight. Once that is in hand, the direction of friction is just a question of whether the string alone holds up more or less than the weight.

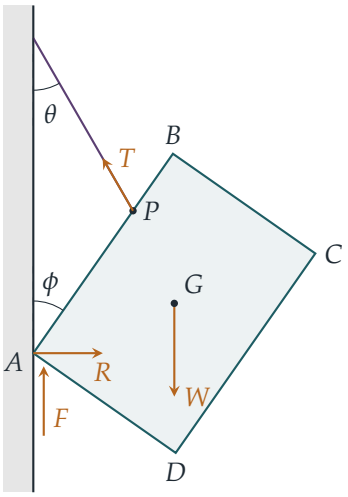
Solution

We model the lamina as a uniform rigid body of weight W with its centre of mass G at the centre of the rectangle, and the string as light and inextensible, so the tension T is the same along it. At A the wall exerts a normal reaction R (horizontal, away from the wall) and a friction force F (vertical). We take F to be positive upwards, so a negative F means the friction actually acts downwards.

Set up axes with A at the origin, x horizontal away from the wall and y vertically up. The side AB makes angle ϕ with the upward vertical, so the unit vector along AB is $(\sin\phi, \cos\phi)$. The side AD is perpendicular to it and points away from the wall, so its unit vector is $(\cos\phi, -\sin\phi)$. Hence

$$G = b(\sin\phi, \cos\phi) + a(\cos\phi, -\sin\phi), \qquad P = d(\sin\phi, \cos\phi),$$

where P is the point where the string is attached. The string runs from P up to the wall at angle θ to the downward vertical, so the tension on the lamina is $T(-\sin\theta, \cos\theta)$.



Part (i)

Resolving horizontally and vertically,

$$R = T\sin\theta,$$
$$F + T\cos\theta = W.$$

For moments about A, R and F contribute nothing. The anticlockwise moment of a force (f_x, f_y) at (x, y) is $xf_y - yf_x$. For the tension at P this is

$$d\sin\phi \cdot T\cos\theta - d\cos\phi \cdot (-T\sin\theta) = Td\sin(\theta+\phi),$$

and for the weight $(0, -W)$ at G it is $-W(a\cos\phi + b\sin\phi)$. So

$$Td\sin(\theta+\phi) = W(a\cos\phi + b\sin\phi). \tag{1}$$

The quantity $a\cos\phi + b\sin\phi$ is just the horizontal distance of G from the wall, which is a quick way to see the weight's moment.

In limiting equilibrium with friction upwards, $F = \mu R = \mu T\sin\theta$, so the vertical equation becomes

$$T(\cos\theta + \mu\sin\theta) = W.$$

Substituting W into (1) and cancelling T (which is positive, as the string is taut),

$$d\sin(\theta+\phi) = (\cos\theta + \mu\sin\theta)(a\cos\phi + b\sin\phi),$$

which is $(*)$.

Part (ii)

If the friction is limiting and acts downwards, then $F = -\mu R = -\mu T\sin\theta$ and the vertical equation is $T(\cos\theta - \mu\sin\theta) = W$. Nothing else changes, so $(*)$ becomes

$$d\sin(\theta+\phi) = (\cos\theta - \mu\sin\theta)(a\cos\phi + b\sin\phi);$$

that is, μ is replaced by $-\mu$.

Part (iii)

Now we don't assume the friction is limiting. Equation (1) holds in any equilibrium and gives

$$T = \frac{W(a\cos\phi + b\sin\phi)}{d\sin(\theta+\phi)},$$

and $\sin(\theta+\phi) > 0$ because θ and ϕ are both acute. The vertical equation then gives

$$F = W - T\cos\theta = W\left(1 - \frac{\cos\theta\,(a\cos\phi + b\sin\phi)}{d\sin(\theta+\phi)}\right).$$

? **Ask yourself** How do we turn "$F > 0$" into a condition involving only tangents?

The friction acts upwards exactly when $F > 0$, that is, when

$$d\sin(\theta+\phi) > \cos\theta\,(a\cos\phi + b\sin\phi),$$

since $d\sin(\theta+\phi) > 0$. Expand $\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$ and divide both sides by $\cos\theta\cos\phi$, which is positive because both angles are acute, so the inequality keeps its direction:

$$d(\tan\theta + \tan\phi) > a + b\tan\phi.$$

As $\tan\theta + \tan\phi > 0$, the necessary and sufficient condition is

$$d > \frac{a + b\tan\phi}{\tan\theta + \tan\phi}.$$

(When d equals this value there is no friction at all, and when d is smaller the friction acts downwards.)

Physically, the string is attached to the side AB, so $d \le 2b$. If the condition holds, then

$$\frac{a + b\tan\phi}{\tan\theta + \tan\phi} < d \le 2b,$$

so $a + b\tan\phi < 2b\tan\theta + 2b\tan\phi$, that is, $a < b(2\tan\theta + \tan\phi)$. Taking the contrapositive: if $b(2\tan\theta + \tan\phi) < a$, then no allowed position of the attachment point makes the friction act upwards, so the condition cannot be satisfied. This completes the proof.

Checks

- With $\mu = 0$ the wall is smooth, and $(*)$ and its part (ii) version coincide, as they should. They both reduce to the borderline case $d\sin(\theta+\phi) = \cos\theta\,(a\cos\phi + b\sin\phi)$ of part (iii).
- Lamina with AB vertical ($\phi \to 0$): the borderline becomes $d\tan\theta = a$. This says the string's line of action passes through the vertical line through G, at horizontal distance a from the wall. Then the string and weight can balance alone, with the wall only pushing horizontally.

- Numerically, with $a = 1$, $b = 1.5$, $\theta = 30^\circ$, $\phi = 40^\circ$: the critical value is $d \approx (1 + 1.259)/(0.577 + 0.839) \approx 1.595 < 3 = 2b$. Solving the three equilibrium equations directly with moments about G instead of A gives $F > 0$ just above this value and $F < 0$ just below it. A computer check of thousands of random configurations confirmed $(*)$, its part (ii) form and the condition in (iii).

Common slips

- Taking moments about P or G: it works, but brings R and F into the moment equation and makes the algebra much messier.
- Getting the position of G wrong: D lies on the side away from the wall, so AD points along $(\cos\phi, -\sin\phi)$ and G is at horizontal distance $a\cos\phi + b\sin\phi$ from the wall.
- Forgetting the constraint $d \leq 2b$ in the last part; it is the only place the length of AB enters.

Beyond STEP

This is a small statically determinate problem: three unknown forces and three independent equations of equilibrium for a rigid body in a plane. Engineers check this counting before anything else; with more supports the forces can't be found from statics alone, and you need to know how the body deforms. The trick of taking moments about the point where the unknowns act is the planar version of choosing a clever origin for torques. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2014, Question 10

Mechanics • Envelope of Trajectories

STEP II 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.9, *The Safety Parabola*: https://beyondstep.keithmcnulty.org/ch11/safety-parabola.html

The key idea

Once we've removed the time, the trajectory equation is a *quadratic in* $\lambda = \tan\theta$, because $\sec^2\theta = 1 + \lambda^2$. Fix a point (x, y) and ask whether some launch angle passes through it. That's the same as asking whether this quadratic in λ has a real root, so the discriminant gives the boundary of the reachable region. For the final part, the boundary curve turns out to be a parabola with its focus at O, and that makes distances from O very easy to work out.

Solution

The trajectory

Model the projectile as a particle moving freely under gravity, with no air resistance. After time t,

$$x = ut\cos\theta, \qquad y = ut\sin\theta - \tfrac{1}{2}gt^2.$$

Since $\cos\theta \neq 0$ (the angle is less than a right angle), $t = x/(u\cos\theta)$, and substituting gives

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}.$$

Using $\sec^2\theta = 1 + \tan^2\theta = 1 + \lambda^2$, the Cartesian equation is

$$\boxed{y = \lambda x - \frac{gx^2}{2u^2}\left(1 + \lambda^2\right).}$$

The maximum height Y above a given x

Fix u and fix $x > 0$. Now y is a function of λ alone, and we can rewrite it as

$$\frac{gx^2}{2u^2}\lambda^2 - x\lambda + \left(\frac{gx^2}{2u^2} + y\right) = 0.$$

Ask yourself Why think of this as an equation for λ rather than for y?

A value y occurs above this x exactly when some launch angle produces it, that is, when this quadratic has a real root λ. Its discriminant must be non-negative:

$$x^2 - 4 \cdot \frac{gx^2}{2u^2}\left(\frac{gx^2}{2u^2} + y\right) \geq 0.$$

Dividing by $x^2 > 0$ and then multiplying by $u^2/(2g) > 0$, which doesn't change the direction of the inequality,

$$\frac{u^2}{2g} \geq \frac{gx^2}{2u^2} + y \iff y \leq \frac{u^2}{2g} - \frac{gx^2}{2u^2}.$$

So

$$\boxed{Y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}}$$

and no trajectory goes higher than Y above this x.

We also need every value *up to* Y to be reached, using a genuine launch angle with $\lambda > 0$. As a function of λ, y is a downward-opening quadratic. The vertex is at $\lambda_0 = u^2/(gx) > 0$, where $y = Y$ (substitute and check). For $\lambda \geq \lambda_0$, y is continuous and decreasing, and $y \to -\infty$ as $\lambda \to \infty$. So by continuity every value $y \leq Y$ is taken for some $\lambda \geq \lambda_0 > 0$, that is, for some $\theta \in (0, \frac{\pi}{2})$. Hence the values of y form the whole interval $y \leq Y$, as required. (At $x = 0$ we have to throw straight up. Then y goes from 0 to $u^2/(2g)$, which is again $Y(0)$.)

If we're allowed to project in either direction, the case $x < 0$ is the mirror image, and the same formula holds because Y depends only on x^2.

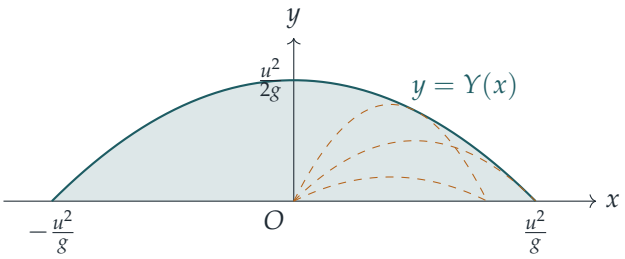
The graph and the reachable region

The graph of Y is an inverted parabola. Its vertex is at $(0, u^2/2g)$, the greatest possible height, and it meets the ground at $x = \pm u^2/g$, the greatest possible range. Every individual trajectory lies under this curve and touches it once, at $\lambda = \lambda_0$. For this reason it's often called the *bounding parabola* or *parabola of safety*.

The particle stops once it hits the ground. Take a point with $y \geq 0$ on a trajectory with $\lambda > 0$ and $x > 0$. That trajectory is above the ground for all x between O and the point, so the particle really does reach the point before landing. So the reachable set is the region

$$0 \leq y \leq \frac{u^2}{2g} - \frac{gx^2}{2u^2}, \qquad |x| \leq \frac{u^2}{g},$$

shaded below. The figure uses units of u^2/g, and the dashed curves are three sample trajectories.



The furthest distance from O

Any reachable point P lies on the segment from O to the point Q where the ray OP meets the boundary. The region is bounded by the bounding parabola and the ground, and it contains every point between O and its boundary along each ray. So $OP \le OQ$, and we only need to maximise the distance over the boundary. On the ground the furthest points are at distance u^2/g. On the curved part, $y = Y(x)$, so

$$\begin{aligned} OQ^2 = x^2 + Y^2 &= x^2 + \left(\frac{u^2}{2g}\right)^2 - 2 \cdot \frac{u^2}{2g} \cdot \frac{gx^2}{2u^2} + \left(\frac{gx^2}{2u^2}\right)^2 \\ &= \left(\frac{u^2}{2g}\right)^2 + \frac{x^2}{2} + \left(\frac{gx^2}{2u^2}\right)^2 = \left(\frac{u^2}{2g} + \frac{gx^2}{2u^2}\right)^2. \end{aligned}$$

So $OQ = \dfrac{u^2}{2g} + \dfrac{gx^2}{2u^2} = \dfrac{u^2}{g} - Y$. In other words, the distance from O equals the distance down from the line $y = u^2/g$: O is the focus of the bounding parabola and $y = u^2/g$ is its directrix. This distance is largest when Y is smallest, and on the reachable part of the curve the least value is $Y = 0$, at $x = \pm u^2/g$. Hence the furthest distance from O that a projectile can reach is

$$\boxed{\frac{u^2}{g}},$$

which is the maximum horizontal range, achieved at $\theta = \frac{\pi}{4}$. Every point at a height $y > 0$ on the boundary is closer, by exactly y.

Checks

- Put $x = 0$: $Y = u^2/(2g)$, the height reached when throwing straight up (from $v^2 = u^2 - 2gh$). Put $Y = 0$: $x = u^2/g$, the familiar maximum range $u^2 \sin 2\theta/g$ at $\theta = \frac{\pi}{4}$.
- Take $u^2/g = 10$ (say $u = \sqrt{98}\ \mathrm{m\,s^{-1}}$ with $g = 9.8$). At $x = 4$ the formula gives $Y = 5 - 0.8 = 4.2$, reached at $\lambda_0 = 10/4 = 2.5$: indeed $2.5 \cdot 4 - \frac{1}{20} \cdot 16 \cdot 7.25 = 10 - 5.8 = 4.2$.

- A computer search over many launch angles and times, for $u = 10\ \mathrm{m\,s^{-1}}$, found the furthest point from O at distance $10.204\ \mathrm{m} = u^2/g$, on the ground.

Common slips

- Stopping at the discriminant. It shows that $y \leq Y$ is *necessary*, but the question also wants every such value to be *attained* with a genuine launch angle above the ground.
- Shading the whole region under $y = Y$, including points below the ground. The particle stops when it lands, so the reachable set also has $y \geq 0$.
- Assuming the furthest point is the top of the parabola of safety. That point is only $u^2/(2g)$ from O, half the range.

Beyond STEP

The bounding parabola is the *envelope* of the family of trajectories: the curve that touches every member of the family. One standard way to find an envelope is to eliminate the parameter between $F(x, y, \lambda) = 0$ and $\partial F/\partial\lambda = 0$, which is just the double-root condition we used. There is an energy link too: every trajectory has the same directrix, the line $y = u^2/(2g)$, which is the height at which all of the kinetic energy would have turned into potential energy. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers projectile motion and energy at first-year level.

WORKED SOLUTION

STEP II 2014, Question 11

Mechanics • Circular Motion

STEP II 2014, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q11

The key idea

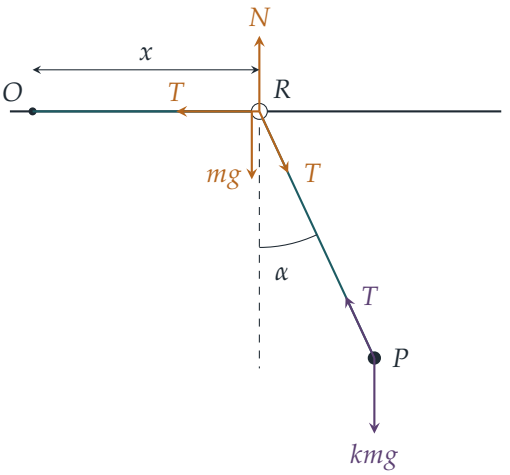
Everything is driven by one variable, x. Because the string is inextensible and α is fixed, the position of P is a simple function of x, so both of P's acceleration components are multiples of $\ddot{x}$. That gives three equations of motion (two for P, one horizontal one for the ring) in just two unknowns, $\ddot{x}$ and T. An overdetermined system is only consistent if something special holds, and that consistency condition is what ties α to k.

Solution

Model. The ring and P are particles, the string is light and inextensible, and the ring and rail are smooth. Since the ring is smooth, the tension T is the same in both parts of the string. Take O as origin, the x-axis along the rail in the direction from O to the ring, and y vertically up.

Coordinates of P. The part of the string from O to the ring lies along the rail and has length x, so the part from the ring to P has length $L - x$. It makes angle α with the downward vertical, and P hangs on the side of the ring away from O (we justify this choice at the end of part (i)). So

$$P = \big(x + (L - x)\sin\alpha,\ -(L - x)\cos\alpha\big).$$



Part (i)

Differentiating the coordinates of P twice, with α constant, gives

$$\ddot{x}_P = \ddot{x}\,(1 - \sin\alpha), \qquad \ddot{y}_P = \ddot{x}\cos\alpha.$$

The forces on P are its weight kmg downwards and the tension T along the string towards the ring, whose components are $-T\sin\alpha$ horizontally and $+T\cos\alpha$ vertically.

P, vertically: $km\,\ddot{y}_P = T\cos\alpha - kmg$, that is

$$km\ddot{x}\cos\alpha = T\cos\alpha - kmg,$$

as required.

P, horizontally:

$$km\ddot{x}\,(1 - \sin\alpha) = -T\sin\alpha. \tag{1}$$

Ring, horizontally. The ring feels the string on both sides: the part along the rail pulls it towards O with force T, and the part going down to P pulls it with horizontal component $T\sin\alpha$ away from O. Its weight and the normal reaction of the rail are vertical (the rail is smooth). So

$$m\ddot{x} = T\sin\alpha - T = -T(1 - \sin\alpha). \tag{2}$$

Ask yourself Why must P be on the far side of the ring from O?

If P hung on the O side, its x-coordinate would be $x - (L - x)\sin\alpha$, and the two horizontal equations would become $km\ddot{x}(1 + \sin\alpha) = T\sin\alpha$ and $m\ddot{x} = -T(1 + \sin\alpha)$. The first makes $\ddot{x} \geq 0$ and the second makes $\ddot{x} \leq 0$, so $\ddot{x} = 0$ and $T = 0$; but then the vertical equation gives $0 = -kmg$, which is impossible. So that configuration can't move with constant α, and the coordinates above are the right ones.

Part (ii)

First, $T \neq 0$: if $T = 0$ then (2) gives $\ddot{x} = 0$, and the vertical equation gives $0 = -kmg$, a contradiction.

From (2), $m\ddot{x} = -T(1 - \sin\alpha)$. Substituting this into (1):

$$k\big(-T(1-\sin\alpha)\big)(1-\sin\alpha) = -T\sin\alpha.$$

Dividing by $-T \neq 0$,

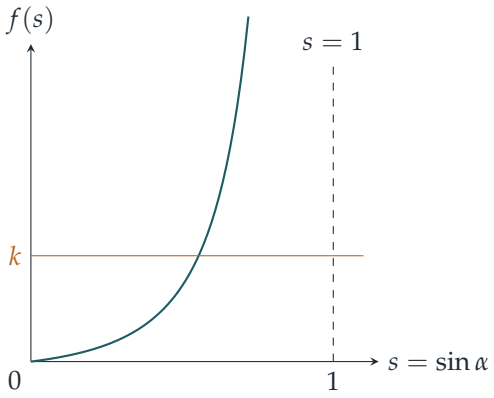
$$\frac{\sin\alpha}{(1-\sin\alpha)^2} = k,$$

as required. (Since α is acute, $0 < \sin\alpha < 1$, so the division by $(1-\sin\alpha)^2$ is fine.)

Every k is possible. Put $s = \sin\alpha$ and $f(s) = \dfrac{s}{(1-s)^2}$ on $0 < s < 1$, which covers exactly the acute angles α with $\alpha \neq 0$. By the quotient rule,

$$f'(s) = \frac{(1-s)^2 + 2s(1-s)}{(1-s)^4} = \frac{1+s}{(1-s)^3} > 0,$$

so f is strictly increasing. Also $f(s) \to 0$ as $s \to 0^+$ and $f(s) \to \infty$ as $s \to 1^-$, and f is continuous. So f takes every positive value exactly once, and for each $k > 0$ there is exactly one acute angle α satisfying the condition.



Whatever the height of the horizontal line $y = k$, it meets the curve exactly once.

Part (iii)

We now eliminate T using (2) and the vertical equation. From (2), $T = -\dfrac{m\ddot{x}}{1-\sin\alpha}$. Substituting into the vertical equation and dividing by m:

$$k\ddot{x}\cos\alpha = -\frac{\ddot{x}\cos\alpha}{1-\sin\alpha} - kg \quad\Longrightarrow\quad \ddot{x}\cos\alpha\left(k + \frac{1}{1-\sin\alpha}\right) = -kg.$$

Now use part (ii) to replace k inside the bracket:

$$k + \frac{1}{1-\sin\alpha} = \frac{\sin\alpha + (1-\sin\alpha)}{(1-\sin\alpha)^2} = \frac{1}{(1-\sin\alpha)^2}.$$

Hence

$$\ddot{x} = -\frac{kg(1-\sin\alpha)^2}{\cos\alpha} = -\frac{g\sin\alpha}{\cos\alpha} = -g\tan\alpha,$$

using $k(1-\sin\alpha)^2 = \sin\alpha$ once more. This is the required result.

As a bonus, $T = -\dfrac{m\ddot{x}}{1-\sin\alpha} = \dfrac{mg\tan\alpha}{1-\sin\alpha} > 0$, so the string really is taut, as the model needs.

Checks

- With $k = 1$, the condition $s = (1-s)^2$ gives $\sin\alpha = \frac{1}{2}(3-\sqrt{5}) \approx 0.382$, so $\alpha \approx 22.46^\circ$ and $\ddot{x} \approx -0.413g$. Substituting into (1), (2) and the vertical equation, all three balance.
- We simulated the full two-variable motion (in x and the string angle) numerically, starting at this angle with the ring moving: the angle stayed constant to within 10^{-10} rad and the ring decelerated at exactly $g\tan\alpha$. Starting at a slightly different angle, the angle immediately begins to change, so the special value of α really matters.
- Limiting cases: as $k \to 0$ (a very light particle), $\alpha \to 0$ and $\ddot{x} \to 0$; as $k \to \infty$, $\alpha \to 90^\circ$ and the deceleration grows without bound, because the light ring is being hauled by a heavy particle.
- Signs: $\ddot{x} < 0$, so the ring is always accelerating towards O. That fits: the net horizontal force on the ring, $-T(1-\sin\alpha)$, points towards O.

Common slips

- Forgetting that the ring feels the string on *both* sides; the pull along the rail towards O is what makes (2) work.
- Writing the horizontal acceleration of P as $\ddot{x}$; the length $L - x$ of the hanging part also changes, which contributes $-\ddot{x}\sin\alpha$.
- Treating α as a variable when differentiating. It is constant here, which is exactly what makes the problem tractable.

Beyond STEP

This is a system with two degrees of freedom (the ring's position and the string's angle), and the question looks for a special solution in which one of them stays fixed, in the spirit of a relative equilibrium. In a first-year course you would write down the Lagrangian in x and the angle and find that constant-angle motion is possible only for

the one value of α found here; the natural next question is whether that motion is stable. Kleppner and Kolenkow, *An Introduction to Mechanics*, is good on constrained systems of this kind.

WORKED SOLUTION

STEP II 2014, Question 12

Probability and Statistics • Hazard Functions

STEP II 2014, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q12

The key idea

The hazard function is the rate of dying *for those still alive*. Everything in the later parts follows from one observation: since $F'(t) = f(t)$,

$$h(t) = \frac{f(t)}{1 - F(t)} = -\frac{\mathrm{d}}{\mathrm{d}t} \ln\bigl(1 - F(t)\bigr).$$

So knowing h means knowing the derivative of $\ln(1 - F)$. Integrate, use the fact that $F = 0$ where the lifetime starts, and differentiate F to get f.

Solution

Part (i)

Let A be the event $T \geq t$ (the fly is alive at time t) and B the event $t \leq T \leq t + \delta t$ (it dies in the next δt). Since $B \cap A = B$ (for a continuous variable $\mathrm{P}(T = t) = 0$, so it doesn't matter whether the endpoints are included),

$$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(B)}{\mathrm{P}(A)} = \frac{F(t + \delta t) - F(t)}{1 - F(t)}.$$

For small δt, $F(t + \delta t) - F(t) \approx F'(t)\,\delta t = f(t)\,\delta t$, so

$$\mathrm{P}(B \mid A) \approx \frac{f(t)\,\delta t}{1 - F(t)} = h(t)\,\delta t,$$

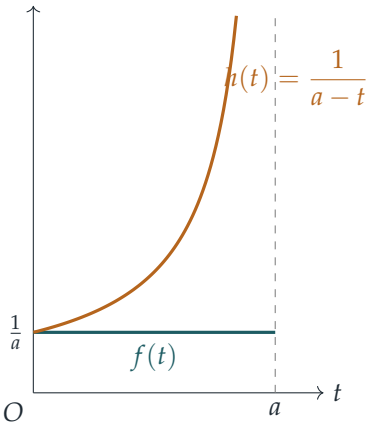
as required. (We need $F(t) < 1$ so that we are not conditioning on an impossible event; this is exactly where h is defined.)

Part (ii)

Here $f(t) = F'(t) = \frac{1}{a}$ for $0 < t < a$: T is uniform on $(0, a)$. Then

$$h(t) = \frac{1/a}{1 - t/a} = \frac{1}{a - t}, \qquad 0 < t < a.$$

The density is flat, but the hazard starts at $h(0) = \frac{1}{a}$ and increases, with $h(t) \to \infty$ as $t \to a^-$. That makes sense: a fly that has survived to almost time a is certain to die in the tiny time remaining.



(The diagram is drawn with $a = 2$.)

Part (iii)

Using the key idea, $-\dfrac{\mathrm{d}}{\mathrm{d}t}\ln\big(1 - F(t)\big) = \dfrac{1}{t}$ for $t > a$. Integrating from a to t, and using $F(a) = 0$ because T takes values only above a,

$$-\ln\big(1 - F(t)\big) = \ln t - \ln a = \ln\frac{t}{a}, \qquad \text{so} \qquad 1 - F(t) = \frac{a}{t}.$$

Hence $F(t) = 1 - \dfrac{a}{t}$ and

$$f(t) = \frac{a}{t^2}, \qquad t > a$$

(and $f(t) = 0$ for $t \le a$). As a check, $\int_a^\infty at^{-2}\,\mathrm{d}t = 1$, and $f/(1-F) = (a/t^2)/(a/t) = 1/t$.

Part (iv)

"If". Suppose $f(t) = k\mathrm{e}^{-k(t-b)}$ for $t > b$ with $k > 0$. Since $\int_b^\infty k\mathrm{e}^{-k(t-b)}\,\mathrm{d}t = 1$, all the probability lies in $t > b$, so $f(t) = 0$ and $F(t) = 0$ for $t \le b$. Hence $h(t) = 0/1 = 0$ for $t \le b$. For $t > b$,

$$F(t) = \int_b^t k\mathrm{e}^{-k(s-b)}\,\mathrm{d}s = 1 - \mathrm{e}^{-k(t-b)}, \qquad h(t) = \frac{k\mathrm{e}^{-k(t-b)}}{\mathrm{e}^{-k(t-b)}} = k,$$

which is constant.

"Only if". Suppose $h(t) = 0$ for $t \le b$ and $h(t) = k$ (a constant) for $t > b$.

Ask yourself What does $h(t) = 0$ for $t \le b$ tell us about $F(b)$?

Where $h(t) = 0$ we have $f(t) = h(t)\big(1 - F(t)\big) = 0$. So f vanishes for all $t \le b$, giving $F(t) = \int_{-\infty}^{t} f = 0$ for $t \le b$; in particular $F(b) = 0$. (This also confirms $F < 1$ there, so h really is defined on $t \le b$.)

For $t > b$, $-\dfrac{\mathrm{d}}{\mathrm{d}t}\ln\big(1 - F(t)\big) = k$. Integrating from b to t and using $F(b) = 0$,

$$-\ln\big(1 - F(t)\big) = k(t - b), \qquad F(t) = 1 - \mathrm{e}^{-k(t-b)},$$

and differentiating, $f(t) = k\mathrm{e}^{-k(t-b)}$ for $t > b$. Finally, k must be positive: $k \ge 0$ because $f \ge 0$, and if $k = 0$ then F would stay at 0 for ever, so $F(t) \not\to 1$ and T would not be a genuine random variable. This completes both directions.

Part (v)

Again $-\dfrac{\mathrm{d}}{\mathrm{d}t}\ln\big(1 - F(t)\big) = \dfrac{\lambda}{\theta^{\lambda}}t^{\lambda-1}$ for $t > 0$. Integrating from 0 to t, with $F(0) = 0$ since $T > 0$, and noting that $\lambda > 0$ makes $t^{\lambda} \to 0$ as $t \to 0$,

$$-\ln\big(1 - F(t)\big) = \frac{\lambda}{\theta^{\lambda}} \cdot \frac{t^{\lambda}}{\lambda} = \left(\frac{t}{\theta}\right)^{\lambda}.$$

So $F(t) = 1 - \exp\big(-(t/\theta)^{\lambda}\big)$, and differentiating,

$$f(t) = \frac{\lambda}{\theta^{\lambda}}\, t^{\lambda-1} \exp\left(-\left(\frac{t}{\theta}\right)^{\lambda}\right), \qquad t > 0.$$

This is a genuine density: it is positive, and $F(t) \to 1$ as $t \to \infty$.

Checks

- Each answer can be checked by computing $f/(1 - F)$ directly. For example, in (v) the exponential factors cancel and leave $\lambda\theta^{-\lambda}t^{\lambda-1}$.
- Part (v) with $\lambda = 1$ gives constant hazard $1/\theta$ and $f(t) = \frac{1}{\theta}\mathrm{e}^{-t/\theta}$, which is part (iv) with $b = 0$, $k = 1/\theta$.
- Numerically, with $\lambda = \frac{1}{2}$, $\theta = 2$, integrating f over $(0, \infty)$ gives 1.0000, and $F(1) = 1 - \mathrm{e}^{-1/\sqrt{2}} \approx 0.5069$ agrees with direct integration of f from 0 to 1.

Common slips

- Forgetting the condition $F = 0$ at the start of the lifetime, and leaving an arbitrary constant of integration in F.
- In (iv), proving only one direction of the "if and only if", or not using $h = 0$ for $t \le b$ to deduce $F(b) = 0$.
- Losing the minus sign: $\frac{\mathrm{d}}{\mathrm{d}t}\ln(1 - F) = -f/(1 - F)$.

Beyond STEP

The relation $1 - F(t) = \exp\left(-\int_0^t h(s)\,\mathrm{d}s\right)$ is the foundation of *survival analysis* and reliability theory. The distribution in (v) is the Weibull distribution: $\lambda < 1$ gives a falling hazard (early failures), $\lambda = 1$ the memoryless exponential, and $\lambda > 1$ a rising hazard (wearing out). Survival analysis is used everywhere from medical trials to engineering. For the underlying probability, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP II 2014, Question 13

Probability and Statistics • Discrete Random Variables

STEP II 2014, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q13

The key idea

This is the birthday problem in disguise. The event $X \geq k$ just says that the first $k-1$ integers are all different, and $X = r$ says that the first $r-1$ are different and the rth matches one of them. Both probabilities are products of simple factors. The two curious identities then come from computing the same quantity in two ways: total probability equals 1 in part (i), and two formulas for $\mathrm{E}(X)$ agree in part (iv).

Solution

Throughout, write

$$Q_m = \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right) \qquad (m \geq 1),$$

with $Q_1 = 1$ (an empty product). Since each integer is independent of the earlier ones and uniform on $\{1, \ldots, n\}$, Q_m is the probability that $I_1, \ldots, I_m$ are all different: given that the first j are distinct, the next avoids all of them with probability $1 - \frac{j}{n}$.

The value $\mathrm{P}(X = 4)$

We need I_1, I_2, I_3 distinct and then I_4 equal to one of those three:

$$\mathrm{P}(X = 4) = 1 \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdot \frac{3}{n} = \frac{3(n-1)(n-2)}{n^3}.$$

Part (i)

In the same way, $X = r$ means that $I_1, \ldots, I_{r-1}$ are distinct (probability Q_{r-1}) and I_r is one of these $r-1$ values (probability $\frac{r-1}{n}$, independently). So

$$\mathrm{P}(X = r) = \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{r-2}{n}\right)\frac{r-1}{n} = Q_{r-1}\,\frac{r-1}{n}, \qquad 2 \leq r \leq n+1.$$

Ask yourself Why do these probabilities add up to 1?

X certainly exists and lies between 2 and $n+1$: the first integer can't be a repeat, so $X \geq 2$; and among $I_1, \ldots, I_{n+1}$ there are $n+1$ integers taking only n values, so by the pigeonhole principle some value has appeared twice by time $n+1$, so $X \leq n+1$. (The chance that the sequence never repeats is zero, so nothing is missing.) Hence

$$\sum_{r=2}^{n+1} \mathrm{P}(X = r) = 1.$$

Writing out the terms $r = 2, 3, 4, \ldots$ gives

$$\frac{1}{n} + \left(1 - \frac{1}{n}\right)\frac{2}{n} + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\frac{3}{n} + \cdots = 1,$$

which is the required identity. The series stops at $r = n+1$, and any further terms are zero because they contain the factor $1 - \frac{n}{n}$. Done. With $m = r - 1$, this reads

$$\sum_{m=1}^{n} \frac{m}{n} Q_m = 1. \qquad (*)$$

Part (ii)

By definition of expectation,

$$\mathrm{E}(X) = \sum_{r=2}^{n+1} r \, \frac{r-1}{n}\left(1 - \frac{1}{n}\right)\cdots\left(1 - \frac{r-2}{n}\right) = \sum_{m=1}^{n} \frac{m(m+1)}{n} Q_m.$$

Part (iii)

$X \geq k$ happens exactly when there is no repeat among $I_1, \ldots, I_{k-1}$, that is, when these $k-1$ integers are all different. So

$$\mathrm{P}(X \geq k) = Q_{k-1} = \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{k-2}{n}\right), \qquad 2 \leq k \leq n+2,$$

and $\mathrm{P}(X \geq 1) = \mathrm{P}(X \geq 2) = 1$. (For $k = n+2$ the product contains $1 - \frac{n}{n} = 0$, as it should.)

Part (iv)

Write $p_r = \mathrm{P}(Y = r)$. Then $r = \sum_{k=1}^{r} 1$, so

$$\mathrm{E}(Y) = \sum_{r=1}^{N} r\, p_r = \sum_{r=1}^{N}\sum_{k=1}^{r} p_r = \sum_{k=1}^{N}\sum_{r=k}^{N} p_r = \sum_{k=1}^{N} \mathrm{P}(Y \geq k).$$

Swapping the order of summation is fine because the sum is finite: both double sums run over all pairs (k, r) with $1 \leq k \leq r \leq N$. Done.

Ask yourself Which two expressions for $\mathrm{E}(X)$ should we compare?

Apply this result to $Y = X$, which takes values in $\{1, \ldots, N\}$ with $N = n+1$ (the value 1 just has probability 0). Using part (iii),

$$\mathrm{E}(X) = \sum_{k=1}^{n+1} \mathrm{P}(X \geq k) = 1 + \sum_{k=2}^{n+1} Q_{k-1} = 1 + \sum_{m=1}^{n} Q_m.$$

Now set this equal to part (ii), splitting $\frac{m(m+1)}{n} = \frac{m^2}{n} + \frac{m}{n}$ and using $(*)$:

$$\sum_{m=1}^{n} \frac{m(m+1)}{n} Q_m = \sum_{m=1}^{n} \frac{m^2}{n} Q_m + \underbrace{\sum_{m=1}^{n} \frac{m}{n} Q_m}_{=1 \text{ by } (*)}.$$

So $1 + \sum_{m=1}^{n} Q_m = \sum_{m=1}^{n} \frac{m^2}{n} Q_m + 1$, that is,

$$\sum_{m=1}^{n} Q_m \left(1 - \frac{m^2}{n}\right) = 0.$$

The terms $m = 1, 2, 3$ are $\left(1 - \frac{1^2}{n}\right)$, $\left(1 - \frac{1}{n}\right)\left(1 - \frac{2^2}{n}\right)$ and $\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\left(1 - \frac{3^2}{n}\right)$, so this is exactly the required identity. As in part (i), every term beyond $m = n$ has a factor $1 - \frac{n}{n} = 0$. Done.

Checks

- For $n = 1$, $X = 2$ for certain. Part (i) gives $\frac{1}{1} = 1$, and the final sum is the single term $1 - 1 = 0$.
- For $n = 4$, the final sum is $\frac{3}{4} + \frac{3}{4} \cdot 0 + \frac{3}{4} \cdot \frac{1}{2} \cdot \left(-\frac{5}{4}\right) + \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{1}{4} \cdot (-3) = \frac{24-15-9}{32} = 0$, and both formulas give $\mathrm{E}(X) = \frac{103}{32}$.
- For $n = 365$ (birthdays), $\mathrm{E}(X) = 1 + \sum Q_m \approx 24.62$. A computer simulation of 10^5 sequences gave an average of 24.61, and exact enumeration for $n \leq 5$ confirmed every formula above.

Common slips

- Writing $\mathrm{P}(X = r)$ with $r - 1$ factors of the form $1 - \frac{j}{n}$ instead of $r - 2$. Only the second to the $(r-1)$th integers have to avoid earlier values.
- Forgetting the $k = 1$ term in $\sum \mathrm{P}(X \geq k)$. It contributes 1, which is exactly what cancels with $(*)$.

Beyond STEP

The tail-sum formula $\mathrm{E}(Y) = \sum_k \mathrm{P}(Y \geq k)$ holds for any random variable taking non-negative integer values, and its continuous version $\mathrm{E}(T) = \int_0^\infty \mathrm{P}(T > t)\,\mathrm{d}t$ is a standard tool in first-year probability. For large n, $\mathrm{E}(X) \approx \sqrt{\pi n/2}$, which is why collisions in hash functions turn up after only about $\sqrt{n}$ attempts (the "birthday attack"). Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

WORKED SOLUTION

STEP III 2014, Question 1

Pure • Logarithmic Series Expansion

STEP III 2014, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q1

The key idea

The cubic $1 + qx^2 + rx^3$ factorises as $(1 + ax)(1 + bx)(1 + cx)$, so its logarithm splits into three logarithms, each with a series we know. This gives the coefficient of x^n in terms of a, b, c. We can also expand the same logarithm directly as $\ln(1 + u)$ with $u = qx^2 + rx^3$, and that gives the coefficients in terms of q and r. Comparing the two expansions turns facts about q and r into identities between the S_n.

Solution

Multiplying out,

$$(1 + ax)(1 + bx)(1 + cx) = 1 + (a + b + c)x + (ab + bc + ca)x^2 + abc\,x^3.$$

Since $a + b + c = 0$, the x term vanishes. Two polynomials that agree for all real x have equal coefficients, so comparing the coefficients of x^2 and x^3 with $1 + qx^2 + rx^3$ gives

$$q = bc + ca + ab, \qquad r = abc,$$

as required.

Part (i)

For $|x|$ small enough (specifically $|x| < 1/\max(|a|, |b|, |c|)$, or any x if $a = b = c = 0$), each factor is positive and each logarithm has a convergent series. Using $\ln(1 + t) = \sum_{n\geq 1}(-1)^{n+1}t^n/n$ for $|t| < 1$,

$$\begin{aligned}\ln(1 + qx^2 + rx^3) &= \ln(1 + ax) + \ln(1 + bx) + \ln(1 + cx)\\ &= \sum_{n\geq 1}(-1)^{n+1}\frac{a^n x^n}{n} + \sum_{n\geq 1}(-1)^{n+1}\frac{b^n x^n}{n} + \sum_{n\geq 1}(-1)^{n+1}\frac{c^n x^n}{n}\\ &= \sum_{n\geq 1}(-1)^{n+1}\,\frac{a^n + b^n + c^n}{n}\,x^n.\end{aligned}$$

So the coefficient of x^n is $(-1)^{n+1}S_n$, as required. (For $n = 1$ this is $S_1 = a + b + c = 0$, which matches the absence of an x term.)

Part (ii)

Now expand the same function a second way. Put $u = qx^2 + rx^3$, so

$$\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4} + \cdots.$$

Every term of u has degree at least 2, so u^k involves only powers x^{2k} up to x^{3k}. To find the coefficient of x^n we only need the u^k with $2k \le n \le 3k$:

$$u = qx^2 + rx^3,$$
$$u^2 = q^2x^4 + 2qr\,x^5 + r^2x^6.$$

Hence:

- x^2: only u contributes, giving q;
- x^3: only u contributes, giving r;
- x^5: only $-\frac{1}{2}u^2$ contributes, giving $-\frac{1}{2}(2qr) = -qr$.

Comparing with part (i), where the coefficients are $-S_2$, S_3 and S_5,

$$S_2 = -q, \qquad S_3 = r, \qquad S_5 = -qr.$$

Therefore $S_2S_3 = (-q)(r) = -qr = S_5$, as required.

> **Ask yourself** Why are we allowed to compare coefficients of two different series for the same function?

Both expansions are power series in x that converge to $\ln(1 + qx^2 + rx^3)$ on some interval around 0. A power series that converges near 0 has unique coefficients (they are the derivatives at 0 divided by $n!$, as in Maclaurin's theorem), so the coefficients must agree.

Part (iii)

For x^7 we need $2k \le 7 \le 3k$, which forces $k = 3$. Expanding $u^3 = (qx^2 + rx^3)^3$ by the binomial theorem,

$$u^3 = q^3x^6 + 3q^2r\,x^7 + 3qr^2x^8 + r^3x^9,$$

so the coefficient of x^7 in $\ln(1+u)$ is $\frac{1}{3} \cdot 3q^2r = q^2r$. From part (i) it is $(-1)^8S_7 = S_7$, so $S_7 = q^2r$. Using part (ii),

$$S_2S_5 = (-q)(-qr) = q^2r = S_7,$$

as required.

Part (iv)

For x^9 we need $2k \le 9 \le 3k$, so $k = 3$ or $k = 4$.

- From $\frac{1}{3}u^3$: the x^9 term of u^3 is r^3x^9, giving $\frac{1}{3}r^3$.
- From $-\frac{1}{4}u^4$: in $(qx^2 + rx^3)^4$, the x^9 term needs three factors of qx^2 and one of rx^3, giving $\binom{4}{1}q^3r\,x^9 = 4q^3r\,x^9$. This contributes $-q^3r$.

So the coefficient of x^9 is $\frac{1}{3}r^3 - q^3r$, and by part (i) this equals $(-1)^{10}S_9 = S_9$. On the other hand,

$$S_2S_7 = (-q)(q^2r) = -q^3r.$$

Hence

$$S_9 - S_2S_7 = \tfrac{1}{3}r^3 = \tfrac{1}{3}(abc)^3,$$

and the claim $S_2S_7 = S_9$ holds exactly when $abc = 0$. It is false in general.

For a concrete counterexample take $a = b = 1$, $c = -2$ (so $a + b + c = 0$ and $abc = -2 \neq 0$). Then

$$S_2 = \frac{1+1+4}{2} = 3, \qquad S_7 = \frac{1+1-128}{7} = -18, \qquad S_9 = \frac{1+1-512}{9} = -\frac{170}{3},$$

so $S_2S_7 = -54 \neq -\frac{170}{3} = S_9$. The claim is false.

Ask yourself Parts (ii) and (iii) worked, so why does the pattern break at x^9?

For x^5 and x^7 only one power of u contributes, so each coefficient is a single product of qs and rs that happens to factorise as the right product of earlier Ss. At x^9 two powers of u contribute, and the extra r^3 term from u^3 spoils the factorisation.

Checks

- With $a = b = 1$, $c = -2$ we have $q = -3$ and $r = -2$. Then $S_2 = 3 = -q$, $S_3 = (1 + 1 - 8)/3 = -2 = r$ and $S_5 = (1 + 1 - 32)/5 = -6 = -qr$, so $S_2S_3 = S_5$. Also $S_7 = -18 = q^2r$, and $S_9 - S_2S_7 = -\frac{170}{3} + 54 = -\frac{8}{3} = \frac{1}{3}r^3$, exactly as predicted.
- If $c = 0$ (so $b = -a$), all odd S_n vanish and both sides of the claim in (iv) are 0, consistent with the claim holding when $abc = 0$.
- The identities in (ii)–(iv) and the formula for $S_9 - S_2S_7$ were also confirmed by computer, in exact arithmetic, for 200 random rational choices of a and b.

Common slips

- Missing the u^4 contribution to x^9: it is easy to stop at u^3, giving the wrong coefficient $\frac{1}{3}r^3$.
- Sign errors from $(-1)^{n+1}$: the coefficient of x^2 is $-S_2$, so $S_2 = -q$, not q.

- Forgetting the binomial coefficient: the x^7 term of u^3 is $3q^2r$, not q^2r.

Beyond STEP

The S_n are (up to the factor $1/n$) the *power sums* of the roots of a cubic, and q, r are its *elementary symmetric polynomials*. Taking logarithms and comparing coefficients is a generating-function proof of Newton's identities, which express every power sum in terms of the elementary symmetric polynomials; the theorem behind it is that every symmetric polynomial can be written in terms of the elementary ones. You can read about symmetric polynomials and roots of polynomials in Allenby, *Rings, Fields and Groups*, and find more STEP-style practice with series in Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2014, Question 2

Pure • Hyperbolic Integration

STEP III 2014, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q2

The key idea

Both integrands have $\cosh 2x$ underneath, and the double-angle formula can be written either in terms of $\cosh^2 x$ or in terms of $\sinh^2 x$. Choosing the version that matches the numerator turns each integral into a standard rational integral. For the final part, the trick is to go back to exponentials: with $u = \mathrm{e}^x$ both integrands become rational functions of u with $1 + u^4$ in the denominator, and a suitable combination of them gives exactly $1/(1+u^4)$.

Solution

Part (i)

We use $\cosh 2x = 2\cosh^2 x - 1$. With $u = \cosh x$ we have $\mathrm{d}u = \sinh x \, \mathrm{d}x$, so

$$\int \frac{\sinh x}{\cosh 2x} \, \mathrm{d}x = \int \frac{\mathrm{d}u}{2u^2 - 1}.$$

The denominator factorises as $(\sqrt{2}u - 1)(\sqrt{2}u + 1)$, and the two factors differ by 2, so

$$\frac{1}{2u^2 - 1} = \frac{1}{2}\left(\frac{1}{\sqrt{2}u - 1} - \frac{1}{\sqrt{2}u + 1}\right).$$

Each term integrates to a logarithm, with a factor $\frac{1}{\sqrt{2}}$ from the coefficient of u:

$$\begin{aligned}\int \frac{\mathrm{d}u}{2u^2 - 1} &= \frac{1}{2\sqrt{2}}\left(\ln|\sqrt{2}u - 1| - \ln|\sqrt{2}u + 1|\right) + C \\ &= \frac{1}{2\sqrt{2}} \ln\left|\frac{\sqrt{2}\cosh x - 1}{\sqrt{2}\cosh x + 1}\right| + C,\end{aligned}$$

which is the required result. (In fact $\cosh x \geq 1$ makes $\sqrt{2}\cosh x - 1 > 0$, so the modulus signs aren't needed for real x.)

Part (ii)

Now the numerator is $\cosh x$, which is the derivative of $\sinh x$, so the similar substitution is $u = \sinh x$, $\mathrm{d}u = \cosh x\,\mathrm{d}x$. Writing $\cosh 2x = 1 + 2\sinh^2 x$,

$$\int \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x = \int \frac{\mathrm{d}u}{1+2u^2} = \frac{1}{\sqrt{2}}\arctan\left(\sqrt{2}\,u\right) + C$$
$$= \frac{1}{\sqrt{2}}\arctan\left(\sqrt{2}\sinh x\right) + C,$$

using $\int \mathrm{d}u/(1+a^2u^2) = \frac{1}{a}\arctan(au)$ with $a = \sqrt{2}$.

Part (iii)

At first sight there is no cosh anywhere in $\int_0^1 \frac{\mathrm{d}u}{1+u^4}$.

Ask yourself What does $\cosh x/\cosh 2x$ look like if we write everything in terms of $u = \mathrm{e}^x$?

Since $\cosh x = \frac{1}{2}(u + u^{-1})$ and $\cosh 2x = \frac{1}{2}(u^2 + u^{-2})$, multiplying top and bottom by u^2 gives

$$\frac{\cosh x}{\cosh 2x} = \frac{u+u^{-1}}{u^2+u^{-2}} = \frac{u(u^2+1)}{u^4+1}, \qquad \frac{\sinh x}{\cosh 2x} = \frac{u(u^2-1)}{u^4+1}.$$

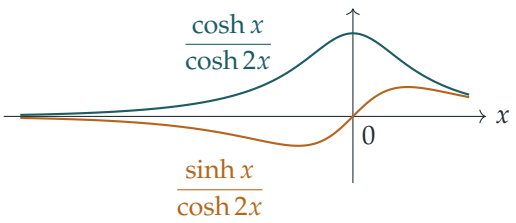
Also $\mathrm{d}x = \mathrm{d}u/u$, so the extra factor of u cancels:

$$\frac{\cosh x}{\cosh 2x}\,\mathrm{d}x = \frac{1+u^2}{1+u^4}\,\mathrm{d}u, \qquad \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{u^2-1}{1+u^4}\,\mathrm{d}u.$$

Subtracting the second from the first leaves $\dfrac{2}{1+u^4}\,\mathrm{d}u$. Finally, u running from 0 to 1 means $x = \ln u$ running from $-\infty$ to 0. Hence

$$\int_0^1 \frac{\mathrm{d}u}{1+u^4} = \frac{1}{2}\int_{-\infty}^0 \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x - \frac{1}{2}\int_{-\infty}^0 \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x.$$

Both integrals on the right converge, because each integrand is $O(\mathrm{e}^x)$ as $x \to -\infty$. (The left-hand side is an ordinary integral, so this is really just a change of variable, with the infinite end coming from $u \to 0$.)



The cosh integral. By part (ii), as $x \to -\infty$ we have $\sinh x \to -\infty$ and so $\arctan(\sqrt{2}\sinh x) \to -\frac{\pi}{2}$. Therefore

$$\int_{-\infty}^{0} \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\left(\arctan 0 - \left(-\tfrac{\pi}{2}\right)\right) = \frac{\pi}{2\sqrt{2}}.$$

The sinh integral. By part (i), as $x \to -\infty$ we have $\cosh x \to \infty$, so the fraction inside the logarithm tends to 1 and the logarithm to 0. At $x = 0$, $\cosh 0 = 1$, so

$$\int_{-\infty}^{0} \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2}-1}{\sqrt{2}+1}.$$

Rationalising, $\dfrac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2$, and $\sqrt{2}-1 = \dfrac{1}{\sqrt{2}+1}$, so

$$\int_{-\infty}^{0} \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\ln(\sqrt{2}-1) = -\frac{1}{\sqrt{2}}\ln(\sqrt{2}+1).$$

This is negative, as it should be: $\sinh x < 0$ for $x < 0$ (see the amber curve).

Putting it together.

$$\int_0^1 \frac{\mathrm{d}u}{1+u^4} = \frac{1}{2}\left(\frac{\pi}{2\sqrt{2}} + \frac{1}{\sqrt{2}}\ln(\sqrt{2}+1)\right) = \frac{\pi + 2\ln(\sqrt{2}+1)}{4\sqrt{2}},$$

as required.

Checks

- Differentiate the answers to (i) and (ii) and recover the integrands; for (ii), $\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\sinh x) = \dfrac{\cosh x}{1+2\sinh^2 x}$.
- The final value is ≈ 0.86697, which agrees with a Simpson's-rule evaluation of $\int_0^1 \mathrm{d}u/(1+u^4)$. It sits sensibly between $\frac{\pi}{4} \approx 0.785$ (the integral of the smaller function $1/(1+u^2)$) and 1.
- Separately, the two infinite integrals were confirmed numerically: $\pi/(2\sqrt{2}) \approx 1.1107$ and $-\ln(1+\sqrt{2})/\sqrt{2} \approx -0.6232$.

Common slips

- Using $\cosh 2x = 2\cosh^2 x - 1$ in part (ii) as well: with $u = \sinh x$ you need the $\sinh^2$ form, $1 + 2\sinh^2 x$.
- Forgetting that $\mathrm{d}x = \mathrm{d}u/u$ when $u = \mathrm{e}^x$; it is exactly this factor that cancels the stray u in the numerators.
- Getting the limits wrong: $u \in [0,1]$ corresponds to $x \in (-\infty, 0]$, not $[0,1]$.

Beyond STEP

The same integral can be done directly with partial fractions, since $1+u^4 = (u^2+\sqrt{2}u+1)(u^2-\sqrt{2}u+1)$; the combination $\frac{1+u^2}{1+u^4}$ and $\frac{1-u^2}{1+u^4}$ is a neat shortcut that avoids most of that algebra. At university, integrals like $\int_0^\infty \mathrm{d}u/(1+u^4) = \pi/(2\sqrt{2})$ are done in one line by contour integration and the residue theorem, and improper integrals are handled with full rigour in analysis. Needham's *Visual Complex Analysis* is a beautiful place to see the complex-analytic side.

WORKED SOLUTION

STEP III 2014, Question 3

Pure • Distance to a Parabola

STEP III 2014, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q3

The key idea

Every point of the parabola can be written as $(at^2, 2at)$, so each distance we need becomes a function of the single variable t, and in both parts that function turns out to be a quadratic (in t for the line, in t^2 for the point). Minimising a quadratic is just completing the square, but we must respect where the variable is allowed to live: t^2 can't be negative, and that restriction is exactly what splits part (ii) into cases. The circle then comes for free, because the distance from a point to a circle is its distance to the centre minus the radius.

Solution

Part (i)

Write L as $mx - y + c = 0$. The perpendicular distance from the point $P = (at^2, 2at)$ of the parabola to L is

$$\frac{|mat^2 - 2at + c|}{\sqrt{m^2+1}}.$$

Let $f(t) = mat^2 - 2at + c$. Since $ma > 0$, completing the square gives

$$f(t) = ma\left(t - \frac{1}{m}\right)^2 + c - \frac{a}{m} = ma\left(t - \frac{1}{m}\right)^2 + \frac{mc - a}{m},$$

so f has minimum value $\dfrac{mc-a}{m}$, attained at $t = 1/m$.

Case $mc > a$**.** Then $f(t) \geq \frac{mc-a}{m} > 0$ for every t, so the whole parabola lies on one side of L and never meets it. The shortest distance between L and the parabola is the least distance from a point of the parabola to L, which is the least value of $|f(t)|/\sqrt{m^2+1} = f(t)/\sqrt{m^2+1}$:

$$\frac{mc-a}{m\sqrt{m^2+1}},$$

as required. The nearest point is $t = 1/m$, where the tangent to the parabola has gradient $\mathrm{d}y/\mathrm{d}x = 1/t = m$: it is parallel to L, just as geometry suggests.

Case $mc \leq a$**.** Now $f(1/m) \leq 0$, while $f(t) \to +\infty$ as $t \to \infty$. Since f is continuous, $f(t) = 0$ for some t, so L meets the parabola and the shortest distance is 0.

Part (ii): the point $(p, 0)$

The square of the distance from $C = (p, 0)$ to $P = (at^2, 2at)$ is

$$(at^2 - p)^2 + 4a^2t^2.$$

This depends only on $u = t^2$, and u takes every value in $[0, \infty)$. So we minimise

$$g(u) = (au - p)^2 + 4a^2u = a^2u^2 - 2a(p - 2a)u + p^2 \qquad (u \geq 0).$$

Completing the square,

$$g(u) = a^2\left(u - \frac{p - 2a}{a}\right)^2 + p^2 - (p - 2a)^2 = a^2\left(u - \frac{p - 2a}{a}\right)^2 + 4a(p - a).$$

Ask yourself The unrestricted minimum of g is at $u = (p - 2a)/a$. Is that value always available?

No: $u = t^2$ must be non-negative, and $(p - 2a)/a \geq 0$ only when $p \geq 2a$.

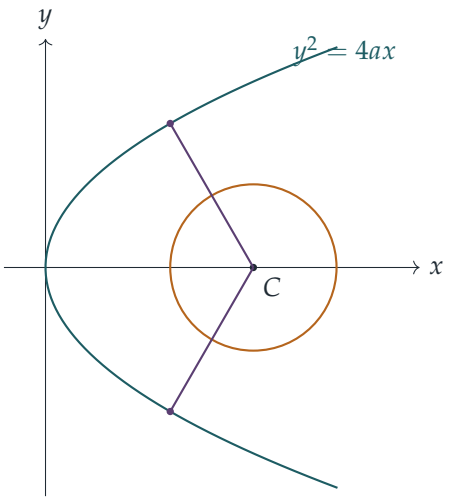
Case $p/a > 2$**.** The vertex $u = p/a - 2$ is positive, so it is allowed, and the least value of g is $4a(p - a)$, which is positive since $p > 2a > a$. The shortest distance is

$$2\sqrt{a(p - a)},$$

achieved at the two points $\left(p - 2a,\ \pm 2\sqrt{a(p - 2a)}\right)$.

Case $p/a \leq 2$**.** The vertex is at $u \leq 0$, so g is increasing on $u \geq 0$ and its least value there is $g(0) = p^2$. The shortest distance is p, achieved at the vertex $(0, 0)$ of the parabola.

The two answers agree at $p = 2a$, where both give $2a$.



The picture shows $a = 1$, $p = 5$: the nearest points of the parabola to C (purple) are at distance $2\sqrt{1 \cdot 4} = 4$, and a circle of radius $b = 2$ about C (amber) stays a distance 2 away.

Part (ii): the circle

Let d be the distance just found from $C = (p, 0)$ to the parabola, so

$$d = \begin{cases} p & \text{if } p \leq 2a, \\ 2\sqrt{a(p-a)} & \text{if } p > 2a. \end{cases}$$

The circle has centre C and radius b.

If $d > b$. Take any point P on the parabola and Q on the circle. By the triangle inequality,

$$|PQ| \geq |PC| - |CQ| = |PC| - b \geq d - b > 0.$$

Equality holds when P is a nearest point of the parabola to C and Q is where the segment CP crosses the circle (it does cross, as $|CP| = d > b$). So the shortest distance is $d - b$.

If $d \leq b$. As P moves along the parabola, $|PC|$ varies continuously from its least value $d \leq b$ to arbitrarily large values, so $|PC| = b$ somewhere. That point lies on both curves, and the shortest distance is 0.

Putting this together, the shortest distance between the circle and the parabola is

$$\begin{cases} p - b & \text{if } p \leq 2a \text{ and } b < p, \\ 2\sqrt{a(p-a)} - b & \text{if } p > 2a \text{ and } b < 2\sqrt{a(p-a)}, \\ 0 & \text{otherwise (that is, if } b \geq d). \end{cases}$$

Checks

- Take $m = 1$, $c = 2$, $a = 1$, so $mc = 2 > a$. The formula gives $\frac{1}{\sqrt{2}} \approx 0.7071$; directly, the point $t = 1$ is $(1, 2)$, and its distance to $x - y + 2 = 0$ is $|1 - 2 + 2|/\sqrt{2} = 1/\sqrt{2}$. If instead $c = \frac{1}{2}$ (so $mc < a$), the line $y = x + \frac{1}{2}$ meets $y^2 = 4x$ where $x^2 - 3x + \frac{1}{4} = 0$, which has real roots, confirming distance 0.
- With $a = 1$, $p = 5$: the points $(3, \pm 2\sqrt{3})$ are at distance $\sqrt{4 + 12} = 4 = 2\sqrt{1 \cdot 4}$ from $(5, 0)$, and a computer search over the parabola found nothing closer.
- As $p \to 0$ the distance $p \to 0$, correctly, since $(0, 0)$ is on the parabola. For large p, $2\sqrt{a(p - a)} < p$, as it must be, because the vertex is at distance p.
- All the results were confirmed by a computer search over hundreds of random choices of m, c, a, p and b.

Common slips

- Setting the derivative of the squared distance with respect to t to zero and forgetting the root $t = 0$, or forgetting that $t^2 = p/a - 2$ has no solution when $p < 2a$: the case split is the whole point of part (ii).
- Giving $(mc - a)/(m\sqrt{m^2 + 1})$ when $mc < a$: this is negative, and the line actually cuts the parabola.
- Writing $d - b$ for the circle without checking its sign; when $b \geq d$ the curves meet and the answer is 0.

Beyond STEP

The nearest point in part (ii) is where CP is normal to the parabola, and the switch at $p = 2a$ is no accident: $(2a, 0)$ is the centre of curvature of the parabola at its vertex. Points beyond it have two nearest points off the axis, and the locus of such switching points, the evolute, is a classic object in the differential geometry of curves. Minimising a function over a restricted range, where the answer may sit at the boundary rather than at a stationary point, is the starting point of constrained optimisation. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2014, Question 4

Pure • Integral Inequalities

STEP III 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 4.7, *The Calculus of Variations*: https://beyondstep.keithmcnulty.org/ch04/variations.html

The key idea

Expand the square in I_1 and compare it with I: the difference is the integral of an exact derivative, $\frac{\mathrm{d}}{\mathrm{d}x}\left(y^2 \tan x\right)$, which vanishes because $y(1) = 0$ and $\tan 0 = 0$. Once I is written as the integral of a square it is obviously non-negative. In part (ii) the same trick works provided we choose b so that the $\sec^2$ term produces exactly a^2y^2, and the only thing that can go wrong is $\tan bx$ blowing up inside $[0,1]$.

Solution

Throughout we assume, as the question intends, that y has a continuous derivative on $[0,1]$.

Part (i)

Since $0 \leq x \leq 1 < \frac{\pi}{2}$, the function $\tan x$ is continuous on $[0,1]$, so I_1 makes sense. Expanding the square,

$$I_1 = \int_0^1 \left((y')^2 + 2yy' \tan x + y^2 \tan^2 x\right) \mathrm{d}x,$$

so

$$\begin{aligned} I - I_1 &= -\int_0^1 \left(y^2 + 2yy' \tan x + y^2 \tan^2 x\right) \mathrm{d}x \\ &= -\int_0^1 \left(2yy' \tan x + y^2 \sec^2 x\right) \mathrm{d}x, \end{aligned}$$

using $1+\tan^2 x = \sec^2 x$. By the product rule, $\frac{\mathrm{d}}{\mathrm{d}x}(y^2\tan x) = 2yy'\tan x + y^2\sec^2 x$, which is exactly the integrand. Hence

$$I - I_1 = -\Big[y^2\tan x\Big]_0^1 = -\big(y(1)^2\tan 1 - y(0)^2\tan 0\big) = 0,$$

because $y(1)=0$ and $\tan 0 = 0$. So $I = I_1$, as required.

Since I_1 is the integral of a square over an interval of positive length, $I_1 \geq 0$, and therefore $I \geq 0$.

Now suppose $I = 0$. Then $I_1 = 0$, and the integrand $(y' + y\tan x)^2$ is continuous and non-negative, so it must vanish everywhere on $[0,1]$ (if it were positive at one point it would be positive on a small interval, making the integral positive). Thus $y' + y\tan x = 0$ for $0 \leq x \leq 1$.

Ask yourself How do we solve $y' + y\tan x = 0$ without dividing by y, which might be zero somewhere?

Use an integrating factor. Since $\cos x > 0$ on $[0,1]$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\cos x}\right) = \frac{y'\cos x + y\sin x}{\cos^2 x} = \frac{y' + y\tan x}{\cos x} = 0,$$

so $y = C\cos x$ for some constant C. The condition $y(1) = 0$ gives $C\cos 1 = 0$, and $\cos 1 \neq 0$, so $C = 0$. Hence $I = 0$ only if $y = 0$ for all x in $[0,1]$.

Part (ii)

Let $K = \displaystyle\int_0^1 (y' + ay\tan bx)^2\,\mathrm{d}x$, with $b > 0$ still to be chosen. Expanding,

$$K = \int_0^1 \left((y')^2 + 2ayy'\tan bx + a^2y^2\tan^2 bx\right)\mathrm{d}x.$$

The derivative we hope to see is

$$\frac{\mathrm{d}}{\mathrm{d}x}(ay^2\tan bx) = 2ayy'\tan bx + aby^2\sec^2 bx = 2ayy'\tan bx + aby^2 + aby^2\tan^2 bx.$$

Comparing this with the integrand of K, the $\tan^2$ terms match exactly when $ab = a^2$, that is, $b = a$. With this choice,

$$(y' + ay\tan ax)^2 = (y')^2 - a^2y^2 + \frac{\mathrm{d}}{\mathrm{d}x}(ay^2\tan ax),$$

and so

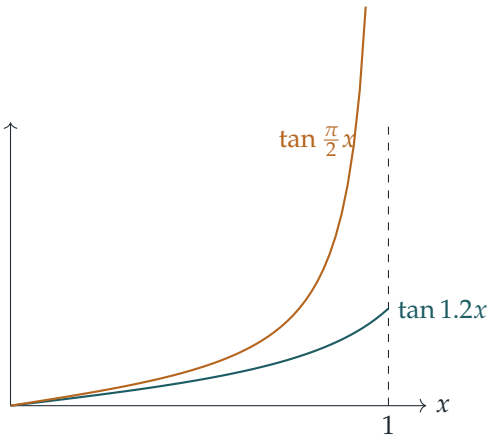
$$K = J + \Big[ay^2\tan ax\Big]_0^1 = J + a\,y(1)^2\tan a = J,$$

since $y(1) = 0$ and $\tan 0 = 0$.

This argument needs $\tan ax$ to be defined and continuous on the whole of $[0,1]$, so that K exists and the fundamental theorem of calculus applies. As $a > 0$, ax runs over $[0,a]$, and tan is continuous there exactly when $a < \frac{\pi}{2}$. So the proof is valid for

$$a < k = \frac{\pi}{2}.$$

For such a we have $J = K \geq 0$, since K is the integral of a square. (In fact, the argument of part (i) with $\cos ax$ in place of $\cos x$ shows that $J = 0$ would force $y = C\cos ax$ with $C\cos a = 0$, so $y \equiv 0$; for y not identically zero we even get $J > 0$.)



The picture shows why $k = \frac{\pi}{2}$: for $a < \frac{\pi}{2}$ the weight $\tan ax$ stays bounded on $[0,1]$, but at $a = \frac{\pi}{2}$ it has an asymptote at the end point $x = 1$.

The case $a = \frac{\pi}{2}$

The proof suggests where to look: $K = 0$ needs $y' + ay\tan ax = 0$, whose solutions are $y = C\cos ax$. For $a < \frac{\pi}{2}$ these fail the condition $y(1) = 0$, but for $a = \frac{\pi}{2}$ they satisfy it. So try

$$y = \cos\tfrac{\pi x}{2}, \qquad y' = -\tfrac{\pi}{2}\sin\tfrac{\pi x}{2},$$

which is not everywhere zero and has $y(1) = \cos\frac{\pi}{2} = 0$. Then

$$J = \frac{\pi^2}{4}\int_0^1 \left(\sin^2\tfrac{\pi x}{2} - \cos^2\tfrac{\pi x}{2}\right)\mathrm{d}x = -\frac{\pi^2}{4}\int_0^1 \cos \pi x\,\mathrm{d}x$$
$$= -\frac{\pi^2}{4}\left[\frac{\sin \pi x}{\pi}\right]_0^1 = 0.$$

So $y = \cos\frac{\pi x}{2}$ (or any non-zero multiple of it) gives $J = 0$ when $a = k$.

Checks

- Take $y = 1 - x$ in part (i): $I = \int_0^1 \left(1 - (1-x)^2\right)\mathrm{d}x = 1 - \frac{1}{3} = \frac{2}{3} > 0$, and evaluating I_1 numerically also gives 0.6667.

- For many random cubic multiples of $(1-x)$ and random a in $(0, \frac{\pi}{2})$, a computer check confirmed $J = K > 0$ to high accuracy.
- The bound $k = \frac{\pi}{2}$ is sharp: with $y = \cos\frac{\pi x}{2}$ one finds $J = \frac{1}{2}\left(\frac{\pi^2}{4} - a^2\right)$, which is negative for every $a > \frac{\pi}{2}$ (for example $J \approx -0.766$ when $a = 2$).

Common slips

- Forgetting that the boundary term at $x = 0$ vanishes because $\tan 0 = 0$, not because of any condition on $y(0)$; nothing is assumed about $y(0)$.
- Concluding $y' + y\tan x = 0$ from $I_1 = 0$ without saying why: it is the continuity of the integrand that matters.
- Choosing b without checking that $\tan bx$ stays finite on $[0,1]$; that is exactly where the condition $a < \frac{\pi}{2}$ comes from.

Beyond STEP

The inequality $\int_0^1 (y')^2\,\mathrm{d}x \geq \frac{\pi^2}{4}\int_0^1 y^2\,\mathrm{d}x$ for $y(1) = 0$ is a form of Wirtinger's (or Poincaré's) inequality, and the constant $\frac{\pi^2}{4}$ is the smallest eigenvalue of $-y'' = \lambda y$ with $y'(0) = 0$, $y(1) = 0$; the extremal function $\cos\frac{\pi x}{2}$ is its eigenfunction. In the calculus of variations, the trick of completing the square with a well-chosen derivative is the idea behind Legendre's and Jacobi's tests for whether a stationary path really is a minimum. For more questions in this spirit, see Siklos, *Advanced Problems in Mathematics*; for the underlying eigenvalue problems, Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 2014, Question 5

Pure • Complex Numbers in Geometry

STEP III 2014, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q5

The key idea

In the complex plane, "parallel and equal in length" just means "equal as displacement vectors", and a quarter turn anticlockwise is multiplication by i. So every geometric condition in this question becomes a short linear equation in the vertices. For the squares on the sides, we find each centre by walking to the midpoint of the side and then stepping outwards by half the side length, at right angles to it.

Solution

The parallelogram criterion

Suppose first that $a + c = b + d$. Then

$$b - a = c - d \qquad \text{and} \qquad d - a = c - b,$$

so the displacement $\overrightarrow{AB}$ equals $\overrightarrow{DC}$ and $\overrightarrow{AD}$ equals $\overrightarrow{BC}$. Equal displacements are parallel and of equal length, so both pairs of opposite sides are parallel and equal: $ABCD$ is a parallelogram.

Conversely, suppose $ABCD$ is a parallelogram. Two segments that are parallel and of equal length have displacements that are equal or opposite, so

$$b - a = \varepsilon(c - d), \qquad d - a = \eta(c - b), \qquad \varepsilon, \eta \in \{1, -1\}.$$

If $\varepsilon = 1$ or $\eta = 1$, either equation rearranges to $a + c = b + d$. The only other case is $\varepsilon = \eta = -1$, that is $b - a = d - c$ and $d - a = b - c$. Subtracting the second from the first gives $b - d = d - b$, so $b = d$, which is impossible for distinct vertices. Hence $a + c = b + d$, and the equivalence is proved.

The square criterion

Let $ABCD$ be a parallelogram, and let $m = \frac{1}{2}(a+c) = \frac{1}{2}(b+d)$, the common midpoint of the diagonals (its centre). Then

$$a-m = \tfrac{1}{2}(a-c), \qquad b-m = \tfrac{1}{2}(b-d), \qquad c-m = -(a-m), \qquad d-m = -(b-m).$$

If $ABCD$ is a square, its diagonals are equal in length and bisect each other at right angles, so B is obtained from A by a quarter turn about m. That turn is anticlockwise because the vertices are labelled anticlockwise. So $b-m = \mathrm{i}(a-m)$, which is $\frac{1}{2}(b-d) = \frac{1}{2}\mathrm{i}(a-c)$, that is, $\mathrm{i}(a-c) = b-d$.

Conversely, if $\mathrm{i}(a-c) = b-d$, write $w = a-m \neq 0$. Then

$$a-m = w, \quad b-m = \mathrm{i}w, \quad c-m = -w, \quad d-m = -\mathrm{i}w,$$

so the four vertices are w turned successively through 90° about m. They are the vertices of a square (all four sides are $|w|\sqrt{2}$, and each side is the previous one multiplied by i). This proves the second equivalence.

Part (i)

The vertices P, Q, R, S run anticlockwise, so as we walk from P to Q the interior of $PQRS$ is on our left. The external square on PQ therefore lies on our right.

The centre X is reached from the midpoint $\frac{1}{2}(p+q)$ of PQ by moving a distance $\frac{1}{2}|q-p|$ (half the side) perpendicular to PQ, to the right. Multiplying $q-p$ by $-\mathrm{i}$ turns it through 90° clockwise, i.e. to the right, so this step is $-\frac{1}{2}\mathrm{i}(q-p)$. Hence

$$x = \tfrac{1}{2}(p+q) - \tfrac{1}{2}\mathrm{i}(q-p) = \tfrac{1}{2}\big(p(1+\mathrm{i}) + q(1-\mathrm{i})\big),$$

as required.

Ask yourself Why $-\mathrm{i}$ rather than $+\mathrm{i}$?

Multiplying by $+\mathrm{i}$ would turn $\overrightarrow{PQ}$ to the left, which points into the quadrilateral and would give the centre of the square built on the inside. Because every internal angle is less than 180°, the interior lies entirely on the left of each side, so the right is genuinely outside.

Part (ii)

Applying part (i) to each side in turn (the same reasoning works for every side, since each is traversed with the interior on the left),

$$2x = p(1+\mathrm{i}) + q(1-\mathrm{i}), \qquad 2y = q(1+\mathrm{i}) + r(1-\mathrm{i}),$$
$$2z = r(1+\mathrm{i}) + s(1-\mathrm{i}), \qquad 2t = s(1+\mathrm{i}) + p(1-\mathrm{i}).$$

We compare the two expressions in the criteria.

Sums of opposite vertices.

$$\begin{aligned} 2(x+z)-2(y+t) &= (p+r)(1+\mathrm{i})+(q+s)(1-\mathrm{i})-(q+s)(1+\mathrm{i})-(p+r)(1-\mathrm{i}) \\ &= 2\mathrm{i}\,(p+r-q-s), \end{aligned}$$

so

$$x+z-(y+t) = \mathrm{i}\big(p+r-(q+s)\big). \tag{1}$$

Diagonals.

$$\begin{aligned} 2\mathrm{i}(x-z) &= \mathrm{i}(p-r)(1+\mathrm{i})+\mathrm{i}(q-s)(1-\mathrm{i}) = (p-r)(\mathrm{i}-1)+(q-s)(1+\mathrm{i}), \\ 2(y-t) &= (q-s)(1+\mathrm{i})+(r-p)(1-\mathrm{i}) = (q-s)(1+\mathrm{i})+(p-r)(\mathrm{i}-1). \end{aligned}$$

These are identical, so

$$\mathrm{i}(x-z) = y-t \tag{2}$$

for *every* such quadrilateral $PQRS$: the segments XZ and YT are always equal in length and perpendicular.

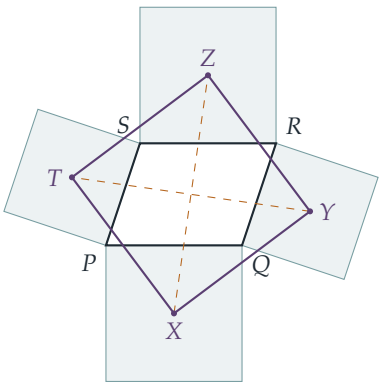
Ask yourself Since (2) always holds, what decides whether $XYZT$ is a square?

Only the parallelogram condition. Now we put the pieces together.

If $PQRS$ is a parallelogram, then $p+r = q+s$, so by (1) $x+z = y+t$ and $XYZT$ is a parallelogram by the first result. Together with (2), the square criterion shows that $XYZT$ is a square. (The converse half of that criterion does not need any assumption about the labelling, and $X \neq Z$, since $x = z$ would mean $r-p = \mathrm{i}(s-q)$, so that $\overrightarrow{PR}$ would be $\overrightarrow{QS}$ turned a quarter turn *anticlockwise*; but in a convex quadrilateral labelled anticlockwise, $\overrightarrow{PR}$ is always $\overrightarrow{QS}$ turned *clockwise* through an angle between $0°$ and $180°$ and rescaled.)

If $XYZT$ is a square, then in particular it is a parallelogram, so $x+z = y+t$ by the first result. By (1), $\mathrm{i}(p+r-q-s) = 0$, so $p+r = q+s$ and $PQRS$ is a parallelogram.

Hence $XYZT$ is a square if and only if $PQRS$ is a parallelogram.



Checks

Take the parallelogram $p = 0$, $q = 4$, $r = 5 + 3\mathrm{i}$, $s = 1 + 3\mathrm{i}$ (drawn above). The formula gives $x = 2 - 2\mathrm{i}$, $y = 6 + \mathrm{i}$, $z = 3 + 5\mathrm{i}$, $t = -1 + 2\mathrm{i}$. The sides of $XYZT$ are $4 + 3\mathrm{i}$, $-3 + 4\mathrm{i}$, $-4 - 3\mathrm{i}$, $3 - 4\mathrm{i}$, each i times the previous one: a square of side 5. As a check on part (i), $x = 2 - 2\mathrm{i}$ is indeed the centre of the square with vertices 0, 4, $4 - 4\mathrm{i}$, $-4\mathrm{i}$ below PQ. A computer check on thousands of random convex quadrilaterals confirmed identity (2) every time, and that $XYZT$ was a square exactly when $PQRS$ was a parallelogram.

Common slips

- Using $+\mathrm{i}$ in part (i), which puts the squares inside the quadrilateral.
- In the parallelogram criterion, forgetting that "parallel and equal" allows $b - a = -(c - d)$, which has to be ruled out.
- Proving only one direction of an "if and only if". In part (ii) the direction "square $\Rightarrow$ parallelogram" needs only (1).

Beyond STEP

Identity (2) is van Aubel's theorem: for *any* quadrilateral, the segments joining the centres of opposite external squares are equal and perpendicular. It is a small example of a general principle: geometric constructions built from rotations and midpoints are linear maps on the vertices, so theorems about them reduce to algebraic identities (Napoleon's theorem on equilateral triangles is another). Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully.

WORKED SOLUTION

STEP III 2014, Question 6

Pure • Inequalities via Convexity

STEP III 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q6

The key idea

A function that starts at zero with zero slope, and whose graph bends upwards, must rise above zero. Integrating the given fact twice turns that picture into a proof. Each inequality in the question is then a matter of moving everything to one side to get a function f with $f(0) = f'(0) = 0$, and checking that f'' is positive. The real work is choosing the right arrangement, so that f'' comes out simple.

Solution

The opening result

Suppose $f''(t) > 0$ for $0 < t < x_0$ and $f(0) = f'(0) = 0$. Fix x with $0 < x < x_0$.

Take $h = f''$. Since $f''(t) > 0$ for $0 < t < x$, the given result gives

$$f'(x) = f'(x) - f'(0) = \int_0^x f''(t)\,\mathrm{d}t > 0.$$

This holds for every x in $(0, x_0)$, so $f'(t) > 0$ for $0 < t < x_0$.

Now apply the given result again, this time with $h = f'$. For $0 < x < x_0$ we have $f'(t) > 0$ for $0 < t < x$, so

$$f(x) = f(x) - f(0) = \int_0^x f'(t)\,\mathrm{d}t > 0.$$

So $f(t) > 0$ for $0 < t < x_0$, as required. We will use this with $x_0 = \frac{1}{2}\pi$ throughout.

Part (i)

Let $f(x) = 1 - \cos x \cosh x$. Then $f(0) = 1 - 1 = 0$, and by the product rule

$$f'(x) = \sin x \cosh x - \cos x \sinh x, \qquad f'(0) = 0 - 0 = 0.$$

Differentiating again,

$$\begin{aligned} f''(x) &= (\cos x \cosh x + \sin x \sinh x) - (-\sin x \sinh x + \cos x \cosh x) \\ &= 2 \sin x \sinh x. \end{aligned}$$

For $0 < x < \frac{1}{2}\pi$ both $\sin x > 0$ and $\sinh x > 0$, so $f''(x) > 0$. By the opening result, $f(x) > 0$, that is,

$$\cos x \cosh x < 1 \qquad \text{for } 0 < x < \tfrac{1}{2}\pi.$$

Part (ii)

Ask yourself Which function should we apply the opening result to? Using $\frac{\sin x}{x} - \frac{1}{\cosh x}$ directly looks messy.

Clear the fractions first. For $0 < x < \frac{1}{2}\pi$, the quantities x, $\sin x$, $\sinh x$ and $\cosh x$ are all positive, so multiplying through by positive quantities does not change the direction of an inequality. Then

$$\frac{1}{\cosh x} < \frac{\sin x}{x} \iff x < \sin x \cosh x, \qquad \frac{\sin x}{x} < \frac{x}{\sinh x} \iff \sin x \sinh x < x^2.$$

(Multiply the first by $x \cosh x > 0$, the second by $x \sinh x > 0$.) The products $\sin x \cosh x$ and $\sin x \sinh x$ differentiate neatly, which is why this is the right form.

Left-hand inequality. Let $g(x) = \sin x \cosh x - x$. Then $g(0) = 0$ and

$$g'(x) = \cos x \cosh x + \sin x \sinh x - 1, \qquad g'(0) = 1 + 0 - 1 = 0.$$

Differentiating again, the $\sin x \cosh x$ terms cancel:

$$\begin{aligned} g''(x) &= (-\sin x \cosh x + \cos x \sinh x) + (\cos x \sinh x + \sin x \cosh x) \\ &= 2 \cos x \sinh x, \end{aligned}$$

which is positive for $0 < x < \frac{1}{2}\pi$, because $\cos x > 0$ and $\sinh x > 0$ there. (This is where the upper limit $\frac{1}{2}\pi$ matters.) By the opening result, $g(x) > 0$, so $x < \sin x \cosh x$ and hence $\frac{1}{\cosh x} < \frac{\sin x}{x}$.

Right-hand inequality. Let $k(x) = x^2 - \sin x \sinh x$. Then $k(0) = 0$ and

$$k'(x) = 2x - \cos x \sinh x - \sin x \cosh x, \qquad k'(0) = 0.$$

Differentiating again, the $\sin x \sinh x$ terms cancel:

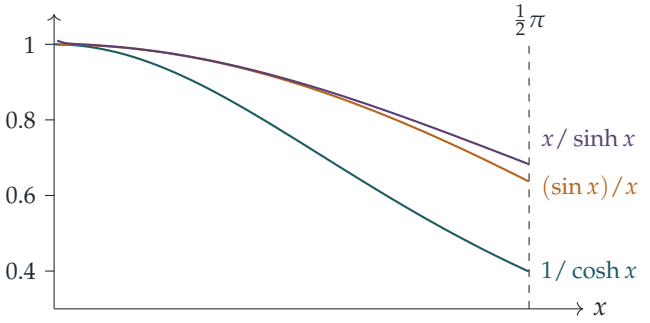
$$\begin{aligned} k''(x) &= 2 - (-\sin x \sinh x + \cos x \cosh x) - (\cos x \cosh x + \sin x \sinh x) \\ &= 2 - 2 \cos x \cosh x. \end{aligned}$$

By part (i), $\cos x \cosh x < 1$ for $0 < x < \frac{1}{2}\pi$, so $k''(x) > 0$ there. By the opening result, $k(x) > 0$, so $\sin x \sinh x < x^2$ and hence $\frac{\sin x}{x} < \frac{x}{\sinh x}$.

Putting the two together, for $0 < x < \frac{1}{2}\pi$,

$$\frac{1}{\cosh x} < \frac{\sin x}{x} < \frac{x}{\sinh x},$$

as required.



Checks

- Series: $\cos x \cosh x = 1 - \frac{1}{6}x^4 + \cdots$, so it does dip just below 1 for small x, consistent with (i). Similarly $\frac{1}{\cosh x} \approx 1 - \frac{1}{2}x^2$, $\frac{\sin x}{x} \approx 1 - \frac{1}{6}x^2$ and $\frac{x}{\sinh x} \approx 1 - \frac{1}{6}x^2$, so the left inequality is clear for small x, while the right one is decided by the next terms (the x^4 coefficients are $\frac{1}{120}$ for $\frac{\sin x}{x}$ and the larger $\frac{7}{360}$ for $\frac{x}{\sinh x}$).
- At $x = 1$: $\cos 1 \cosh 1 \approx 0.834 < 1$, and $\frac{1}{\cosh 1} \approx 0.648 < \sin 1 \approx 0.841 < \frac{1}{\sinh 1} \approx 0.851$.
- Near $x = \frac{1}{2}\pi$, at $x = 1.57$: the three values are about 0.399, 0.637 and 0.683, and $\cos x \cosh x \approx 0.002$.

Common slips

- Applying the opening result to $\frac{\sin x}{x} - \frac{1}{\cosh x}$ itself: this is not even defined at 0, and its derivatives are unpleasant. Clear denominators first, checking that every factor you multiply by is positive.
- Forgetting to check $f'(0) = 0$ as well as $f(0) = 0$; without it the opening result does not apply.
- Not noticing that the second inequality needs part (i): $k'' = 2 - 2\cos x \cosh x$ is positive precisely because of (i).

Beyond STEP

The opening result is a form of the mean value theorem argument: a function with positive derivative is increasing, applied twice. It says that a convex function lies

above its tangent line, here the tangent $y = 0$ at the origin, and this "compare second derivatives" technique is a standard way to prove inequalities between elementary functions. Abbott's *Understanding Analysis* develops the mean value theorem and its consequences carefully.

WORKED SOLUTION

STEP III 2014, Question 7

Pure • Vectors and Cyclic Quadrilaterals

STEP III 2014, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

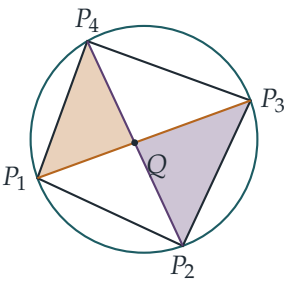
step.maths.org/questions/14-s3-q7

The key idea

Part (i) is the intersecting chords theorem, and it comes from angles in the same segment. The vector parts give the same fact a second time. Four points in a plane are "affinely dependent", and the coefficients in that dependence locate the crossing point of the diagonals, so they divide each diagonal in a known ratio. Putting those ratios into (i) gives the final relation.

Solution

Two facts about the configuration are used throughout. First, no three of the P_i are collinear, because a line meets a circle in at most two points. Second, since the vertices are labelled in order round the circle, P_1 and P_3 lie on the two different arcs cut off by the chord P_2P_4. So they are on opposite sides of the line P_2P_4, and the diagonals cross at a point Q strictly inside both segments P_1P_3 and P_2P_4.



Part (i)

In the triangles P_1QP_4 and P_2QP_3:

- $\angle P_1QP_4 = \angle P_2QP_3$, since they are vertically opposite angles at Q;

- $\angle QP_1P_4 = \angle P_3P_1P_4$ and $\angle QP_2P_3 = \angle P_4P_2P_3$. These are equal because both are angles subtended by the chord P_3P_4 at points P_1, P_2 on the same arc (the arc from P_4 to P_3 through P_1 and P_2).

So the triangles are similar, with $P_1 \leftrightarrow P_2$, $Q \leftrightarrow Q$, $P_4 \leftrightarrow P_3$. Corresponding sides are in proportion:

$$\frac{P_1Q}{P_2Q} = \frac{QP_4}{QP_3}, \qquad \text{so} \qquad (P_1Q)(QP_3) = (P_2Q)(QP_4),$$

as required.

Part (ii)

The cyclic quadrilateral lies in a plane, so the three vectors $\mathbf{p}_2 - \mathbf{p}_1$, $\mathbf{p}_3 - \mathbf{p}_1$, $\mathbf{p}_4 - \mathbf{p}_1$ all lie in a two-dimensional space. Three vectors in two dimensions are linearly dependent, so there are numbers b_2, b_3, b_4, not all zero, with

$$b_2(\mathbf{p}_2 - \mathbf{p}_1) + b_3(\mathbf{p}_3 - \mathbf{p}_1) + b_4(\mathbf{p}_4 - \mathbf{p}_1) = \mathbf{0}.$$

(Concretely: $\mathbf{p}_2 - \mathbf{p}_1$ and $\mathbf{p}_3 - \mathbf{p}_1$ are not parallel, since P_1, P_2, P_3 are not collinear, so they span the plane and $\mathbf{p}_4 - \mathbf{p}_1$ is a combination of them. Then we may take $b_4 = -1$.) Now put $a_1 = -(b_2 + b_3 + b_4)$ and $a_i = b_i$ for $i = 2, 3, 4$. Then $\sum a_i = 0$, not all a_i are zero, and

$$\sum_{i=1}^{4} a_i\mathbf{p}_i = a_1\mathbf{p}_1 + b_2\mathbf{p}_2 + b_3\mathbf{p}_3 + b_4\mathbf{p}_4 = b_2(\mathbf{p}_2 - \mathbf{p}_1) + b_3(\mathbf{p}_3 - \mathbf{p}_1) + b_4(\mathbf{p}_4 - \mathbf{p}_1) = \mathbf{0},$$

so $(*)$ holds.

Part (iii)

$a_1 + a_3 \neq 0$. Suppose instead that $a_1 + a_3 = 0$. Then $a_2 + a_4 = 0$ too, by the first equation of $(*)$, so $a_3 = -a_1$ and $a_4 = -a_2$. The second equation becomes

$$a_1(\mathbf{p}_1 - \mathbf{p}_3) + a_2(\mathbf{p}_2 - \mathbf{p}_4) = \mathbf{0}.$$

The lines P_1P_3 and P_2P_4 are different (otherwise all four points would be collinear) and they meet at Q, so they are not parallel. Hence $\mathbf{p}_1 - \mathbf{p}_3$ and $\mathbf{p}_2 - \mathbf{p}_4$ are non-zero and not parallel, which forces $a_1 = a_2 = 0$. Then all four a_i are zero, a contradiction. So $a_1 + a_3 \neq 0$, and likewise $a_2 + a_4 = -(a_1 + a_3) \neq 0$.

The intersection point. From $(*)$, $a_1\mathbf{p}_1 + a_3\mathbf{p}_3 = -(a_2\mathbf{p}_2 + a_4\mathbf{p}_4)$ and $a_1 + a_3 = -(a_2 + a_4)$. Dividing one by the other,

$$\mathbf{r} = \frac{a_1\mathbf{p}_1 + a_3\mathbf{p}_3}{a_1 + a_3} = \frac{a_2\mathbf{p}_2 + a_4\mathbf{p}_4}{a_2 + a_4}.$$

The first form shows $\mathbf{r} = \mathbf{p}_1 + \dfrac{a_3}{a_1 + a_3}(\mathbf{p}_3 - \mathbf{p}_1)$, a point of the line P_1P_3. In the same way the second form shows it is a point of the line P_2P_4. Two non-parallel lines meet in only one point, so $\mathbf{r}$ is the position vector of Q.

Ask yourself How do we turn this formula for Q into the lengths P_1Q and QP_3?

Subtract. Using $\mathbf{q} = \mathbf{r}$,

$$\mathbf{q} - \mathbf{p}_1 = \frac{a_3(\mathbf{p}_3 - \mathbf{p}_1)}{a_1 + a_3}, \qquad \mathbf{p}_3 - \mathbf{q} = \frac{a_1(\mathbf{p}_3 - \mathbf{p}_1)}{a_1 + a_3}.$$

Taking lengths gives $P_1Q = \dfrac{|a_3|}{|a_1 + a_3|}P_1P_3$ and $QP_3 = \dfrac{|a_1|}{|a_1 + a_3|}P_1P_3$. In the same way, from the second form of $\mathbf{r}$, $P_2Q = \dfrac{|a_4|}{|a_2 + a_4|}P_2P_4$ and $QP_4 = \dfrac{|a_2|}{|a_2 + a_4|}P_2P_4$.

The signs. Since Q lies strictly between P_1 and P_3, the fraction $\frac{a_3}{a_1+a_3}$ lies strictly between 0 and 1. Then $\frac{a_1}{a_1+a_3} = 1 - \frac{a_3}{a_1+a_3}$ is also strictly between 0 and 1. Two positive numbers have a positive product, so

$$\frac{a_1a_3}{(a_1 + a_3)^2} > 0, \qquad \text{that is, } a_1a_3 > 0.$$

The same argument on P_2P_4 gives $a_2a_4 > 0$. So $|a_1a_3| = a_1a_3$ and $|a_2a_4| = a_2a_4$.

The deduction. Substitute into part (i):

$$\frac{a_1a_3}{(a_1 + a_3)^2}(P_1P_3)^2 = (P_1Q)(QP_3) = (P_2Q)(QP_4) = \frac{a_2a_4}{(a_2 + a_4)^2}(P_2P_4)^2.$$

Since $(a_2 + a_4)^2 = (a_1 + a_3)^2 \neq 0$, multiplying through by it gives

$$a_1a_3(P_1P_3)^2 = a_2a_4(P_2P_4)^2,$$

which is the required result.

Checks

- Take the square with vertices $(1,0), (0,1), (-1,0), (0,-1)$. Then $\mathbf{p}_1 + \mathbf{p}_3 = \mathbf{0} = \mathbf{p}_2 + \mathbf{p}_4$, so $a = (1,-1,1,-1)$ satisfies $(*)$. The formula gives Q at the origin, and $a_1a_3(P_1P_3)^2 = 4 = a_2a_4(P_2P_4)^2$.
- For an unsymmetrical case, take $P_1, \ldots, P_4$ on the unit circle at angles $200°, 290°, 20°, 120°$, as in the diagram. All three claims of (iii), including $a_1a_3 > 0$ and $a_2a_4 > 0$, were confirmed by a computer check on this and on thousands of random cyclic quadrilaterals, to within 10^{-10}.
- The answer survives rescaling all the a_i by the same factor λ, as it must: both sides of the final relation scale by λ^2.

Common slips

- Pairing the wrong angles in (i): the equal angles at P_1 and P_2 stand on the chord P_3P_4, and this fixes which sides correspond.

- Dropping the modulus signs in (iii) without comment. The relation with absolute values is easy; the work is in showing $a_1a_3 > 0$ and $a_2a_4 > 0$, which uses the fact that Q is inside both diagonals.
- Forgetting why the diagonals are not parallel when proving $a_1 + a_3 \neq 0$.

Beyond STEP

The quantity $(P_1Q)(QP_3)$ is the *power* of the point Q with respect to the circle; it is the same for every chord through Q. Part (ii) is the statement that four points in a plane are affinely dependent. The numbers a_i are barycentric-style coordinates, and the formula for Q is a first look at how affine geometry handles points, lines and ratios without any coordinates. Strang's *Introduction to Linear Algebra* covers linear dependence and the dimension count behind (ii).

WORKED SOLUTION

STEP III 2014, Question 8

Pure • Convergence of Series

STEP III 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 2.3, *A Divergent Series and Its Convergent Cousin*: https://beyondstep.keithmcnulty.org/ch02/kempner.html

The key idea

Chop the positive integers into blocks $[k^n, k^{n+1})$. Every term in a block lies between the value at the start of the block and the value at the start of the next one, so each block sum is squeezed between the number of terms times those two values. Adding up the blocks turns an awkward series into a geometric series, or into one with a constant term in each block, and we can see at once whether it diverges. The same counting works for the integers with no digit 2, as long as we count how many of them lie in each block.

Solution

The block inequality

The block $k^n \le r \le k^{n+1} - 1$ contains

$$(k^{n+1} - 1) - k^n + 1 = k^{n+1} - k^n = k^n(k-1)$$

integers. Since f is strictly decreasing, for every r in the block,

$$\mathrm{f}(k^{n+1}) < \mathrm{f}(r) \le \mathrm{f}(k^n),$$

the first because $r < k^{n+1}$ and the second because $r \ge k^n$ (with equality only at $r = k^n$). Adding these over the $k^n(k-1)$ values of r gives

$$k^n(k-1)\,\mathrm{f}(k^{n+1}) \le \sum_{r=k^n}^{k^{n+1}-1} \mathrm{f}(r) \le k^n(k-1)\,\mathrm{f}(k^n),$$

as required. (The left-hand inequality is in fact strict.)



The blocks for $k = 2$, with the number of terms in each.

Part (i)

Take $k = 2$ and $f(r) = \frac{1}{r}$, which is strictly decreasing. The block inequality becomes

$$\frac{2^n}{2^{n+1}} \leq \sum_{r=2^n}^{2^{n+1}-1} \frac{1}{r} \leq \frac{2^n}{2^n}, \qquad \text{that is} \qquad \tfrac{1}{2} \leq \sum_{r=2^n}^{2^{n+1}-1} \frac{1}{r} \leq 1.$$

The blocks for $n = 0, 1, \ldots, N$ fit together exactly to cover $1 \leq r \leq 2^{N+1} - 1$, since each block ends just before the next one starts. Adding the $N + 1$ inequalities,

$$\frac{N+1}{2} \leq \sum_{r=1}^{2^{N+1}-1} \frac{1}{r} \leq N + 1.$$

For the deduction: the partial sums $H_m = \sum_{r=1}^{m} \frac{1}{r}$ increase with m, and $H_{2^{N+1}-1} \geq \frac{N+1}{2}$, which can be made larger than any number we like by choosing N large. So the partial sums are unbounded and $\sum \frac{1}{r}$ does not converge.

Part (ii)

Take $k = 2$ and $f(r) = \frac{1}{r^3}$, again strictly decreasing. Now we need only the right-hand inequality:

$$\sum_{r=2^n}^{2^{n+1}-1} \frac{1}{r^3} \leq \frac{2^n}{2^{3n}} = \frac{1}{4^n}.$$

Adding over $n = 0, 1, \ldots, N$ as before,

$$\sum_{r=1}^{2^{N+1}-1} \frac{1}{r^3} \leq \sum_{n=0}^{N} \frac{1}{4^n} < \sum_{n=0}^{\infty} \frac{1}{4^n} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}.$$

Ask yourself This bounds only some of the partial sums. Why does it bound the infinite sum?

All the terms are positive, so the partial sums increase. Any partial sum $\sum_{r=1}^{m} r^{-3}$ is at most the one with $2^{N+1} - 1 \geq m$, and so is less than $\frac{4}{3}$. An increasing sequence that is bounded above converges, and its limit is at most the bound. Hence

$$\sum_{r=1}^{\infty} \frac{1}{r^3} \leq \frac{4}{3} = 1\tfrac{1}{3}.$$

Part (iii)

Counting $S(1000)$. Write each integer from 0 to 999 as a string of three digits, padding with leading zeros (so 7 becomes 007). Padding zeros never introduce a 2, so an integer has no 2 exactly when its string has no 2. Each of the three places can be any of the nine digits $0, 1, 3, 4, \ldots, 9$, which gives 9^3 strings, all for different integers. One of them is 000, which is not positive, so $S(1000)$ contains $9^3 - 1$ numbers.

Counting by blocks. The same argument with $n+1$ places shows that there are $9^{n+1} - 1$ integers in $S(10^{n+1})$. So the block $10^n \le r \le 10^{n+1} - 1$ of $(n+1)$-digit numbers contains

$$(9^{n+1} - 1) - (9^n - 1) = 8 \cdot 9^n$$

integers with no 2. (Equivalently, the leading digit has 8 choices, as it cannot be 0 or 2, and each of the other n digits has 9.)

Bounding $\sigma(n)$. This is the block argument with $k = 10$, except that we count only the integers with no 2. Every r in the block has $\frac{1}{r} \le \frac{1}{10^n}$, so the contribution of the block to the sum is at most

$$8 \cdot 9^n \cdot \frac{1}{10^n} = 8\left(\tfrac{9}{10}\right)^n.$$

Now fix n and choose M with $n \le 10^{M+1}$. Every element of $S(n)$ lies in one of the blocks for $m = 0, 1, \ldots, M$, and the reciprocals are all positive. So

$$\sigma(n) \le \sum_{m=0}^{M} 8\left(\tfrac{9}{10}\right)^m < \sum_{m=0}^{\infty} 8\left(\tfrac{9}{10}\right)^m = \frac{8}{1 - \frac{9}{10}} = 80.$$

The second inequality is strict because we left out the positive terms with $m > M$. Hence $\sigma(n) < 80$ for all n.

Checks

- Part (ii): $\sum r^{-3} = 1.2020\ldots$ (Apéry's constant), comfortably below $1\frac{1}{3}$.
- Part (i) with $N = 2$: $\sum_{r=1}^{7} \frac{1}{r} \approx 2.593$, which lies between 1.5 and 3.
- Part (iii): a direct count confirms $728 = 9^3 - 1$ numbers below 1000 and $8 \cdot 9^n$ in each block. Summing directly by computer gives $\sigma(10^7) \approx 10.28$, far below 80. The bound is generous, but it is all we were asked for.

Common slips

- Miscounting the block. It runs from k^n to $k^{n+1} - 1$, which is $k^{n+1} - k^n$ terms, not $k^{n+1} - k^n - 1$.
- Counting all 9^n no-2 strings in the n-th block for (iii). The leading digit cannot be 0, so there are only $8 \cdot 9^n$. Using 9^{n+1} gives a bound of 90, which is not enough.

- Stopping at a bound for the partial sums up to $2^{N+1} - 1$ in (ii) without saying why the whole infinite sum is bounded.

Beyond STEP

This is Cauchy's condensation test: for a decreasing positive sequence, $\sum a_r$ converges exactly when $\sum 2^n a_{2^n}$ does. It shows at once that $\sum r^{-p}$ converges if and only if $p > 1$. Part (iii) is the Kempner series: delete every integer with a given digit and the harmonic series, which diverges, becomes convergent, because almost all large integers contain every digit. Abbott's *Understanding Analysis* covers condensation and the comparison tests carefully.

WORKED SOLUTION

STEP III 2014, Question 9

Mechanics • Projectiles with Resistance

STEP III 2014, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q9

The key idea

We're given the position, so the velocity is one differentiation away, and checking the equation of motion is just a matter of putting $\mathbf{v}$ and $\dot{\mathbf{v}}$ into $\dot{\mathbf{v}} = \mathbf{g} - k\mathbf{v}$. After that, everything is about components: the $\mathbf{j}$-component of $\mathbf{r}$ fixes T, and the ratio of the components of $\mathbf{v}$ at time T gives $\tan\beta$. The final inequality looks messy until we write $x = kT$ and clear the denominators. Then it collapses to exactly the hint $\sinh x > x$.

Solution

The velocity, and checking the motion

We model the body as a particle. Newton's second law gives $m\dot{\mathbf{v}} = m\mathbf{g} - mk\mathbf{v}$, so the equation of motion is

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{g} - k\mathbf{v},$$

and the initial conditions are $\mathbf{r} = \mathbf{0}$ and $\mathbf{v} = \mathbf{u}$ at $t = 0$.

Here $\mathbf{g}$ and $\mathbf{u}$ are constant vectors, so we differentiate the given $\mathbf{r}$ coefficient by coefficient:

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{k - k\mathrm{e}^{-kt}}{k^2}\,\mathbf{g} + \frac{k\mathrm{e}^{-kt}}{k}\,\mathbf{u} = \frac{1 - \mathrm{e}^{-kt}}{k}\,\mathbf{g} + \mathrm{e}^{-kt}\,\mathbf{u}.$$

Differentiating again gives $\dot{\mathbf{v}} = \mathrm{e}^{-kt}\mathbf{g} - k\mathrm{e}^{-kt}\mathbf{u}$. On the other hand,

$$\mathbf{g} - k\mathbf{v} = \mathbf{g} - \left(1 - \mathrm{e}^{-kt}\right)\mathbf{g} - k\mathrm{e}^{-kt}\mathbf{u} = \mathrm{e}^{-kt}\mathbf{g} - k\mathrm{e}^{-kt}\mathbf{u},$$

which is the same, so the equation of motion holds. At $t = 0$ we have $\mathrm{e}^0 = 1$, so $\mathbf{r} = \frac{0-1+1}{k^2}\mathbf{g} + 0\,\mathbf{u} = \mathbf{0}$ and $\mathbf{v} = 0\,\mathbf{g} + \mathbf{u} = \mathbf{u}$. Both initial conditions hold.

The time of flight

Now take $\mathbf{u} = u\cos\alpha\,\mathbf{i} + u\sin\alpha\,\mathbf{j}$ and $\mathbf{g} = -g\,\mathbf{j}$. The $\mathbf{j}$-component of $\mathbf{r}$ is

$$\mathbf{r}\cdot\mathbf{j} = -\frac{kt - 1 + \mathrm{e}^{-kt}}{k^2}\,g + \frac{1-\mathrm{e}^{-kt}}{k}\,u\sin\alpha.$$

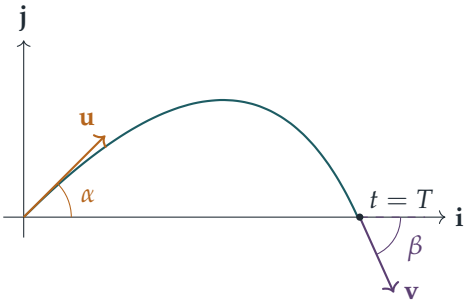
Setting this to zero at $t = T$ and multiplying by k^2:

$$uk\sin\alpha\left(1-\mathrm{e}^{-kT}\right) = \left(kT - 1 + \mathrm{e}^{-kT}\right)g.$$

Since $T > 0$ and $k > 0$, we have $1 - \mathrm{e}^{-kT} > 0$, so we may divide by it. Writing $kT - 1 + \mathrm{e}^{-kT} = kT - \left(1 - \mathrm{e}^{-kT}\right)$,

$$uk\sin\alpha = \left(\frac{kT}{1-\mathrm{e}^{-kT}} - 1\right)g,$$

as required.



The angle at time T

At $t = T$ the components of $\mathbf{v} = \frac{1-\mathrm{e}^{-kT}}{k}\mathbf{g} + \mathrm{e}^{-kT}\mathbf{u}$ are

$$v_x = \mathrm{e}^{-kT}u\cos\alpha, \qquad v_y = -\frac{1-\mathrm{e}^{-kT}}{k}\,g + \mathrm{e}^{-kT}u\sin\alpha.$$

Since $0 < \alpha < 90°$, $v_x > 0$. We expect $v_y < 0$, since the particle is coming back down, and we'll prove it below. So the acute angle between $\mathbf{v}$ and $\mathbf{i}$ satisfies $\tan\beta = -v_y/v_x$. Dividing through by v_x:

$$\tan\beta = \frac{\left(1-\mathrm{e}^{-kT}\right)g}{k\,\mathrm{e}^{-kT}u\cos\alpha} - \frac{\mathrm{e}^{-kT}u\sin\alpha}{\mathrm{e}^{-kT}u\cos\alpha} = \frac{\left(\mathrm{e}^{kT}-1\right)g}{uk\cos\alpha} - \tan\alpha,$$

using $\left(1-\mathrm{e}^{-kT}\right)/\mathrm{e}^{-kT} = \mathrm{e}^{kT} - 1$.

Showing $\tan\beta > \tan\alpha$

From the last result, $\tan\beta > \tan\alpha$ is the same as

$$\frac{\left(\mathrm{e}^{kT}-1\right)g}{uk\cos\alpha} > 2\tan\alpha \iff \left(\mathrm{e}^{kT}-1\right)g > 2uk\sin\alpha,$$

multiplying by $uk\cos\alpha > 0$. Now we use the time-of-flight result to replace $uk\sin\alpha$, which removes u and α altogether.

Ask yourself How do we turn this into something involving $\sinh kT$?

Write $x = kT > 0$. Then $\frac{x}{1-e^{-x}} = \frac{xe^x}{e^x-1}$, and after dividing by $g > 0$ the inequality we need becomes

$$e^x - 1 > \frac{2xe^x}{e^x - 1} - 2.$$

Multiply by $e^x - 1 > 0$ (so the direction is kept) and expand:

$$\begin{aligned}
&\iff e^{2x} - 2e^x + 1 > 2xe^x - 2e^x + 2 \\
&\iff e^{2x} - 1 > 2xe^x \\
&\iff e^x - e^{-x} > 2x \qquad \text{(dividing by } e^x > 0\text{)} \\
&\iff \sinh x > x.
\end{aligned}$$

Every step is reversible, and $\sinh kT > kT$ for $kT > 0$ (we're allowed to assume this). So $\tan\beta > \tan\alpha$.

This also settles the sign of v_y: we've shown $-v_y/v_x > \tan\alpha > 0$ and $v_x > 0$, so $v_y < 0$. The particle really is descending at time T, and the formula for $\tan\beta$ is justified.

Finally, α and β both lie in $(0°, 90°)$, where tan is strictly increasing. So $\tan\beta > \tan\alpha$ gives $\beta > \alpha$. With resistance, the particle comes down more steeply than it went up.

Checks

- As $k \to 0$, $\mathbf{r} \to \frac{1}{2}t^2\mathbf{g} + t\mathbf{u}$, using $e^{-kt} \approx 1 - kt + \frac{1}{2}k^2t^2$. That's the usual projectile. Also $\frac{x}{1-e^{-x}} - 1 \approx \frac{x}{2}$ for small x, so the time of flight gives $T \to 2u\sin\alpha/g$. Similarly $\tan\beta \to \frac{gT}{u\cos\alpha} - \tan\alpha = \tan\alpha$, the symmetric parabola.
- Take $k = 1\,\mathrm{s}^{-1}$, $g = 10\,\mathrm{m\,s}^{-2}$, $u = 10\,\mathrm{m\,s}^{-1}$ and $\alpha = 45°$. Solving $\mathbf{r}\cdot\mathbf{j} = 0$ numerically gives $T \approx 1.185\,\mathrm{s}$, shorter than the $1.414\,\mathrm{s}$ with no resistance. The landing angle is $\beta \approx 65.7° > 45°$, and both formulae check out to many decimal places. A step-by-step numerical integration of $\dot{\mathbf{v}} = \mathbf{g} - k\mathbf{v}$ by computer reproduces the given $\mathbf{r}$ and $\mathbf{v}$.

Common slips

- Taking $\tan\beta = v_y/v_x$ without the minus sign: v_y is negative at landing, and β is acute.
- Dividing or multiplying by $1 - e^{-kT}$ or $e^{kT} - 1$ without saying that it's positive. That's what keeps the inequality the right way round.
- Stopping at $\tan\beta > \tan\alpha$: the deduction $\beta > \alpha$ needs the fact that tan is increasing on the acute angles.

Beyond STEP

The given $\mathbf{r}$ solves the linear differential equation $\ddot{\mathbf{r}} + k\dot{\mathbf{r}} = \mathbf{g}$, which you can solve directly with an integrating factor e^{kt}. Linear drag is the one resistance law where the $\mathbf{i}$- and $\mathbf{j}$-motions separate. With the more realistic quadratic drag $-mk|\mathbf{v}|\mathbf{v}$ they become coupled, and no closed form exists. The lopsided trajectory, steeper on the way down, holds in both cases. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion with resistive forces in this spirit.

WORKED SOLUTION

STEP III 2014, Question 10

Mechanics • Coupled Oscillations

STEP III 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.2, *Normal Modes*: https://beyondstep.keithmcnulty.org/ch11/normal-modes.html

The key idea

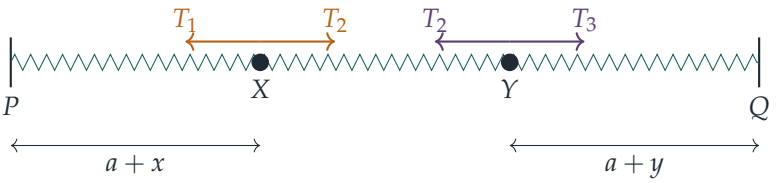
Put everything on one line with a single coordinate, work out the extension of each spring in terms of x and y, and Newton's second law gives two coupled equations. Coupled equations with this symmetry separate when you add and subtract them: $x - y$ and $x + y$ each obey their own simple harmonic equation, with different frequencies. Whether Y ever comes back then turns on whether those two frequencies have a rational ratio.

Solution

The equations of motion

Model X and Y as particles and the springs as light, obeying Hooke's law; the table is smooth, so the only horizontal forces are the spring tensions. Every force acts along the line PQ, and the motion stays on it.

Measure positions along the line from P, in the direction of Q, so P is at 0 and Q at $3a$. Then X is at $a + x$ and Y is at $3a - (a + y) = 2a - y$.



The three springs have lengths and tensions (by Hooke's law, tension $= \lambda \times$ extension$/a$,

negative meaning thrust):

$$PX:\ \text{length } a+x, \qquad T_1 = \frac{\lambda x}{a},$$
$$XY:\ \text{length } (2a-y)-(a+x) = a-x-y, \qquad T_2 = -\frac{\lambda(x+y)}{a},$$
$$YQ:\ \text{length } a+y, \qquad T_3 = \frac{\lambda y}{a}.$$

A spring in tension pulls each end towards the other. So X feels T_1 towards P and T_2 towards Y; taking the direction $P \to Q$ as positive,

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -T_1 + T_2 = -\frac{\lambda}{a}x - \frac{\lambda}{a}(x+y) = -\frac{\lambda}{a}(2x+y),$$

as required. (The signs take care of themselves: if $T_2 < 0$ the spring XY is compressed and pushes X towards P.)

For Y, its position is $2a-y$, so its acceleration in the positive direction is $-\ddot{y}$. It feels T_2 towards X (negative direction) and T_3 towards Q (positive direction):

$$-m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -T_2 + T_3 = \frac{\lambda}{a}(x+y) + \frac{\lambda}{a}y, \qquad \text{so} \qquad m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\frac{\lambda}{a}(x+2y).$$

This is the first equation with x and y swapped, as the symmetry of the set-up (reflect in the midpoint of PQ) says it must be.

The combination $x-y$

Subtracting the two equations,

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}(x-y) = -\frac{\lambda}{a}\left[(2x+y)-(x+2y)\right] = -\frac{\lambda}{a}(x-y).$$

With $\omega^2 = \lambda/(ma)$ this is $\dfrac{\mathrm{d}^2}{\mathrm{d}t^2}(x-y) = -\omega^2(x-y)$, simple harmonic motion, so

$$x - y = A\cos\omega t + B\sin\omega t.$$

Initially $XP = a$ and $YQ = \frac{1}{2}a$, so $x = 0$ and $y = -\frac{1}{2}a$ at $t = 0$, giving $x - y = \frac{1}{2}a$ and hence $A = \frac{1}{2}a$. The particles are released from rest, so $\dot{x} = \dot{y} = 0$ at $t = 0$; differentiating, $\omega B = 0$, so $B = 0$. Thus

$$x - y = \tfrac{1}{2}a\cos\omega t.$$

The combination $x+y$

Adding the equations instead,

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}(x+y) = -\frac{\lambda}{a}(3x+3y), \qquad \text{so} \qquad \frac{\mathrm{d}^2}{\mathrm{d}t^2}(x+y) = -3\omega^2(x+y).$$

This is simple harmonic motion with angular frequency $\sqrt{3}\,\omega$, so $x + y = C\cos\sqrt{3}\,\omega t + D\sin\sqrt{3}\,\omega t$. At $t = 0$, $x + y = -\frac{1}{2}a$ and its derivative is 0, so $C = -\frac{1}{2}a$ and $D = 0$:

$$x + y = -\tfrac{1}{2}a\cos\sqrt{3}\,\omega t.$$

Y never returns

Half the difference of the two results gives

$$y = \tfrac{1}{2}\big[(x+y)-(x-y)\big] = -\tfrac{1}{4}a\big(\cos\omega t + \cos\sqrt{3}\,\omega t\big).$$

(Similarly $x = \tfrac{1}{4}a(\cos\omega t - \cos\sqrt{3}\,\omega t)$.) Y is back at its initial position exactly when $y = -\tfrac{1}{2}a$, that is, when

$$\cos\omega t + \cos\sqrt{3}\,\omega t = 2.$$

? **Ask yourself** Both cosines are periodic, so surely they line up again eventually?

Each cosine is at most 1, so the sum equals 2 only if both equal 1 at the same time $t > 0$. That needs

$$\omega t = 2\pi n \quad \text{and} \quad \sqrt{3}\,\omega t = 2\pi k$$

for integers n, k, with $n \geq 1$ because $t > 0$. Dividing, $\sqrt{3} = k/n$ would be rational, which it is not. So there is no such $t > 0$, and Y never returns to its initial position. The two periods $2\pi/\omega$ and $2\pi/(\sqrt{3}\,\omega)$ are incommensurable, so the two oscillations never get back in step, although they come arbitrarily close to doing so.

Checks

- At $t = 0$ the formulae give $x = \tfrac{1}{4}a(1-1) = 0$ and $y = -\tfrac{1}{4}a(1+1) = -\tfrac{1}{2}a$, matching the starting positions, and both derivatives vanish there.
- The length of spring XY is $a - x - y = a + \tfrac{1}{2}a\cos\sqrt{3}\,\omega t$, which stays between $\tfrac{1}{2}a$ and $\tfrac{3}{2}a$, so the particles never collide; likewise $PX = a + x \geq \tfrac{1}{2}a$ and $YQ = a + y \geq \tfrac{1}{2}a$. The model stays physically sensible throughout.
- Stepping the motion numerically with $m = a = 1$, $\lambda = 2$ reproduces the formulae for x and y to better than 10^{-10} up to $t = 20$. A computer search over $0 < \omega t \leq 20\,000$ finds that $\cos\omega t + \cos\sqrt{3}\,\omega t$ gets as close as 1.999995 to 2, but never reaches it.

Common slips

- Getting the sign of the Y equation wrong: y is measured from Q, so increasing y moves Y in the *negative* direction.
- Writing the length of XY as $a + x + y$ instead of $a - x - y$; both x and y increasing make the middle spring shorter.
- Arguing only that the two periods differ. Different periods alone are not enough (periods 1 and 2 realign at $t = 2$); what matters is that their ratio $\sqrt{3}$ is irrational.

Beyond STEP

The combinations $x - y$ and $x + y$ are the *normal modes* of this system: the eigenvectors of the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with eigenvalues 1 and 3 giving the squared frequencies ω^2 and $3\omega^2$. Every small oscillation of a coupled system splits into normal modes like this, and the motion is periodic only when all the frequency ratios are rational; otherwise it is quasi-periodic, never repeating but returning arbitrarily close to its start. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers coupled oscillators, and Strang, *Introduction to Linear Algebra*, explains the eigenvector method behind them.

WORKED SOLUTION

STEP III 2014, Question 11

Mechanics • Circular Motion

STEP III 2014, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q11

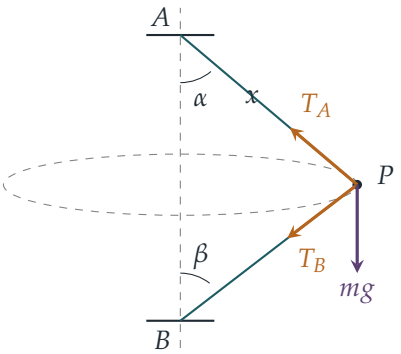
The key idea

Three forces act on the particle (two tensions and its weight), and together they must supply exactly the centripetal force needed for circular motion. Resolving vertically and horizontally gives two linear equations for the two tensions. A string can only pull, so $T_B \geq 0$, and that gives the inequality. In the boundary case the lower string goes slack, and the cosine rule in triangle ABP tells us which angles α can occur.

Solution

Setting up

Model P as a particle and the strings as light and inextensible. P moves in a horizontal circle centred on the line AB, with radius $r = x\sin\alpha$, so its acceleration is $\omega^2 x\sin\alpha$ horizontally towards the axis. The tension T_A acts along PA, making angle α with the upward vertical. The tension T_B acts along PB, making angle β with the downward vertical.



The tensions

Resolving vertically (there is no vertical acceleration) and horizontally towards the axis:

$$T_A \cos\alpha - T_B \cos\beta = mg, \tag{1}$$

$$T_A \sin\alpha + T_B \sin\beta = m\omega^2 x \sin\alpha. \tag{2}$$

These equations still hold if β is obtuse, that is, if P is below B, because then $\cos\beta < 0$ and the vertical component of T_B points upwards, as it should.

To eliminate T_B, multiply (1) by $\sin\beta$ and (2) by $\cos\beta$ and add. Using $\sin\alpha\cos\beta + \cos\alpha\sin\beta = \sin(\alpha+\beta)$,

$$T_A = \frac{m\left(g\sin\beta + \omega^2 x \sin\alpha\cos\beta\right)}{\sin(\alpha+\beta)}.$$

To eliminate T_A, multiply (2) by $\cos\alpha$ and (1) by $\sin\alpha$ and subtract:

$$T_B = \frac{m\sin\alpha\left(\omega^2 x\cos\alpha - g\right)}{\sin(\alpha+\beta)}.$$

Here α and β are two angles of the triangle ABP, so $0 < \alpha+\beta < \pi$ and $\sin(\alpha+\beta) > 0$. Also $\sin\alpha > 0$. A taut string has $T_B \geq 0$, so

$$\omega^2 x\cos\alpha \geq g,$$

as required.

The case $\omega^2 x\cos\alpha = g$

Now $T_B = 0$: the lower string is just taut but carries no tension. Equation (1) then gives

$$T_A = \frac{mg}{\cos\alpha},$$

which agrees with (2), since $T_A = m\omega^2 x$ there. Note also that $\cos\alpha = g/(\omega^2 x) > 0$, so α is acute.

? **Ask yourself** Where does an inequality involving only h, d and α come from?

It comes from the geometry, not the mechanics. Apply the cosine rule to triangle ABP at the vertex A:

$$d^2 = h^2 + x^2 - 2hx\cos\alpha \quad\Longleftrightarrow\quad x^2 - (2h\cos\alpha)\,x + (h^2 - d^2) = 0.$$

The length x is a real root of this quadratic, so its discriminant is non-negative:

$$4h^2\cos^2\alpha - 4(h^2 - d^2) \geq 0 \quad\Longleftrightarrow\quad h^2\cos^2\alpha \geq h^2 - d^2.$$

Both $h\cos\alpha$ and $\sqrt{h^2-d^2}$ are positive (since α is acute and $h > d$), so we may take square roots and get

$$h\cos\alpha \geq \sqrt{h^2 - d^2}.$$

Bounds on T_A

The particle is genuinely off the axis, so $0 < \alpha < \frac{\pi}{2}$ and $0 < \cos\alpha < 1$. Hence $T_A = mg/\cos\alpha > mg$. For the other bound, the previous result gives $\cos\alpha \geq \sqrt{h^2 - d^2}/h$, so

$$mg < T_A = \frac{mg}{\cos\alpha} \leq \frac{mgh}{\sqrt{h^2 - d^2}}.$$

When the upper bound is attained

Equality needs $h\cos\alpha = \sqrt{h^2 - d^2}$, so the discriminant is zero and the quadratic has the repeated root

$$x = h\cos\alpha = \sqrt{h^2 - d^2}.$$

Then $x^2 + d^2 = h^2$, that is, $AP^2 + BP^2 = AB^2$. By the converse of Pythagoras, the angle APB is a right angle: the two strings are perpendicular, and P lies on the circle with diameter AB. Equivalently, AP is tangent to the circle of radius d about B, which is why α is as large as it can be here. In this position $\sin\alpha = d/h$, and P is a height d^2/h above B. For any other acute α allowed by the inequality there are two possible triangles (two values of x), and they merge at this position.

Checks

- If ω is very large, $T_B \approx m\omega^2 x\sin\alpha/\sin(\alpha+\beta)$ grows like ω^2, as both strings must then supply the large centripetal force.
- Take $h = 3$, $d = 2$ at the extreme position: $\cos\alpha = \sqrt{5}/3$, $x = \sqrt{5}$, and $x^2 + d^2 = 5 + 4 = 9 = h^2$. Then $T_A/(mg) = 3/\sqrt{5} \approx 1.342$, equal to $h/\sqrt{h^2 - d^2}$.
- Substituting the formulas for T_A and T_B back into the vector equation of motion, in many random configurations checked by computer, gives zero residual.

Common slips

- Taking the radius of the circle to be x rather than $x\sin\alpha$.
- Giving T_B the wrong vertical sense. With β measured from BA, the string PB pulls P downwards (when β is acute).
- Squaring or square-rooting an inequality without checking that both sides are positive.

Beyond STEP

The boundary case $T_B = 0$ is a constraint becoming *inactive*: a string is a one-sided constraint (it can pull but not push). Problems of this kind, with inequality constraints, lead to Lagrange multipliers with sign conditions in mechanics and optimisation. The

conical pendulum and its relatives are treated well in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2014, Question 12

Probability and Statistics • Lognormal Distribution

STEP III 2014, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q12

The key idea

Because $y = \mathrm{e}^x$ is strictly increasing, statements about Y turn into statements about X simply by taking logarithms: $Y \le y$ exactly when $X \le \ln y$. That gives the median and the density of Y at once. For the mean, the trick is to complete the square in the exponent so that the integrand becomes another normal density, whose integral we already know is 1.

Solution

Part (i)

Since e^x is strictly increasing, the events $\{Y \le \mathrm{e}^{x_m}\}$ and $\{X \le x_m\}$ are the same event. Hence

$$\mathrm{P}\big(Y \le \mathrm{e}^{x_m}\big) = \mathrm{P}(X \le x_m) = \tfrac{1}{2},$$

so the median of Y is

$$y_m = \mathrm{e}^{x_m}.$$

(Medians pass through any increasing function in this way; means do not, as part (iv) shows.)

Part (ii)

Y takes only positive values. For $y > 0$, let G be the cumulative distribution function of Y. Using the same equivalence as before,

$$G(y) = \mathrm{P}(Y \le y) = \mathrm{P}(X \le \ln y) = F(\ln y).$$

Differentiating with the chain rule, and using $F' = f$,

$$g(y) = G'(y) = f(\ln y) \cdot \frac{1}{y} = \frac{f(\ln y)}{y}, \qquad y > 0,$$

as required (and $g(y) = 0$ for $y \leq 0$).

The mode λ is where g takes its maximum. It lies in the open interval $(0, \infty)$, where g is differentiable, so $g'(\lambda) = 0$. By the quotient rule,

$$g'(y) = \frac{f'(\ln y) \cdot \frac{1}{y} \cdot y - f(\ln y)}{y^2} = \frac{f'(\ln y) - f(\ln y)}{y^2}.$$

Since $y^2 \neq 0$, setting $g'(\lambda) = 0$ gives

$$f'(\ln \lambda) = f(\ln \lambda),$$

which is the required condition.

Part (iii)

The function

$$\frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-(x-\mu-\sigma^2)^2/(2\sigma^2)}$$

is exactly the probability density function of a $\mathrm{N}(\mu + \sigma^2, \sigma^2)$ random variable: it is the given normal density with μ replaced by $\mu + \sigma^2$. Any probability density function integrates to 1 over the whole real line, which explains the stated integral.

Now

$$\mathrm{E}(Y) = \mathrm{E}\left(\mathrm{e}^X\right) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^x \, \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} \, \mathrm{d}x.$$

We complete the square in the combined exponent. Expanding $(x - \mu - \sigma^2)^2 = (x - \mu)^2 - 2\sigma^2 x + 2\mu\sigma^2 + \sigma^4$, we get

$$\begin{aligned} x - \frac{(x-\mu)^2}{2\sigma^2} &= -\frac{(x-\mu)^2 - 2\sigma^2 x}{2\sigma^2} \\ &= -\frac{(x-\mu-\sigma^2)^2 - 2\mu\sigma^2 - \sigma^4}{2\sigma^2} \\ &= -\frac{(x-\mu-\sigma^2)^2}{2\sigma^2} + \mu + \tfrac{1}{2}\sigma^2. \end{aligned}$$

The constant $\mathrm{e}^{\mu + \frac{1}{2}\sigma^2}$ comes outside the integral, so

$$\mathrm{E}(Y) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2} \cdot \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu-\sigma^2)^2/(2\sigma^2)} \, \mathrm{d}x = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2},$$

using the integral explained above. This is the required result.

Part (iv)

We find all three quantities explicitly.

Median. The normal density is symmetric about μ, so $x_m = \mu$, and by part (i) $y_m = \mathrm{e}^{\mu}$.

Mode. Differentiating the normal density,

$$f'(x) = -\frac{x-\mu}{\sigma^2} f(x).$$

Ask yourself How do we use the condition from part (ii) when f is never zero?

Since $f(x) > 0$ for every x, we can divide the condition $f'(\ln\lambda) = f(\ln\lambda)$ by $f(\ln\lambda)$ to get

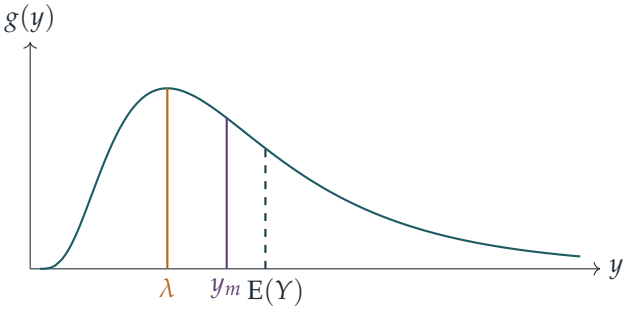
$$-\frac{\ln\lambda - \mu}{\sigma^2} = 1 \quad \Longrightarrow \quad \ln\lambda = \mu - \sigma^2 \quad \Longrightarrow \quad \lambda = \mathrm{e}^{\mu-\sigma^2}.$$

This is the only stationary point of g, and we are told Y has a unique mode, so it is the mode.

Comparison. Since $\sigma > 0$, we have $\mu - \sigma^2 < \mu < \mu + \frac{1}{2}\sigma^2$, and e^t is strictly increasing, so

$$\lambda = \mathrm{e}^{\mu-\sigma^2} \;<\; y_m = \mathrm{e}^{\mu} \;<\; \mathrm{E}(Y) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2},$$

as required. The strict inequalities rely on $\sigma^2 > 0$, which holds for any genuine normal distribution.



The sketch shows the density of Y for $\mu = 0$, $\sigma = 0.6$: the long right-hand tail pulls the mean furthest to the right, the median sits in the middle, and the peak is to the left of both.

Checks

- As $\sigma \to 0$, Y is concentrated at e^{μ} and all three values tend to e^{μ}, as they should.
- With $\mu = 0$, $\sigma = 1$: $\lambda = \mathrm{e}^{-1} \approx 0.368$, $y_m = 1$ and $\mathrm{E}(Y) = \mathrm{e}^{1/2} \approx 1.649$. Numerical integration of $\mathrm{e}^x f(x)$ and a numerical search for the maximum of g agree with these values.
- The ratio $\mathrm{E}(Y)/\lambda = \mathrm{e}^{3\sigma^2/2}$ depends only on σ, which makes sense: changing μ just rescales Y by the factor e^{μ}.

Common slips

- Writing the density of Y as $f(\ln y)$, forgetting the factor $1/y$ from the chain rule. Without it, the mode would come out as e^{μ}, the same as the median.

- Assuming $\mathrm{E}(\mathrm{e}^X) = \mathrm{e}^{\mathrm{E}(X)}$. That is true for the median but not for the mean.
- Sign errors when completing the square: expand $(x - \mu - \sigma^2)^2$ carefully and check the constant left over.

Beyond STEP

Y here has the *lognormal* distribution, which is used to model incomes, share prices and particle sizes. The inequality $\mathrm{E}(\mathrm{e}^X) \geq \mathrm{e}^{\mathrm{E}(X)}$ is a case of Jensen's inequality for convex functions, and the calculation in part (iii) is the moment generating function of the normal distribution, $\mathrm{E}(\mathrm{e}^{tX}) = \mathrm{e}^{\mu t + \frac{1}{2}\sigma^2 t^2}$, evaluated at $t = 1$. Blitzstein and Hwang, *Introduction to Probability*, covers transformations of random variables, moment generating functions and the lognormal distribution.

WORKED SOLUTION

STEP III 2014, Question 13

Probability and Statistics • Probability Generating Functions

STEP III 2014, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q13

The key idea

Condition on what happens in the first round. The game has no memory: after any round in which it carries on, what follows is a brand-new copy of the same game, so the *extra* score from that point on has the same distribution as N itself. That gives an equation for G in terms of itself, which we solve and expand as a geometric series.

Solution

Write $p_n = P(N = n)$, so $G(t) = \sum_{n\geq 0} p_n t^n$. Let E_1, E_2, E_3 be the events that the first round has outcome 1, 2 or 3, so that $P(E_1) = a$, $P(E_2) = b$ and $P(E_3) = c$.

Part (i)

Given E_1, the game stops with score 0, so $P(N = 0 \mid E_1) = 1$ and $P(N = n \mid E_1) = 0$ for $n \geq 1$. The conditional generating function is therefore

$$\sum_{n\geq 0} P(N = n \mid E_1)\, t^n = 1 \cdot t^0 = 1.$$

Part (ii)

Given E_2, the score is still 0 after round one and the game continues. Since the rounds have the same probabilities and do not depend on what came before, the rounds from round two onwards form a fresh game starting from score 0. Let N' be the score that fresh game produces; it has the same distribution as N. The final score is $0 + N' = N'$, so

$$P(N = n \mid E_2) = P(N' = n) = p_n \quad \text{for every } n \geq 0,$$

and the conditional generating function is $\sum p_n t^n = G(t)$.

Part (iii)

In the same way, given E_3 the score is 1 after round one and the rest is a fresh game with score N', so the final score is $1 + N'$. Hence $P(N = 0 \mid E_3) = 0$ and $P(N = n \mid E_3) = p_{n-1}$ for $n \geq 1$; the conditional generating function is $\sum_{n\geq 1} p_{n-1}t^n = tG(t)$.

By the law of total probability,

$$p_n = a\,P(N = n \mid E_1) + b\,P(N = n \mid E_2) + c\,P(N = n \mid E_3).$$

The coefficient of t^n in $G(t)$ is p_n. On the right, the coefficients of t^n in 1, $G(t)$ and $tG(t)$ are exactly $P(N = n \mid E_1)$, $P(N = n \mid E_2)$ and $P(N = n \mid E_3)$ from parts (i)–(iii). So the coefficients of t^n in $G(t)$ and in $a + bG(t) + ctG(t)$ agree for every $n \geq 0$, which proves

$$G(t) = a + bG(t) + ctG(t).$$

Ask yourself Can we divide by $1 - b - ct$, and is it legitimate to expand as a geometric series?

Rearranging, $G(t)(1 - b - ct) = a$. Since $1 - b = a + c > c > 0$, we have $1 - b - ct \neq 0$ whenever $|t| < (1 - b)/c$, and this range includes $|t| \leq 1$ because $(1 - b)/c = 1 + a/c > 1$. For such t,

$$G(t) = \frac{a}{1 - b - ct} = \frac{a}{1 - b} \cdot \frac{1}{1 - \dfrac{ct}{1 - b}} = \frac{a}{1 - b} \sum_{n\geq 0} \left(\frac{c}{1 - b}\right)^n t^n,$$

since $|ct/(1 - b)| < 1$. Reading off the coefficient of t^n,

$$P(N = n) = \frac{ac^n}{(1 - b)^{n+1}}, \qquad n \geq 0.$$

(Equivalently, the coefficient comparison gives $p_0 = a + bp_0$ and $p_n = bp_n + cp_{n-1}$ for $n \geq 1$, i.e. $p_0 = a/(1 - b)$ and $p_n = \frac{c}{1-b} p_{n-1}$, which leads to the same formula.)

Part (iv)

The mean is $\mu = G'(1)$. Differentiating $G(t) = a(1 - b - ct)^{-1}$,

$$G'(t) = \frac{ac}{(1 - b - ct)^2}, \qquad \text{so} \qquad \mu = G'(1) = \frac{ac}{(1 - b - c)^2} = \frac{ac}{a^2} = \frac{c}{a},$$

using $1 - b - c = a$. Then $1 + \mu = \dfrac{a + c}{a} = \dfrac{1 - b}{a}$, and so

$$\frac{\mu^n}{(1 + \mu)^{n+1}} = \frac{c^n}{a^n} \cdot \frac{a^{n+1}}{(1 - b)^{n+1}} = \frac{ac^n}{(1 - b)^{n+1}} = P(N = n),$$

as required.

Checks

- The probabilities sum to 1: $\sum_n p_n = G(1) = a/(1-b-c) = a/a = 1$.
- A direct argument: ignore the "wasted" outcome-2 rounds. Each remaining round is a stop with probability $a/(a+c)$ or a point with probability $c/(a+c)$, so N is the number of points before the first stop, and $P(N=n) = \left(\frac{c}{a+c}\right)^n \frac{a}{a+c}$, which is the same formula since $a + c = 1 - b$.
- With $a = \frac{1}{5}$, $b = \frac{3}{10}$, $c = \frac{1}{2}$ we get $\mu = \frac{5}{2}$ and $P(N=0) = \frac{2}{7} \approx 0.2857$, $P(N=1) = \frac{10}{49} \approx 0.2041$. A computer simulation of 2×10^5 games gave 0.2855, 0.2038 and a mean score of 2.498.

Common slips

- Writing the conditional generating function in case 3 as $G(t)$ instead of $tG(t)$: the point already scored shifts every power of t up by one.
- Forgetting to say *why* the game after round one is a copy of the original: it relies on each round having the same probabilities, independently of the past.

Beyond STEP

Conditioning on the first step is the basic tool of Markov chain theory, where it is called *first-step analysis*, and the answer here is a geometric distribution in disguise. Part (iv) shows that a geometric distribution on $\{0, 1, 2, \dots\}$ is determined by its mean alone. The same equation-for-G trick handles branching processes, where it leads to the famous result about the probability that a family name dies out. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and branching processes very readably.

Beyond STEP

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